Fixed Wing UAV for the Delivery of Emergency Medical Supplies to Remote Areas

Conceptual Design report submitted in fulfillment of the requirements
for the completion of the course

AS5213: Design of MAVs/UAVs

by

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Declaration

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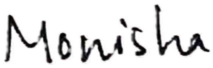
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Group 11: Skunk Works

Abstract

This report presents the conceptual design of a fixed-wing, electrically powered Unmanned Aerial Vehicle (UAV) for the delivery of emergency medical supplies— specifically antivenom and blood products—to remote and rural areas of India where ground-transport delays represent a critical threat to patient survival.

The design is constrained to a maximum wingspan of 2.00 m and targets a 1.00 kg droppable medical payload over a round-trip operational radius of 40.00 km to 50.00 km. Starting from a literature-derived empty-weight regression and an iterative weight-convergence methodology, a Maximum Take-Off Weight (MTOW) of 6.59 kg was established across two design iterations.

Aerodynamic optimisation using a wetted-aspect-ratio regression yielded a maximum lift-to-drag ratio of 12.55 and a zero-lift drag coefficient $C_{D_0} = 0.02873$. The wing employs a Selig S7055 aerofoil in a rectangular, high-wing planform with an aspect ratio of 7.53 and an area of 0.53 m^2. Propulsion is provided by two T-Motor AM480 900 KV motors driving APC 12×6E propellers, powered by a GenX 6S 12 Ah LiPo battery.

Longitudinal static stability was verified analytically, with a static margin of approximately 14.00 % achieved through a NACA 0012 horizontal stabiliser of area 0.13 m^2 on a 0.70 m tail arm. Performance analysis confirms that the design meets all primary mission requirements: a maximum range of 83.15 km, an endurance of 2.00 h 5.00 min, and cruise speeds in the range 18.00 $m\,s^{-1}$ to 22.00 $m\,s^{-1}$.

Index Terms — Fixed-wing UAV, medical logistics, STOL, weight estimation, longitudinal static stability, drag polar, flight envelope.

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Nomenclature

Roman Symbols

$(L/D)_{max}$	Maximum lift-to-drag ratio
$\bar{c}$	Mean aerodynamic chord (m)
AR	Aspect ratio
AR_{wet}	Wetted aspect ratio, b^2/S_{wet}
b	Wing span (m)
C_D	Total drag coefficient
C_f	Skin-friction coefficient
C_L	Lift coefficient
C_m	Pitching moment coefficient
C_{D_0}	Zero-lift (parasitic) drag coefficient
C_{D_i}	Induced drag coefficient
C_{L_α}	Lift-curve slope (rad^{-1})
$C_{L_{max}}$	Maximum lift coefficient
C_{m_0}	Pitching moment coefficient at zero angle of attack
C_{m_α}	Pitching moment slope with angle of attack (rad^{-1})
$C_{m_{ac}}$	Pitching moment coefficient about aerodynamic centre

C_{n_β}	Yawing moment coefficient due to sideslip (rad^{-1})
D	Drag force (N)
E	Endurance (h)
e	Oswald span efficiency factor
FF	Component form factor
g	Gravitational acceleration, $9.81\,\mathrm{m\,s^{-2}}$
i_t	Horizontal tail setting angle (°)
i_w	Wing incidence angle (°)
k	Induced drag factor, $1/(\pi e\,AR)$
K_n	Static margin
L	Lift force (N)
l_t, l_v	Horizontal and vertical tail moment arms (m)
n	Load factor
P	Power (W)
Q	Interference factor
R	Range (km)
Re	Reynolds number
ROC	Rate of climb ($\mathrm{m\,s^{-1}}$)
S	Wing reference area (m^2)
S_t, S_v	Horizontal and vertical tail areas (m^2)
S_{wet}	Wetted area (m^2)
T	Thrust (N)

V	Airspeed ($\mathrm{m\,s^{-1}}$)
V_a	Maneuvering speed ($\mathrm{m\,s^{-1}}$)
V_d	Design dive speed ($\mathrm{m\,s^{-1}}$)
V_g	Negative stall speed ($\mathrm{m\,s^{-1}}$)
V_H	Horizontal tail volume coefficient
V_s	Stall speed ($\mathrm{m\,s^{-1}}$)
W	Weight (N)
W/P	Power loading ($\mathrm{N\,W^{-1}}$)
W/S	Wing loading ($\mathrm{N\,m^{-2}}$)
W_0	Maximum take-off weight (kg)
W_b	Battery mass (kg)
W_e	Empty mass (kg)
W_p	Payload mass (kg)
X_{ac}	Aerodynamic centre location from nose (m)
X_{cg}	Centre of gravity location from nose (m)
X_{np}	Neutral point location from nose (m)

Greek Symbols

α	Angle of attack (° or rad)
δ	Oswald efficiency correction factor for tapered wings
η_b	Battery efficiency
η_m	Motor efficiency
η_p	Propeller efficiency

η_t	Tail aerodynamic efficiency factor
Γ	Dihedral angle (°)
γ	Flight path angle (°)
Λ	Wing sweep angle (°)
λ	Wing taper ratio, c_{tip}/c_{root}
$\mathrm{d}\varepsilon/\mathrm{d}\alpha$	Downwash gradient
μ	Ground friction coefficient; dynamic viscosity ($\mathrm{N\,s\,m^{-2}}$)
ρ	Air density ($\mathrm{kg\,m^{-3}}$)
ε	Downwash angle (rad)

Subscripts

cl	Climb
cr	Cruise
des	Descent
loi	Loiter
req	Required
TO	Take-off

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Chapter 1

Mission Definition and Requirements

1.1 Mission Statement

The primary design objective is to "**Design and develop a fixed-wing Unmanned Aerial Vehicle (UAV) with a maximum wingspan of 2 m that is capable of carrying a 1 kg emergency medical payload over a maximum range of 80 km while ensuring safe, reliable, and efficient delivery**."

1.2 Significance of the idea

Medical drone deliveries address critical gaps in rural healthcare, especially for time-sensitive treatments such as antivenom. Snakebite affects millions of people yearly in India, killing up to 50,000 [2], mostly in villages where antivenom must reach within 6 hours for effectiveness. Antivenom neutralizes venom but has a narrow therapeutic window; post-bite delays cause organ failure and eventually death. In remote areas with poor roads and limited transport, delays can turn survivable snake bites into fatalities. Rural regions often lack reliable roads, ambulances, or pharmacies, leading to hours-long delays in essentials.

Drones bypass terrain issues, delivering from hubs to clinics in minutes. Rwanda's Zipline dramatically cut blood delivery times, boosting survival [3]; similar models could supply rural Tamil Nadu villages. Meghalaya's programme, flying 100 km weekly, further demonstrates feasibility [4]. UAVs like those in the project (with 2–4 hour endurance and 1.5–7 kg payloads) suit 1–2 kg antivenom kits over 50–80 km ranges.

1.3 Motivation

The motivation for this project stems from the growing demand for compact, deployable aerial platforms and the proven effectiveness of fixed-wing UAVs in medical logistics. Compared to multirotor systems, fixed-wing UAVs offer superior endurance, higher cruise speeds, and greater range, making them ideal for long-distance missions.

This project aims to design a fixed-wing UAV, with a dimensional constraint of a maximum 2-meter wingspan, capable of carrying a 1 kg medical payload over a maximum range of 80 km. In doing so, it addresses the urgent need for rapid and reliable medical response in rural and remote areas, particularly for time-sensitive emergencies where delayed access to essential supplies can mean the difference between life and death.

By focusing on aerodynamic efficiency, mission reliability, and ease of deployment, the project seeks to contribute both to practical healthcare delivery solutions and to the academic understanding of UAV design under real-world constraints.

1.4 Mission Requirements

To effectively bridge the gap between centralized medical facilities and remote, underdeveloped and rural regions, the proposed Fixed Wing UAV must meet stringent operational criteria. Unlike general disaster relief platforms, this system focuses on time-critical medical logistics, necessitating good speed, range, and payload protection. The key mission requirements are defined as follows:

- **Faster Cruise Capability:** Given the critical nature of payloads such as anti-venom and emergency blood supplies, the UAV must try to reduce delivery time. The system should aim to drastically reduce travel time compared to ground transportation.
- **Extended Operational Range (Hub-and-Spoke Model):** The aircraft must possess sufficient range to operate effectively in a hub-and-spoke logistics model, connecting a central district hospital to remote villages up to 40-50 km away (one way) without intermediate refueling or recharging.
- **Modular Airframe Design:** Modularity is a core requirement to ensure operational flexibility and ease of maintenance.
  - *Quick-Swap Payload Bay:* The fuselage must accommodate interchangeable payload modules (e.g., a temperature-controlled box for blood vs. a shock-proof container for vials) to suit varied medical missions.
  - *Transportability:* The wings and tail sections should be detachable to allow for easy ground transport of the UAV and rapid replacement of damaged parts.
  - *Rapid Battery Swap:* The power system must feature an easily accessible, quick-release battery compartment. This allows operators to swap depleted batteries for fresh ones instantly, minimizing downtime and enabling continuous mission sorties.
- **Infrastructure-Independent Launch and Recovery:** Lacking complex ground infrastructure like catapults or runways, the aircraft must demonstrate robust STOL performance. It requires the ability to take off and land on short, unprepared surfaces (e.g., village roads, open fields) within a distance of less than 200 meters.

- **Payload Integrity and Climate Control:** Unlike non-perishable relief supplies, medical payloads are sensitive to environmental factors. The payload compartment must minimize vibration transmission to prevent hemolysis (damage to blood cells) and maintain appropriate internal temperatures during flight, regardless of ambient conditions.

- **Effective Communication Systems:** The UAV should be equipped with reliable communication systems to facilitate real-time data exchange between the aircraft and ground control.

- **Precision Air-Drop and Payload Survivability:** Since the UAV operates without landing at the delivery site, the system must utilize a robust air-drop mechanism:

  - *Impact Protection:* The payload module must integrate aerodynamic decelerators (e.g., a parachute or streamer) and shock-absorbing internal structure to limit impact G-forces. This is critical to prevent breakage of glass vials (antivenom) and hemolysis in blood bags upon ground contact.

  - *Target Accuracy:* The drop system must account for wind drift and release altitude to achieve a Circular Error Probable (CEP) of less than 10 meters, ensuring the package lands safely within restricted village clearing zones.

- **Target Performance Parameters:** To meet the demands of rapid medical intervention, the design shall target the following performance envelope:

Parameter	**Requirement**
Cruise Speed	18 m/s – 25 m/s
Take-off Distance (STOL)	< 150 m
Operational Radius	40 km – 50 km
Total Range	80 km – 100 km
Endurance	1 hr – 2 hr
Payload Capacity	1 kg – 2 kg

Table 1.1: Mission Specifications for STOL Medical UAV

1.5 Mission Profile and Phases

- **Take-off:**

  For a fixed-wing UAV, takeoff necessitates a runway or alternative methods such as a catapult launch, hand launch, or assisted VTOL system. In conventional takeoffs, it is crucial to consider the factors influencing performance, including overall weight, required takeoff velocity, and takeoff distance. Importantly, these parameters are interconnected and must be evaluated collectively to ensure safe and efficient lift-off.

- **Climb:**

  After takeoff, the UAV ascends to the designated cruise altitude. This phase requires evaluation of climb angle, climb rate, and climb velocity while ensuring aerodynamic efficiency and structural integrity. Minimizing energy consumption during climb is an important design objective.

- **Cruise (Outbound):**

  Once cruise altitude is achieved, the UAV maintains steady-level flight toward the target location. The cruise phase is optimized for aerodynamic efficiency and minimum power consumption to maximize range, endurance, and speed.

- **Payload Drop:**

  Upon reaching the delivery coordinates, the UAV releases the payload at cruise altitude without performing a descent, enabling efficient parachute deployment. The sudden reduction in weight may induce transient disturbances; however, this effect is minimized by positioning the payload at the center of gravity. Any residual disturbances are countered through an effective flight control system to ensure stable flight.

- **Turn (Loiter):**

  After payload deployment, the UAV executes a coordinated turn to reverse direction. A brief loiter maneuver may be performed to ensure stable reorientation before initiating the return leg. Proper control and stability margins are essential during this maneuver.

- **Cruise (Inbound):**

  The UAV retraces its route back to the base at cruise altitude. Due to the reduced gross weight after payload release, slight adjustments in trim conditions and power settings may be required to maintain optimal performance and efficiency.

- **Descent:**

  As the UAV approaches the home station, it performs a controlled descent from cruise altitude. Descent angle and rate must be predefined considering aerodynamic stability, structural limits, and environmental conditions.

- **Landing:**

  During landing, parameters such as approach speed, glide angle, and touchdown velocity are critical to ensuring a safe and precise touchdown under varying operational and environmental conditions.

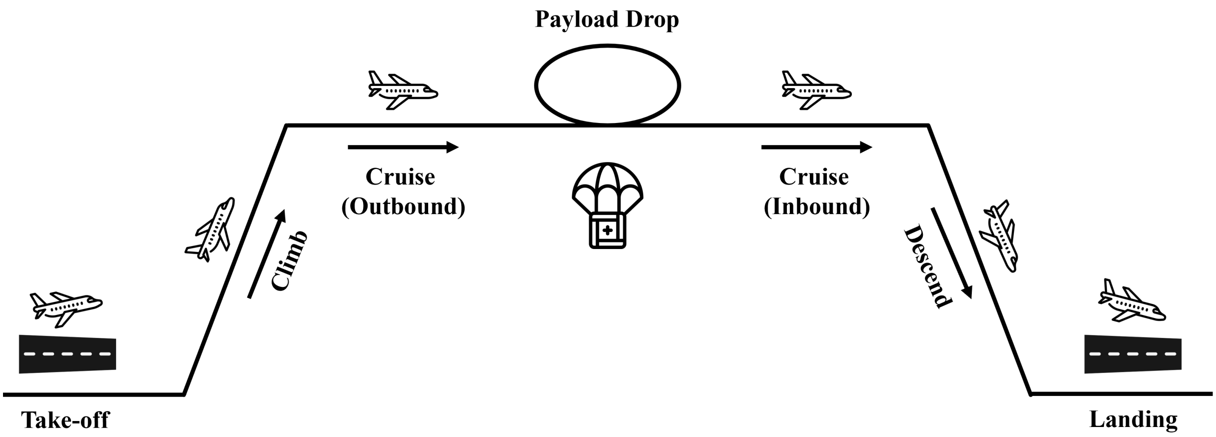


Figure 1.1: Mission Profile

UAV Name	Range (km)	Endurance (hr)	Cruise Speed (m/s)	MTOW (kg)	Payload (kg)	Span (m)	Area (m^2)	AR
Black Swift S2 [5]	110.00	2.00	18.00	9.50	2.30	3.00	0.70	12.86
Albatross UAV [6]	250.00	4.00	18.89	10.00	4.40	3.00	–	–
Puma LE [7]	60.00	6.50	13.00	12.20	2.50	4.60	–	–
3m Ranger [8]	–	1.50	13.89	7.00	2.80	3.00	0.70	12.86
Striver [9]	200.00	2.40	17.00	7.00	1.20	2.10	0.59	7.47
Believer [10]	20.00	2.00	20.00	5.50	0.70	1.96	0.51	7.53
PUMA 3AE [11]	60.00	3.00	13.00	7.10	1.80	2.80	–	–
Desert Hawk 3 [12]	15.00	1.50	15.50	3.72	0.90	1.37	0.33	5.78
Apple White Milo [13]	20.00	3.00	16.67	8.00	2.00	3.30	0.87	12.52
Raven B [14]	10.00	1.25	9.00	2.20	0.38	1.40	–	–
FT 100 Horus [15]	20.00	2.00	20.00	8.00	–	2.70	–	–
Aeromapper Talon [16]	30.00	2.00	16.11	3.50	–	2.00	–	–

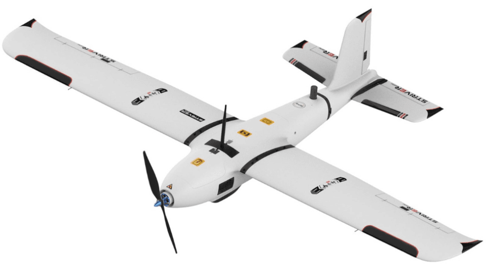
Table 1.3: Technical Specifications of Similar Fixed-Wing UAVs

Black Swift S2

Albatross UAV

Puma LE

3m Ranger



Striver

Believer

FIGURE 1.2: Gallery of Similar UAV Models

PUMA 3AE

Desert Hawk 3

Apple White Milo

Raven B

FT 100 Horus

Aeromapper Talon

FIGURE 1.2: Gallery of Similar UAV Models (Continued)

1.6 Data Collection

1.7 Key Design Parameters

Table 1.4: Key Design Parameters Established — Chapter 1: Mission Definition

Parameter	Symbol	Unit	Requirement / Value
Cruise Speed	V_{cr}	$\mathrm{m\,s^{-1}}$	18 – 25
Take-off Distance	S_{TO}	m	< 150
Operational Radius	—	km	40 – 50
Total Range	R	km	80 – 100
Endurance	E	hr	1 – 2
Drop Payload	$W_{p,\,drop}$	kg	1 – 2
Maximum Wingspan	b	m	≤ 2
Cruise Altitude (AGL)	h_{cr}	m	200
Climb / Glide Angle	γ	deg	8
Cruise Velocity (design)	V_{cr}	$\mathrm{m\,s^{-1}}$	22
Total Mission Payload	W_p	kg	2.1
Drop Payload (design)	$W_{p,\,drop}$	kg	1.2
CEP of Payload Drop	—	m	< 10

Mission requirement

New parameter this chapter

Mission requirement met

1.8 Design Progress

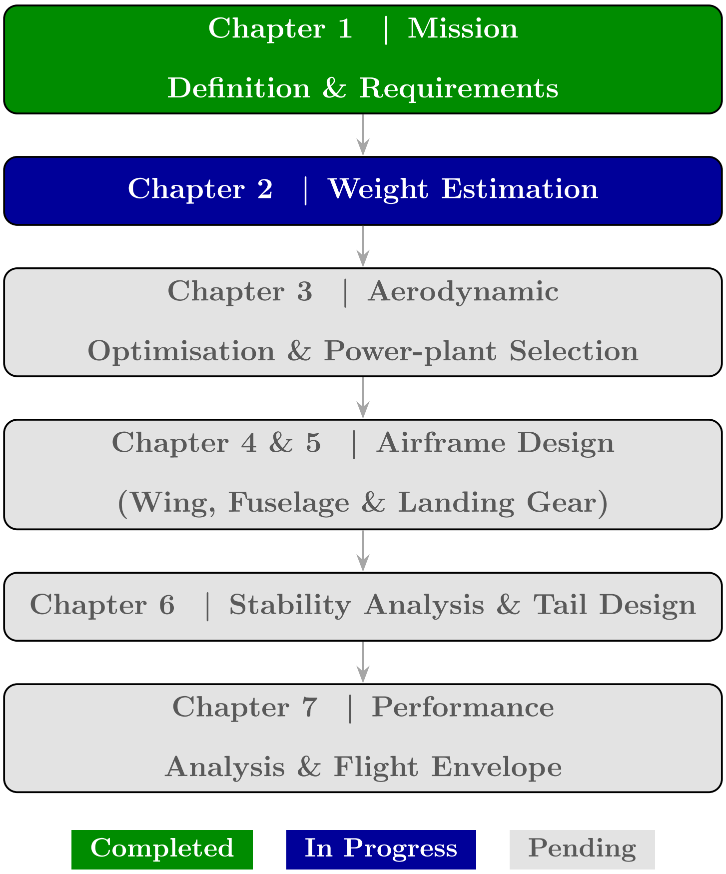


Figure 1.3: Overall UAV Conceptual Design Workflow — Progress after Chapter 1

Chapter 2

Weight Estimation

2.1 Detailed Mission Profile

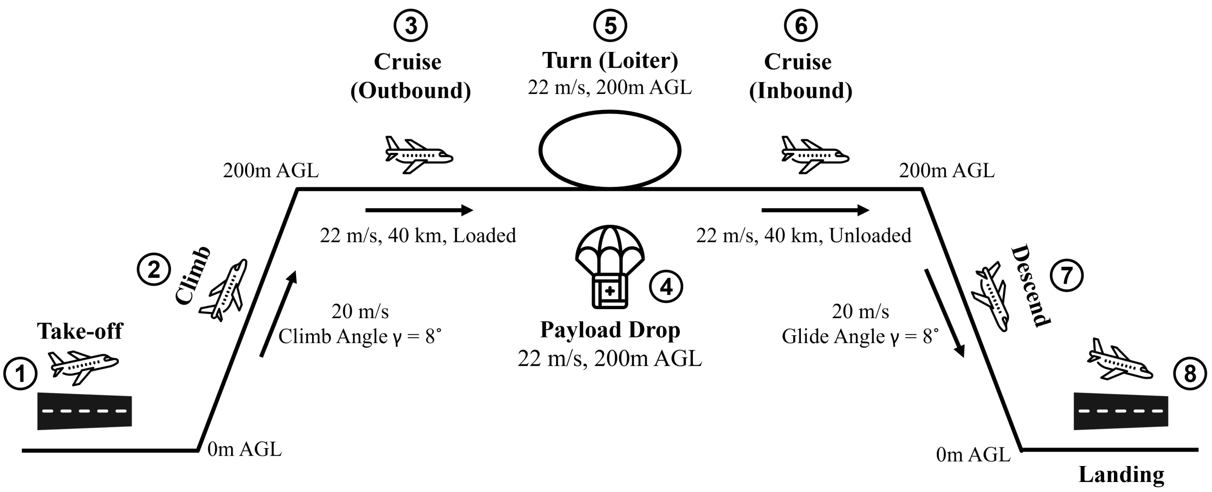


FIGURE 2.1: Phases of the Mission Profile

Mission profile parameters for each flight phase were obtained from a literature survey of UAVs with similar MTOW, payload capacity, and mission requirements. Key performance parameters - cruise, climb, and loiter velocities, take-off distance, climb angle, and glide angle - were adopted as baseline inputs for the present design. [17–21]

S.No	Mission Phase	Velocity	Segment	Altitude	Remarks
	(m/s)	Range (km)	AGL (m)		
1	Takeoff	–	–	0	-
2	Climb	20	1.42	0 – 200	Climb Angle: 8°
3	Outbound Cruise	22	40.0	200	Payload Loaded
4	Payload Drop	22	–	200	-
5	Loiter (Turn)	22	–	200	-
6	Inbound Cruise	22	40.0	200	Payload Empty
7	Descent	20	1.42	200 – 0	Glide Angle: 8°
8	Landing	–	–	0	-

Table 2.1: Mission Profile Parameters

2.2 Payload

2.2.1 Payload Weight Estimation and Configuration

Precise payload estimation is paramount for determining the UAV's Maximum Take-Off Weight (MTOW) and ensuring flight stability. The payload architecture is divided into two categories: the fixed internal avionics required for navigation and control, and the variable drop payload designed for medical logistics.

Based on mission requirements and a market survey of available components, the total payload is estimated at 2.1 kg. The breakdown of these components is detailed in Table 2.2.

Component	Weight (kg)
Variable Drop Payload	1.200
Flight Controller and Telemetry	0.492
Camera and Gimbal (Stabilization & Targeting)	0.183
Electronic Speed Controllers (ESC - 2 units)	0.147
GPS Module	0.042
Total Design Payload	**2.064 $\approx$ 2.1 kg**

Table 2.2: Total Payload Weight Estimate

The Flight Controller and GPS module enable autonomous navigation, while the Camera and Gimbal assembly provides real-time situational awareness and mission recording.

2.2.1.1 Medical Drop Payload Configuration

The drop payload is capped at 1.2 kg and placed at center of gravity (CG) within safe margins after release for the stability post drop. This mass includes the delivery system itself and the medical assets.

The delivery system comprises a servo-actuated drop mechanism and a parachute for safe drop and payload integrity, weighing a cumulative 0.32 kg. This leaves a functional payload capacity of 0.88 kg for medical supplies. A standard mission profile configuration - consisting of a 450 mL blood bag and two units of antivenom - is utilized for this design. The mass allocation is presented in Table 2.3.

Component	**Quantity**	**Mass (kg)**
Blood Bag (Standard 450 mL) [22]	1	0.48
Antivenom Vials [23]	2	0.40
Parachute & Drop Mechanism [24]	1	0.32
Total Drop Mass	**-**	**1.20**

Table 2.3: Medical Drop Payload Mass Allocation

2.3 Empty Weight Calculations

To know the Overall Weight of the aircraft, we need to know the empty weight and battery weight. However, this depends on the aircraft overall weight. Thus, we need to use the iterative process to determine the UAV's weight whose steps are as follows:

$$W_o = W_e + W_b + W_p \tag{2.1}$$

$$W_o = W_o\left(\frac{W_e}{W_o}\right) + W_p + W_o\left(\frac{W_b}{W_o}\right) \tag{2.2}$$

$$W_o = \frac{W_p}{1 - \left(\frac{W_e}{W_o}\right) - \left(\frac{W_b}{W_o}\right)} \tag{2.3}$$

It is observed that Empty weight (W_e) fraction follows a particular trend depending on the Total weight (W_o) which is given by,

$$\left(\frac{W_e}{W_o}\right) = aW_o^L \tag{2.4}$$

From the aircraft data collected during data collection section 1.6 we performed a linear curve fit for empty weight fraction v/s total weight on log scale and obtained the values of a and L.

$$a = 0.56 \quad ; \quad L = -0.0753$$

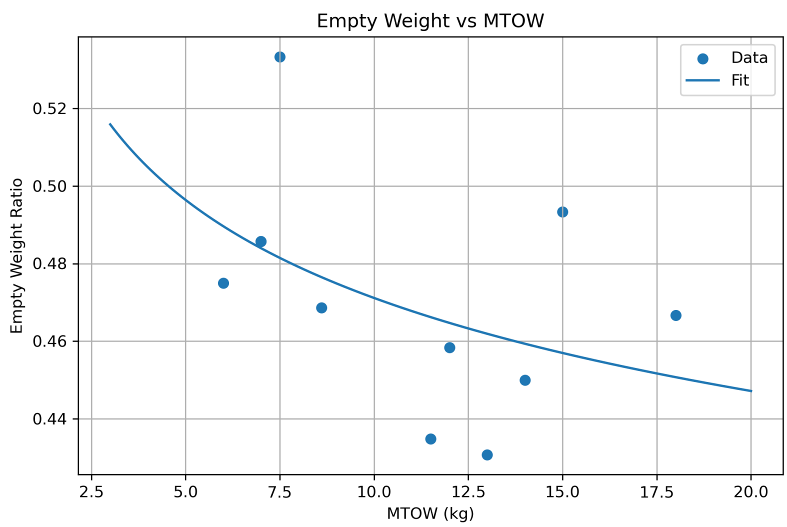


FIGURE 2.2: Variaiton of Empty Weight ratio with MTOW

After obtaining the corresponding relation between empty weight fraction and total weight, we take an initial guess for W_o, W_b for the required payload and find the total weight W_o from equation 2.1. After obtaining the new W_o we use a set of equations to calculate power during each segment of the required mission and thus calculate the new battery weight (W_b) corresponding to the new W_o. We repeat the process of obtaining W_o by using equation 2.1 by updating the values of W_o and W_b after every iteration until the value of W_o converges to a certain value.

2.4 Battery Weight Estimation

Calculating the battery weight for UAVs involves optimizing its weight based on the mission profile to avoid carrying unnecessary dead weight. To accurately estimate the battery weight, the first step is to determine the total energy required for each primary phase of flight. This follows from analyzing the energy consumption during different flight phases such as Takeoff, Climb, Cruise, locator, Descent, and Landing. By understanding the energy demands at each stage of the mission profile, we can calculate the total amount of energy needed for the entire flight. This information is essential for selecting an appropriately sized battery that can provide the required energy without adding excessive weight to the UAV, ensuring optimal performance and efficiency throughout the mission.

2.4.1 Power estimation

2.4.1.1 Cruise

The power required for the cruise condition of a UAV is a critical parameter that ensures the aircraft maintains steady flight during this phase. In essence, during cruise, the UAV is in force and moment equilibrium, where the thrust and drag forces are exactly balanced, or the UAV is in trim condition under unaccelerated flight conditions. The thrust required for cruise at a specific airspeed follows the drag profile (i.e., Parabolic Drag Profile under low speed terminology), ensuring that the total drag does not exceed the available thrust, and hence, the power required for cruise is a crucial factor in determining the energy consumption and performance of the UAV during this phase of flight. For steady level we have,

For steady, level flight, the fundamental force balances are:

$$L = W \tag{2.5}$$

$$T = D \tag{2.6}$$

The lift coefficient C_L can be expressed in terms of the aircraft's weight:

$$W = \frac{1}{2}\rho V^2 S C_L \quad \implies \quad C_L = \frac{2W}{\rho V^2 S} \tag{2.7}$$

Substituting this into the standard drag polar gives the total drag coefficient:

$$C_D = C_{D_0} + \frac{C_L^2}{\pi e AR} \tag{2.8}$$

The power required (P_R) is the product of thrust and velocity. Expanding this using the aerodynamic parameters yields:

$$\begin{aligned} P_R &= T \times V = D \times V \\ &= \left(\frac{1}{2}\rho V^2 S C_D\right) V \\ &= \frac{1}{2}\rho V^3 S C_{D_0} + \frac{2W^2}{\rho V S \pi e AR} \end{aligned} \tag{2.9}$$

This expanded form demonstrates that the power required is a function of several design and environmental variables:

$$P_R = f(W, \rho, V, C_{D_0}, S, e, AR) \tag{2.10}$$

By taking approximations and inputting the specific conditions of our mission profile, most of these parameters become constants. This simplifies the relationship to a direct function of the aircraft's weight:

$$P_R = f(W) \tag{2.11}$$

Evaluating this function for our cruise phase velocity of 22 m/s, the required power is calculated to be **114.58 W**.

2.4.2 Climb Power Requirement

The climb power requirement is determined from steady climbing flight conditions, where the UAV ascends at a constant climb angle γ and velocity V_{cl}. Rather than relying on the simplified approximation $P = W \times \mathrm{ROC} + P_R$, the required power is obtained from a complete aerodynamic force balance.

During a steady climb, the required thrust (T_{cl}) must overcome both the aerodynamic drag (D_{cl}) and the component of weight acting backward along the flight path:

$$T_{cl} = D_{cl} + W \sin\gamma \tag{2.12}$$

Similarly, the lift must balance the perpendicular component of the weight. This defines the required lift coefficient (C_L) for the climb phase:

$$W \cos\gamma = \frac{1}{2}\rho V_{cl}^2 S C_L \quad \Longrightarrow \quad C_L = \frac{2W\cos\gamma}{\rho V_{cl}^2 S} \tag{2.13}$$

Using the standard drag polar, the drag coefficient and the resulting aerodynamic drag force are calculated as:

$$\begin{aligned} C_D &= C_{D_0} + \frac{C_L^2}{\pi e AR} \\ D_{cl} &= \frac{1}{2}\rho V_{cl}^2 S C_D \end{aligned} \tag{2.14}$$

The total power required for the climb (P_{cl}) is the product of the required thrust and the climb velocity. Expanding the thrust term yields:

$$P_{cl} = T_{cl} \times V_{cl} = (D_{cl} + W\sin\gamma)\, V_{cl} \tag{2.15}$$

For the estimated UAV weight and selected climb parameters, the required climb power is evaluated to be **285.45 W**. This power ensures sufficient thrust to overcome aerodynamic drag and the gravitational penalty, enabling efficient and sustained climb performance.

2.4.2.1 Descent

During steady descent, the UAV follows a constant flight path angle γ at a velocity V_{des}. Because gravity assists the motion during this phase, the force balance along the flight path requires less thrust compared to a climb. The required thrust (T_{des}) is defined as:

$$T_{des} = D_{des} - W \sin \gamma \tag{2.16}$$

The corresponding descent power (P_{des}) is the product of this thrust and the descent velocity. To prevent non-physical negative power estimates—which could theoretically occur if the gravitational component exceeds the aerodynamic drag-a lower bound of zero is imposed:

$$\begin{aligned} P_{des} &= T_{des} \times V_{des} \\ &= \max(T_{des} V_{des}, 0) \end{aligned} \tag{2.17}$$

This mathematical condition ensures realistic and physically consistent power predictions throughout the descent phase.

2.4.2.2 Loiter

During the loiter phase, the UAV maintains steady, level flight at a constant velocity V_{loiter} for 15 minutes to allow for payload delivery and the necessary turning maneuvers to return to the starting position. Since the flight path angle is zero ($\gamma = 0$), the aerodynamic force balance simplifies entirely to:

$$L = W \tag{2.18}$$

$$T_{loiter} = D_{loiter} \tag{2.19}$$

This equilibrium dictates the required lift coefficient for the loiter segment:

$$W = \frac{1}{2}\rho V_{loiter}^2 S C_L \quad \Longrightarrow \quad C_L = \frac{2W}{\rho V_{loiter}^2 S} \tag{2.20}$$

Using the drag polar relation, the drag coefficient and the resulting required loiter power (P_{loiter}) are calculated sequentially:

$$C_D = C_{D_0} + \frac{C_L^2}{\pi e AR} \tag{2.21}$$
$$P_{loiter} = D_{loiter} \times V_{loiter}$$

For the selected loiter parameters, the required power is evaluated to be **114.58 W**. This represents the steady aerodynamic power necessary to sustain level flight throughout the entirety of the loiter segment.

2.4.2.3 Take off and Landing

The takeoff and landing segments do not take that long. In addition, estimating the energy spent on takeoff and landing is more complicated due to the need to carry power. So, we will take an excess power of 20percent to account for the errors in different power situations.

2.4.3 Battery energy density

We can calculate the energy by multiplying the power required by the time spent in that phase. From energy, we can obtain the battery weight by dividing it by the energy density. Taking a low bar value, we will assume an energy density of 120 Wh/kg

2.5 Convergence of Total Weight

We took the obtained battery weight (W_b) and substituted it in equation 2.3 to get an updated Total weight (W_o), which will have a corresponding battery weight (to be precise, corresponding total energy consumption). When the updated total and the updated battery weight are again substituted in equation 2.3 this will result in a new W_o and over and over again until the value of W_o converges to a finite non-zero value.

The analysis is done using MATLAB, and the weight obtained at every iteration is plotted against the iteration counter. The convergence plot is shown in Figure 2.3. The MATLAB code used for obtaining this plot and weight estimate is shown in Appendix A.

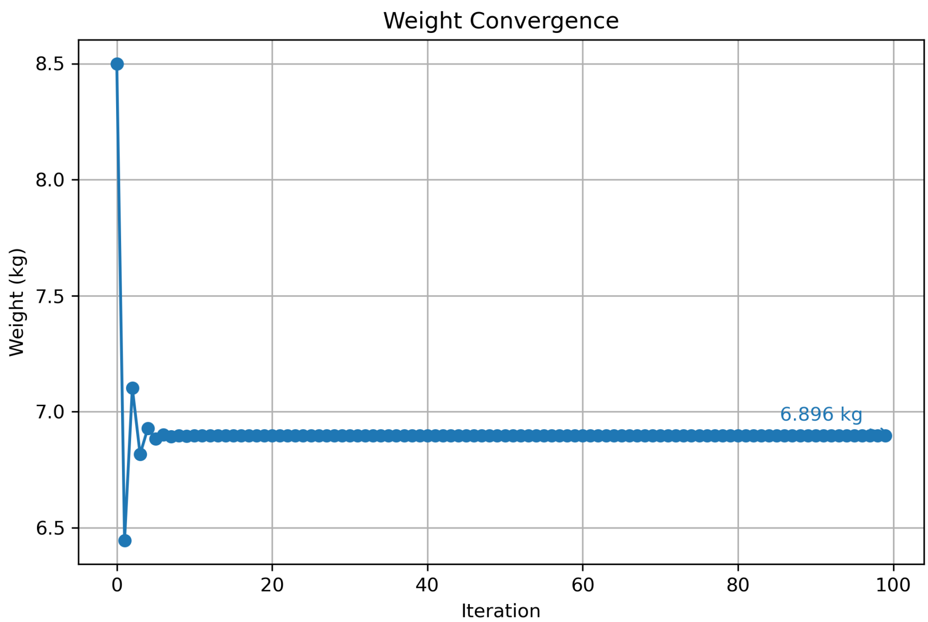


FIGURE 2.3: Weight Convergence

From the plot, we can see that the Weight W_o has converged to $W_o = 6.896$ Kg. This is the First Weight Estimate (MTOW) of the aircraft.

2.5.1 Summary

Table 2.4: Key Design Parameters — Chapter 2: Weight Estimation (First Iteration)

Parameter	Symbol	Unit	Previous	Current
Regression coefficients (empty-weight trend)				
Trend coefficient	a	—	—	0.56
Trend exponent	L	—	—	−0.0753
Assumed energy / battery				
Battery energy density	—	$\mathrm{Wh\,kg^{-1}}$	—	120
Battery (first selection)	—	—	—	GenX 3S 16 Ah
Weight breakdown (first convergence)				
Max Take-off Weight	W_0	kg	—	**6.896**
Empty Weight	W_e	kg	—	3.270
Battery Mass	W_b	kg	—	1.530
Total Payload	W_p	kg	—	2.100
Phase power requirements (at first MTOW)				
Cruise Power	P_{cr}	W	—	114.58
Climb Power	P_{cl}	W	—	285.45
Loiter Power	P_{loi}	W	—	114.58

New parameter this chapter

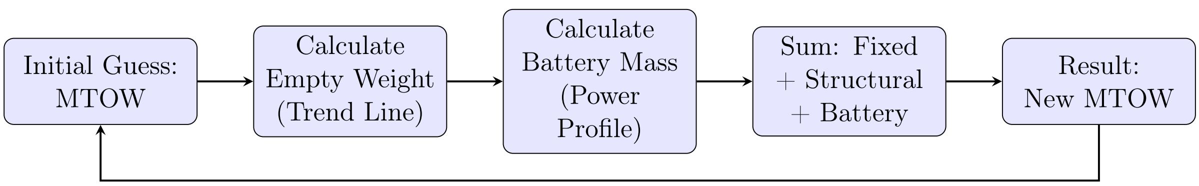


Figure 2.4: Iterative Weight Estimation Process

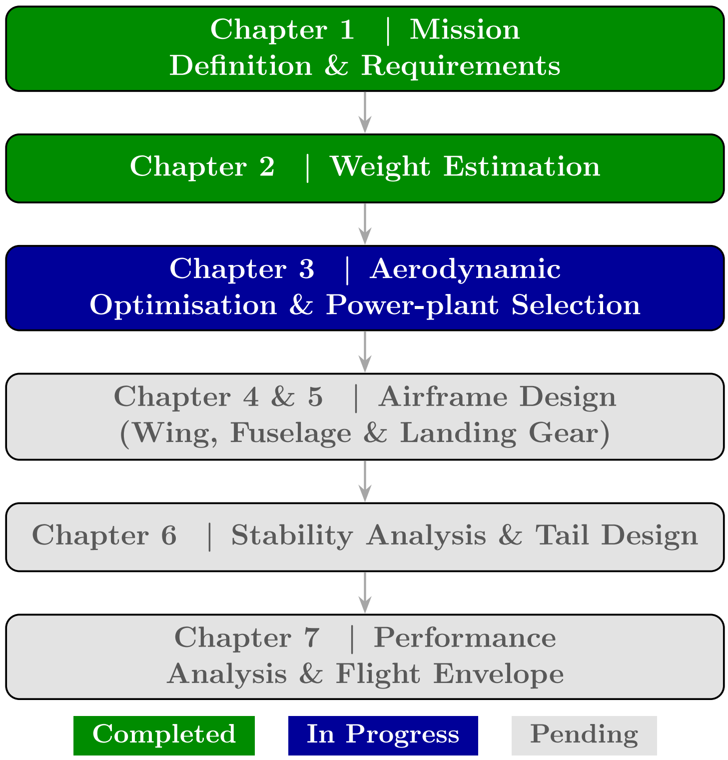


FIGURE 2.5: Overall UAV Conceptual Design Workflow — Progress after Chapter 2

Chapter 3

Optimization And Power-plant Selection

3.1 Propulsion Configuration Strategy

The propulsion system architecture for the medical delivery UAV is currently under active evaluation. While a conventional single-motor tractor configuration remains a primary candidate due to its inherent advantages in generating clean airflow over the control surfaces [25] and improving the static margin, a final decision is pending.

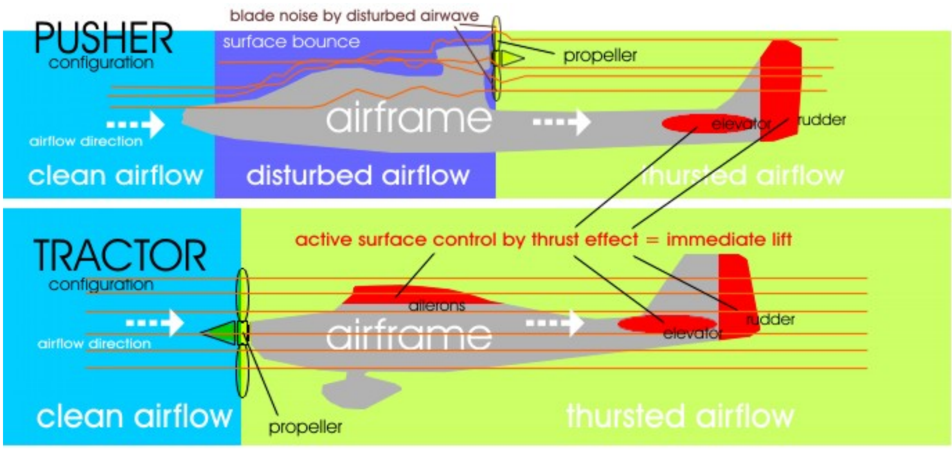


FIGURE 3.1: Comparison of Tractor and Pusher Configurations

Final Propulsion Configuration Decision

Following a comparative evaluation of tractor and pusher configurations, and after modelling both single-motor and twin-motor architectures against the peak power and yaw-authority requirements of the mission, a **twin-tractor configuration** was selected as the definitive propulsion arrangement.

The decision is justified on three grounds. First, the peak climb power demand of 361.50 W can be shared equally between two motors, reducing the thermal load and current draw on each unit and lowering the risk of single-point propulsion failure. Second, the lateral placement of motors on the wing provides differential thrust authority, substantially improving heading control during the autonomous payload drop manoeuvre over restricted village clearings—a phase identified in Chapter 1 as particularly demanding for precision. Third, the T-Motor AM480 900 KV selected in Section 3.4.1 is rated to a maximum output of 650.00 W per unit; a twin installation therefore delivers a combined ceiling of 1300.00 W, yielding a comfortable safety factor of $1300/361.5 \approx 3.6$ over the required climb power and providing adequate reserve for headwind and emergency loiter conditions.

The pusher and single-motor configurations were eliminated primarily because the pusher arrangement degrades control-surface airflow at low speeds—a critical concern for STOL operations—and a single high-power motor of equivalent thrust would require a larger, heavier propeller that conflicts with the ground clearance constraints established in Chapter 5. All subsequent analysis in this report assumes the twin-tractor configuration.

3.2 Aerodynamic Efficiency and maximum L/D Estimation

The maximum aerodynamic efficiency, denoted as $(L/D)_{max}$, is fundamentally influenced by the aircraft's wetted area and wingspan. To capture both geometric effects

FIGURE 3.2: Preliminary CAD of the UAV demonstrating the Tractor Configuration

simultaneously, the Wetted Aspect Ratio (AR_{wet}) is utilized as a comparative metric, defined as $AR_{wet} = (b^2/S_{wet})$.

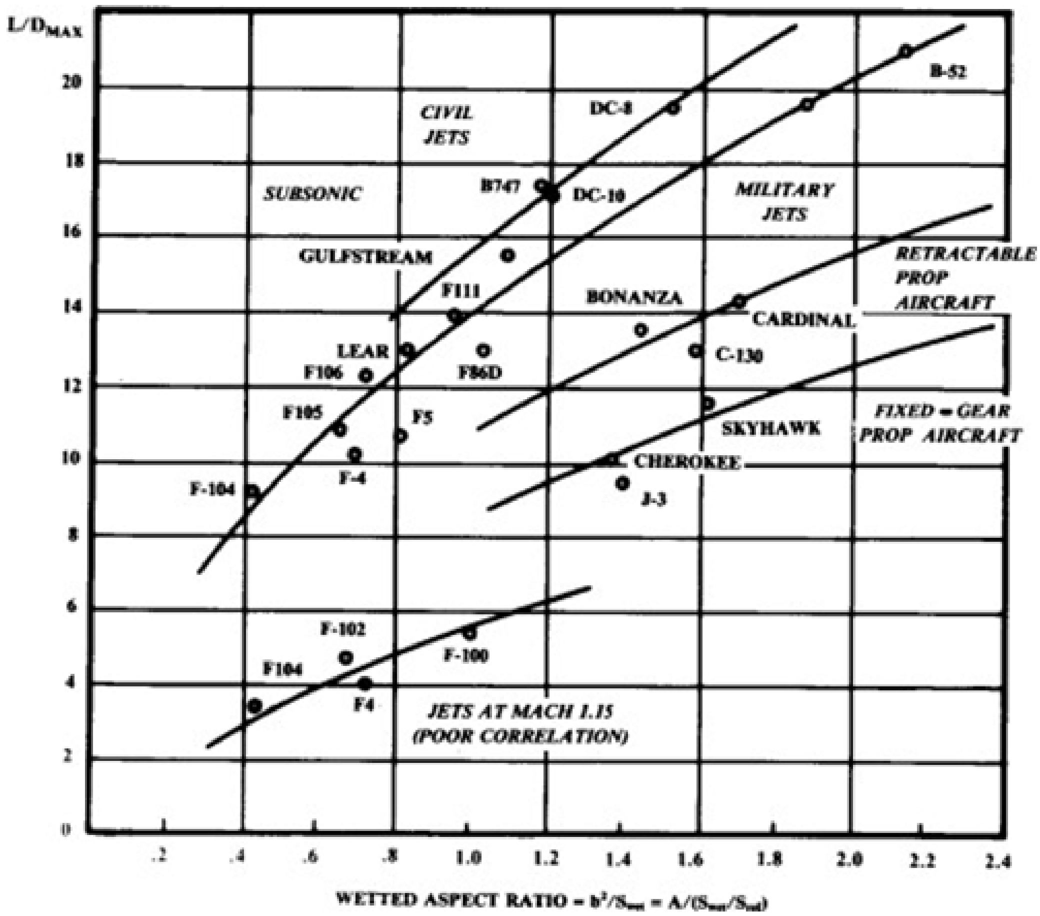


FIGURE 3.3: Maximum Lift to Drag ratio trends based on historical aircraft data [1]

3.2.1 Calculation of S_{wet} and AR_{wet}

By analyzing a database of existing UAV platforms with comparable operational envelopes, a correlational trend between $\sqrt{AR_{wet}}$ and $(L/D)_{max}$ was established. The reference data utilized for this trend analysis is detailed in Table 3.1.

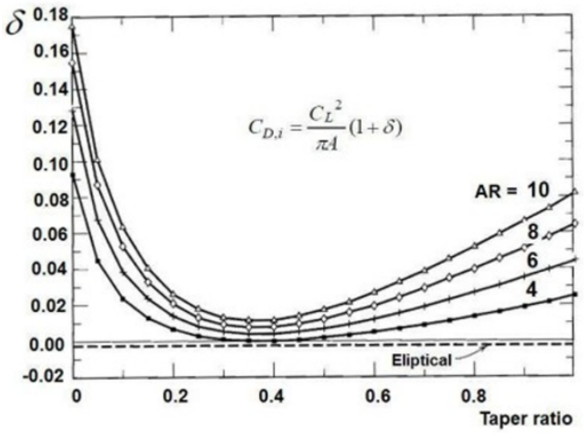


FIGURE 3.4: δ vs Taper ratio for a given AR [1]

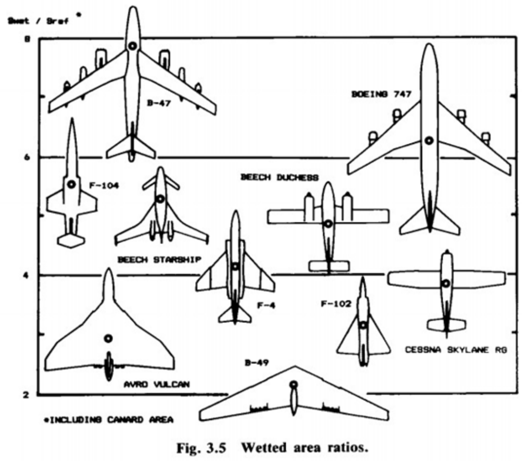


FIGURE 3.5: S_{wet}/S_{ref} ratios for various aircraft planforms [1]

Aircraft	MTOW (kg)	Cruise Vel (m/s)	Cruise Alt (m)	AR_{wet}	$(L/D)_{max}$
Yangda Mapird Plus [26]	11.5	19.44	3500	2.455	10.468
Yangda FW-250 [27]	12.0	23.33	3500	3.535	14.234
Avy Aera [28]	12	26.39	3100	2.727	18.857
Foxtech Baby Shark 260 [29]	13.0	22.00	3000	5.434	12.152
Foxtech AYK-250 [30]	15.0	25.00	2000	5.681	16.612
FlyDragon FLY-2100 [31]	7	18.5	500	2.292	16.063
Black Swift S2 [5]	9.5	18.06	4250	5.590	17.763
Foxtech Nimbus [32]	6	20.0	1500	2.304	14.569
Wingcopter 178 [33]	18.0	24.00	4500	1.517	5.583
Q200	7.0	18.00	4000	1.760	9.916
SkyEye Sierra [34]	12.5	20.00	3000	3.703	6.519

TABLE 3.1: $(L/D)_{max}$ values and parameters of similar reference UAVs

Oswald's or Span efficiency factor 'e' depends on sweep angle and taper ratio. along with AR. The formulae mentioned below are used to find Oswald's efficiency factor for the aircraft.

For Rectangular Wing:

$$e = 1.78\left(1 - 0.045AR^{0.68}\right) - 0.64 \tag{3.1}$$

For Swept Wing:

$$e = 4.61 \left(1 - 0.045 AR^{0.68}\right) \left(\cos \Lambda^{0.15}\right) - 3.1 \tag{3.2}$$

For Tapered Wing: Figure 3.4 is used to find the δ, where

$$e = \frac{1}{1+\delta} \tag{3.3}$$

S_{ref} is measured using Solidworks software, and wetted area ratios for the required platform are taken from Figure 3.5. Then, S_{ref} is found by the known S_{ref} and wetted area ratios. AR_{wet} is then calculated from the acquired S_{wet}.

3.2.2 Calculation of $(L/D)_{max}$

Aerodynamic parameters during cruise are tabulated for the collected aircraft to find $\left(\frac{L}{D}\right)_{\text{max}}$. For a propeller aircraft, the most efficient cruise velocity corresponds to the velocity at $\left(\frac{L}{D}\right)_{\text{max}}$. However, the most efficient loiter speed corresponds to the velocity at 86.66% $\left(\frac{L}{D}\right)_{\text{max}}$. Thus, finding $\left(\frac{L}{D}\right)_{\text{max}}$ for our aircraft is necessary to calculate $\frac{L}{D}$ at different mission segments such as cruise, climb, and loiter.

$$\left(\frac{L}{D}\right)_{\text{max}} \text{ corresponds to } \left(\frac{C_L}{C_D}\right)_{\text{max}} \tag{3.4}$$

$$\frac{C_L}{C_D} = \frac{C_L}{C_{D0} + C_{Di}} \tag{3.5}$$

For $\dfrac{C_L}{C_D}$ to be maximum,

$$C_{D0} = C_{Di} = \frac{C_L^2}{\pi e AR} \tag{3.6}$$

$$\therefore \left(\frac{C_L}{C_D}\right)_{\text{max}} = \left(\frac{L}{D}\right)_{\text{max}} \tag{3.7}$$

$$= \frac{\pi e AR}{2C_L} \tag{3.8}$$

where C_L during cruise is given by

$$C_L = \frac{W_o}{\frac{1}{2}\rho S_{\text{ref}} V_{\text{cr}}^2} \tag{3.9}$$

Using equation 3.8 we obtained $\left(\frac{L}{D}\right)_{\text{max}}$ values for all the aircraft mentioned in Table 3.1 and plotted them against corresponding $\sqrt{AR_{\text{wet}}}$ and drew a linear fit.

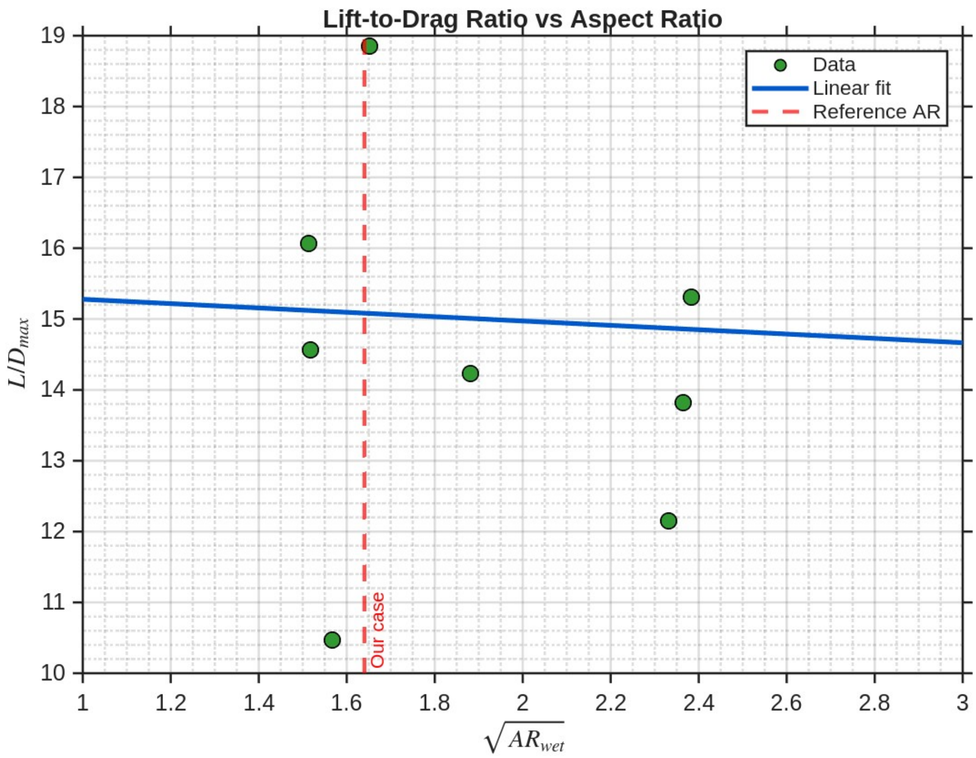


FIGURE 3.6: Regression Curve of $(L/D)_{max}$ and $\sqrt{AR_{wet}}$

Based on the regression curve derived from this historical data and our specific mission profile, the preliminary aerodynamic and geometric parameters for our UAV design have been iteratively defined and are summarized in Table 3.2.

Parameter	Symbol	Value
Aspect Ratio	AR	9.686
Maximum Takeoff Weight	W	68.516 N
Air density	ρ	1.225 kg/m^3
Wing span	b	2.0 m
Wing area	S	0.413 m^2
Oswald efficiency	e	0.7648
Induced drag factor	k	0.0430
Zero-lift drag coefficient	C_{D0}	0.0256
Maximum lift-to-drag ratio	$(L/D)_{max}$	15.0828

TABLE 3.2: Finalized Aircraft Parameters for Analysis

3.3 Estimation of Optimum Velocities

To maximize the efficiency of the battery-powered system, optimal flight velocities must be determined for each mission phase: climb, cruise, and loiter.

3.3.1 Optimum Velocity during Climb

The climbing phase imposes substantial power demands on the propulsion system. During this segment, the total thrust must overcome both the aerodynamic drag and the longitudinal component of the aircraft's weight.

$$L = W \cos\gamma \tag{3.10}$$

$$T = W \sin\gamma + D \tag{3.11}$$

Optimizing the climb trajectory involves determining the velocity that minimizes total energy expenditure for a specified altitude gain, which consequently mandates minimizing aerodynamic drag.

$$D_{min} = \frac{L}{(L/D)_{max}} \tag{3.12}$$

3.3.2 Optimum Velocity during Cruise

The aircraft spends the majority of its flight time in the cruise phase. For battery-powered operations, maximizing the operational radius requires flying at the velocity corresponding to minimum drag, thereby operating at the peak aerodynamic efficiency.

$$D_{min} = \frac{L}{(L/D)_{max}} \quad ; \quad L = W \text{ (for cruise)} \tag{3.13}$$

$$D = \frac{1}{2}\rho_{cr} S_{ref} V_{cr}^2 (C_{D0} + C_{Di}) = \frac{L}{(L/D)_{max}} \tag{3.14}$$

3.3.3 Optimum Velocity during Loiter

To maximize endurance during pre-drop targeting or post-drop observation, the UAV must operate at $(C_L^{3/2}/C_D)_{max}$, which minimizes the power required to overcome drag. Lift here corresponds to the weight multiplied by the load factor during the turning maneuver ($L = nW$).

$$C_{D0} = \frac{1}{3} C_{Di} = \frac{C_L^2}{3\pi e AR} \tag{3.15}$$

Using the established aerodynamic parameters, the equations were solved to determine the optimal velocities and corresponding power requirements for all flight phases, as summarized in Table 3.3.

Phase	**Thrust (N)**	**Power (W)**	**Opt. Velocity (m/s)**	**T/W**
Initial climb	14.7300	361.464	24.53	0.215
Initial cruise	4.5427	85.117	18.74	0.066
Loiter	4.5427	85.117	18.74	0.066
Cruise after drop	3.8415	66.384	17.28	0.066

Table 3.3: Performance Summary Across Mission Phases

3.4 Power Plant and Avionics Selection

Based on the flight envelope analysis, the peak power demand occurs during the initial climb phase, requiring approximately 361.46 W of power and 14.73 N of thrust. To ensure a robust thermal safety margin and reliable operation in varied environments, the selected propulsion system must comfortably exceed these baseline requirements.

3.4.1 Motor and Propeller Comparative Analysis

To ensure optimal performance and cost-effectiveness, a comparative trade study was conducted among several readily available motor and propeller combinations. The parameters assessed included maximum power draw, maximum thrust capability, and localized market pricing.

Motor	Price (INR)	Propeller	Max. Power (W)	Max. Thrust (g)
AT2814 Long Shaft KV900 [35]	4461.76	APC 10x5.5	440.87	1976
AT2814 Long Shaft KV900	4461.76	APC 11x5.5	503.64	2216
AT2814 Long Shaft KV1050	4461.76	APC 10x5.5	582.49	2274
AT2814 Long Shaft KV1050	4461.76	APC 11x5.5	669.24	2553
AT2820 KV1050 [36]	5463.27	APC 11x5.5	755.33	2860
AT2820 KV1050	5463.27	APC 12x6	433.53	2215
T-Motor AM480 900KV	5804.00	APC 12x6	650.75	2779
T-Motor AM480 900KV [37]	5804.00	APC 13x6.5	758.50	3144

Table 3.4: Motor and Propeller Comparative Analysis

From this study, the **T-Motor AM480 900KV** [37] brushless outrunner was selected. This highly efficient fixed-wing motor is 3S LiPo compatible, easily available in the Indian market, and provides a maximum power output exceeding 650 W, offering an excellent safety margin over our 361.5 W climb requirement.

This motor will be paired with an **APC** $12 \times 6E$ [38] electric propeller. The 12-inch diameter and relatively high pitch yield superior cruise efficiency for motors operating in the $300 - 350$ W range. To manage the electrical load, a **Hobbywing Skywalker 50A V2** Electronic Speed Controller (ESC) [39] will be integrated. With a required peak

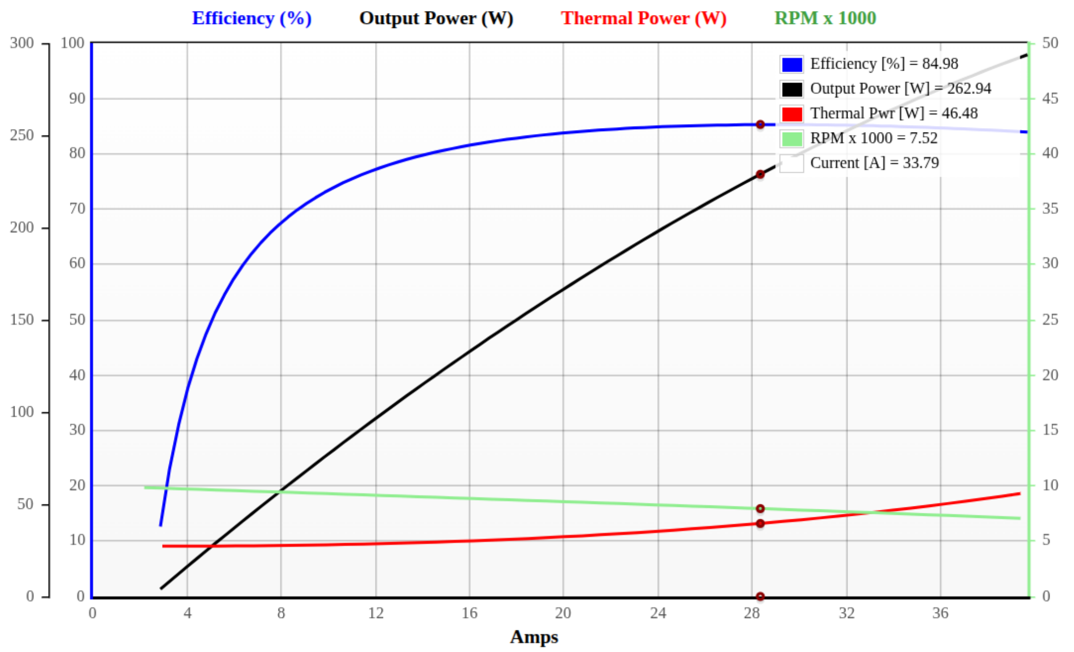


FIGURE 3.7: Motor and Propeller Characteristics: Efficiency, Output Power, Thermal Power, and RPM vs Current

current of approximately 26 A ($I \approx 280\text{W}/11.1\text{V}$), the 50A ESC provides a thermal safety factor of roughly 1.7.

FIGURE 3.8: T-Motor AM480 900KV

FIGURE 3.9: APC $12 \times 6E$ Propeller

3.4.2 Battery Market Survey and Selection

For the energy storage system, a comprehensive power budget was formulated. In addition to the primary propulsion requirements, avionics power draws were accounted for to ensure uninterrupted autonomous operation, including:

- Transmission/Receipt Antennae Module (RFD900A 915MHz ULR) [40]: 5.5 W
- FPV Videography Camera (Skydroid Mini) [41]: 0.9 W
- GPS Module: 0.21 W

FIGURE 3.10: RFD900A 915MHz 40KM ULR Module

FIGURE 3.11: Skydroid FPV Mini Camera

Integrating the total power consumption over the entire mission duration yielded a required energy of **523.7 kJ**. To satisfy this demand while maintaining a 20% reliability margin, an extensive market survey of available lithium-polymer batteries was conducted across various brands and configurations.

Battery Series	Mass (g)	Dimensions (mm)	Voltage	Capacity (mAh)	Price (INR)
GenX LiPo [42]	1812	236x106x34	11.4 (3S)	30000	9445
GenX LiPo [43]	1287	203x91x36	11.4 (3S)	22000	6196
GenX LiPo (Selected) [44]	**987**	**183x5x34**	**11.4 (3S)**	**16000**	**5400**
GenX LiPo [45]	1056	178x75x39	14.8 (4S)	12000	5865
GenX LiPo [46]	648	180x60x31.5	11.1 (3S)	10000	3922
CNHL Black Series [47]	438	147x51x27	11.1 (3S)	5000	4599
Arya Bot 3S LiPo [48]	160	100x30x20	11.1 (3S)	3300	2714
Tattu 3S LiPo [49]	195	103x33x26	11.1 (3S)	2700	2999
Gaoneng GNB [?]	178	106x34x24	11.1 (3S)	2200	2599

TABLE 3.5: Excerpt of Market Survey for LiPo Batteries

The final selected battery is the **GenX 11.4V 3S 16000 mAh HV LiPo**. The total energy provided by this pack is calculated as:

$$E = 11.4 \text{ V} \times 16 \text{ Ah} = 182.4 \text{ Wh} = 657 \text{ kJ} \tag{3.16}$$

This 657 kJ capacity comfortably satisfies our 523.7 kJ mission requirement while providing the necessary power buffer for headwind compensation, avionics draw, and emergency loiter maneuvers.

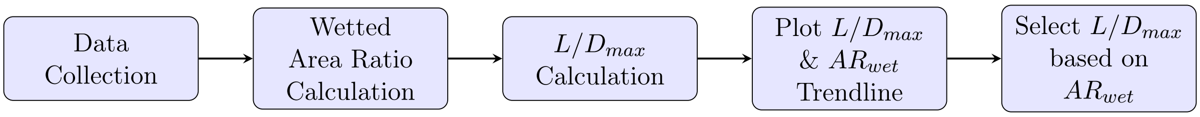
FIGURE 3.12: GenX 11.4V 3S 16000 mAh 25C Premium LiPo Battery

3.4.3 Final System Configuration

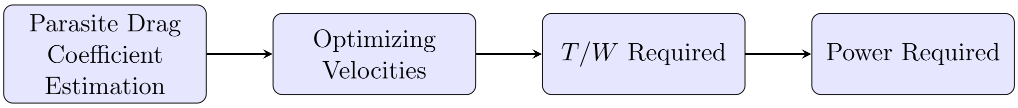
The culmination of the power-plant and avionics sizing results in the following finalised configuration:

- **Propulsion:** Twin-tractor arrangement
- **Motor:** T-Motor AM480 900 KV (×2)
- **Propeller:** APC 12×6E electric propeller (×2)
- **ESC:** Hobbywing Skywalker 50 A V2 (×2)
- **Battery:** GenX 3S 16 Ah HV LiPo (subsequently upgraded to GenX 6S 12 Ah in Chapter 5 following the second weight iteration)

L/D max and Wetted Aspect Ratio



Optimal Speeds and Power Requirements



3.4.4 Summary

Design Progress

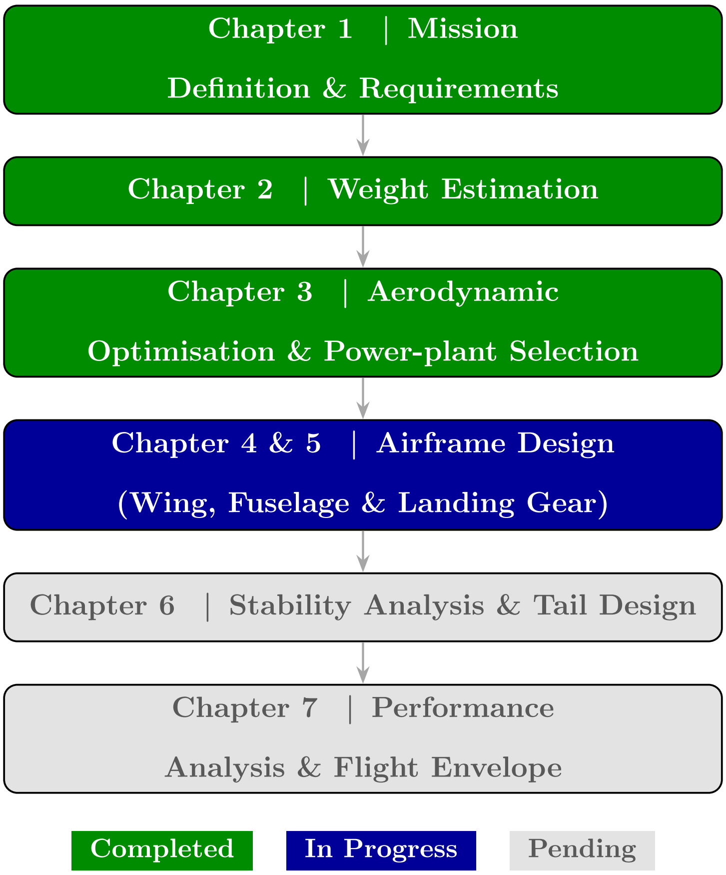


FIGURE 3.13: Overall UAV Conceptual Design Workflow — Progress after Chapter 3

TABLE 3.6: Key Design Parameters — Chapter 3: Aerodynamic Optimisation & Power-plant

Parameter	Symbol	Unit	Previous	Current
Aerodynamic parameters (preliminary)				
Wingspan	b	m	—	2.00
Wing Area	S	m^2	—	0.413
Aspect Ratio	AR	—	—	9.686
Oswald Efficiency	e	—	—	0.7648
Induced Drag Factor	k	—	—	0.0430
Zero-Lift Drag Coeff. (prelim.)	C_{D_0}	—	—	0.0256
Max. Lift-to-Drag Ratio	$(L/D)_{max}$	—	—	15.08
Optimum flight velocities				
Climb velocity	V_{cl}	$\mathrm{m\,s^{-1}}$	—	24.53
Cruise velocity	V_{cr}	$\mathrm{m\,s^{-1}}$	—	18.74
Loiter velocity	V_{loi}	$\mathrm{m\,s^{-1}}$	—	18.74
Peak power demand (initial climb)				
Climb Thrust	T_{cl}	N	—	14.73
Climb Power Required	P_{cl}	W	—	361.46
Power-plant selection				
Motor	—	—	—	T-Motor AM480 900 KV
Propeller	—	—	—	APC 12×6E
ESC	—	—	—	Hobbywing Skywalker 50 A
Propulsion configuration	—	—	—	Twin-tractor
Energy & battery				
Mission Energy Required	—	kJ	—	523.7
Battery Capacity	—	kJ	—	657.0
Battery (confirmed)	—	—	GenX 3S 16 Ah	GenX 3S 16 Ah
Carried forward from Chapter 2				
MTOW	W_0	kg	6.896	6.896

New parameter established this chapter

Chapter 4

Wing Design

4.1 Wing-loading and Power-loading Estimation

4.1.1 Stall Speed

The stall speed of the aircraft can be affixed, from which the wing-loading of the aircraft at the stall speed can be calculated as below:

$$\left(\frac{W}{S}\right)_{V_s} = \frac{\rho V_s^2 C_{L,max}}{2}$$

This can be trivially got by substituting the values:

$$C_L = C_{L,max} \text{ and} \tag{4.1}$$

$$L = W \tag{4.2}$$

in the lift equation, viz.,

$$L = \frac{1}{2}\rho v^2 S$$

Note the fact that as the wing-loading characteristics of the aircraft increase, the stall speed increases as well. Intuitively, this metric can be used to understand the 'glider'-nature. Clearly, then, considering the stall-speed constraint,

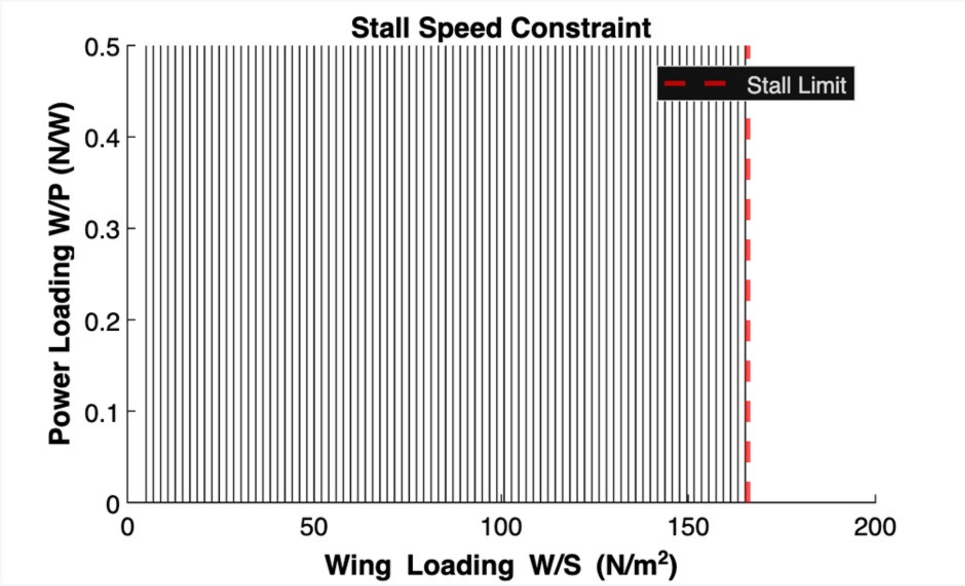


FIGURE 4.1: Stall Speed constraint on the power loading v. Wing-loading curve

4.1.2 Rate of Climb

A quick rate of climb is ideal for UAVs designed on the basis of quick delivery. The drawbacks of the higher rate of climb is the faster draining of the battery, as well as higher power requirement, leading to higher amperage, wasted heat from electronic components – leading to faster degradation of electronics, and a more inefficient use of battery coupled with a shortened battery life-span. However, these trade-offs would be worthwhile keeping in view the application of this UAV specified in the earlier portions of this report.

In the calculation of rate of climb, we use the equation:

$$\left(\frac{W}{P}\right)_{ROC} = \frac{1}{\frac{ROC}{\eta_P} + \sqrt{\frac{2W\sqrt{K}}{\rho S\sqrt{3C_{D_0}}}}\frac{1.155}{(L/D)_{max}\eta_P}}$$

The constraint plot obtained by plotting the graph between the power loading and wing loading up till the stall limit is found to be:

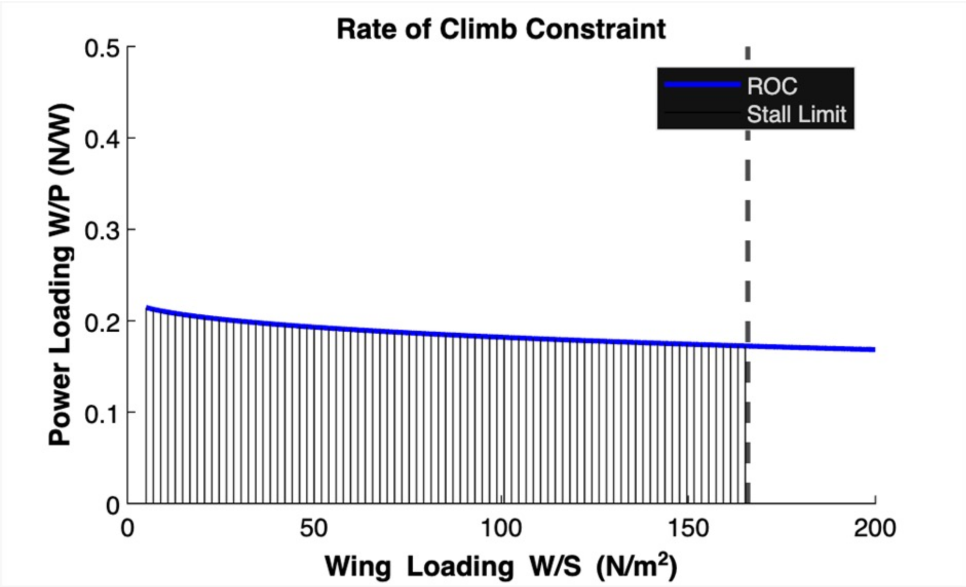


FIGURE 4.2: ROC constraint on the power loading v. Wing-loading curve

4.1.3 Take-off

The take-off distance of the aircraft is the portion of the ground-roll of the aircraft from the initiation of power-input when starting on the ground, up till the clearance of the aircraft of a preset obstacle-height, post rotation and lifting off from the ground/datum. This distance of vital significance in so far as it is indicative of the amount of space required by the aircraft during take-off, which in turn would influence the ease of taking off and landing in more urban/hilly environments where such empty, flat, and regular spaces of land are either unavailable or at a premium. Ideally, we would like the take-off performance to be maximum to ensure a rapid, short, and smooth take-off, but that comes with trade-offs such as power consumption and electronics' heat generation.

$$\left(\frac{W}{P}\right)_{S_{TO}} = \frac{1-\exp\left(0.6\rho g C_{D_G} S_{TO}\frac{1}{W/S}\right)}{\mu-\left(\mu+\frac{C_{D_G}}{C_{L_R}}\right)\exp\left[0.6\rho g C_{D_G} S_{TO}\frac{1}{W/S}\right]V_{TO}}\cdot\eta_P$$

The corresponding constraint plot was found to be

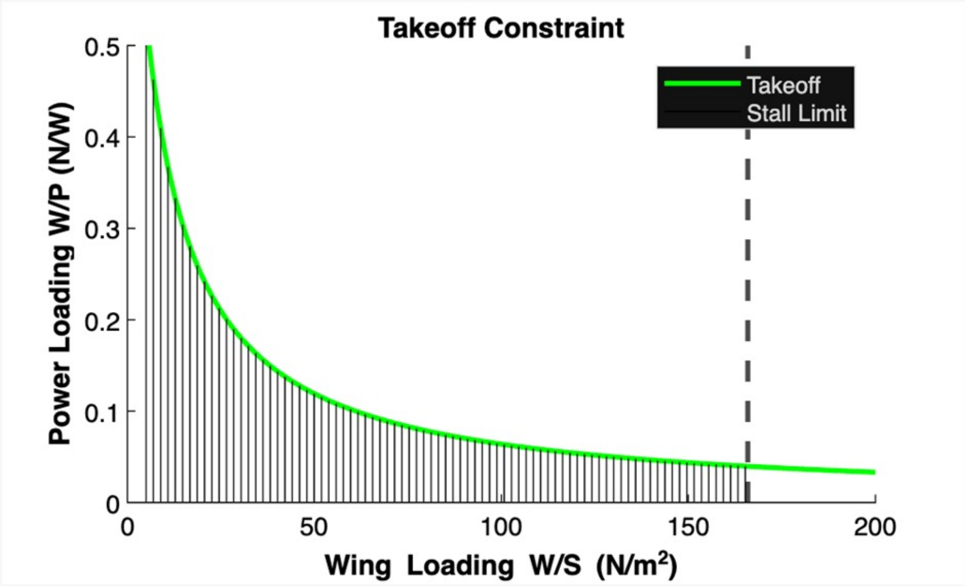


FIGURE 4.3: TO constraint on the power loading v. Wing-loading curve

4.1.4 Minimum Turn Radius

The minimum turn radius of a fixed-wing aeroplane defines the tightest horizontal circle it can fly while turning, directly limiting its manoeuvrability in confined spaces.

The general significance of turn-radius is the intricate interconnection between the turn-radius and the bank angle ϕ given by the following formula:

$$R = \frac{V^2}{g \tan \phi}$$

Indeed, the turn-radius would also determine the load-factor (or more colloquially, g-loading) of the aircraft, and consequently, the lateral/centrifugal forces the aircraft experiences. This in turn can be indicative of the manoeuvrability of the aircraft.

$$\frac{1}{P/W} = \frac{V}{\eta_P} \left[\frac{qC_{C_0}}{W/S} + kn^2 \frac{W/S}{q} \right]$$

Here, the load factor, n is given by

$$n := \sqrt{\frac{V^4}{g^2 R^2} + 1}$$

And the constraint diagram obtained w.r.t. the minimum turn radius is:

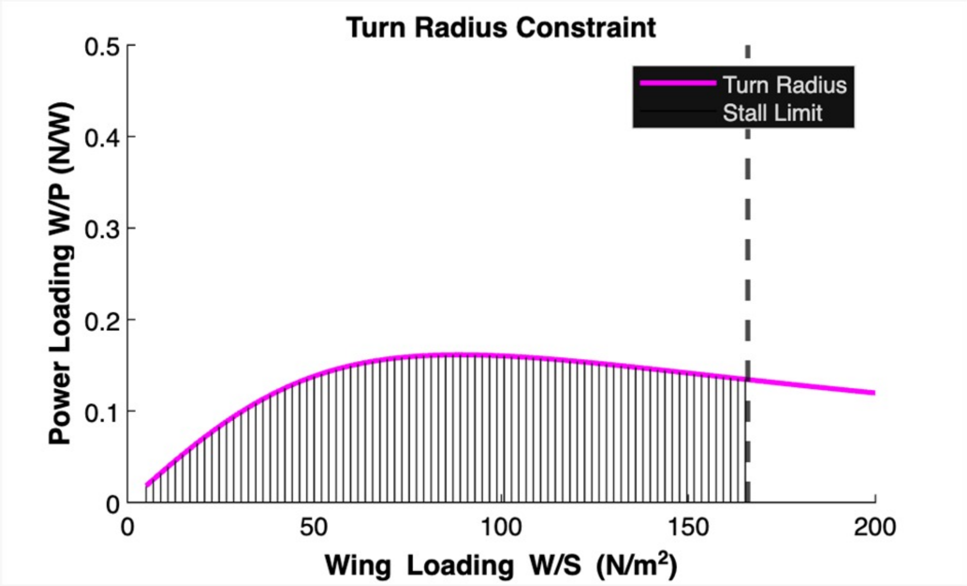


FIGURE 4.4: Turn Radius constraint on the power loading v. Wing-loading curve

4.1.5 Service Ceiling

Service ceiling is the maximum altitude at which a fixed-wing aircraft can maintain a steady climb rate of 100 feet per minute (fpm) in standard atmosphere conditions, marking a practical operational limit [50]. The formula

$$\left(\frac{W}{P}\right)_{ROC} = \frac{1}{\frac{ROC}{\eta_P} + \sqrt{\frac{2W\sqrt{K}}{\rho S\sqrt{3C_{D_0}}}}\frac{1.155}{(L/D)_{max}\eta_P}}$$

is used (with appropriate substitutions for rate of climb) to determine the constraint diagram for obtaining the service ceiling. Then, the constraint diagram for service ceiling would be given by Figure 4.5.

Furthermore, the service ceiling will definitely be lower than the absolute ceiling by virtue of the fact that the aircraft can reach its service ceiling and still keep going, whereas the same aircraft couldn't possibly go above its absolute ceiling during routine operations.

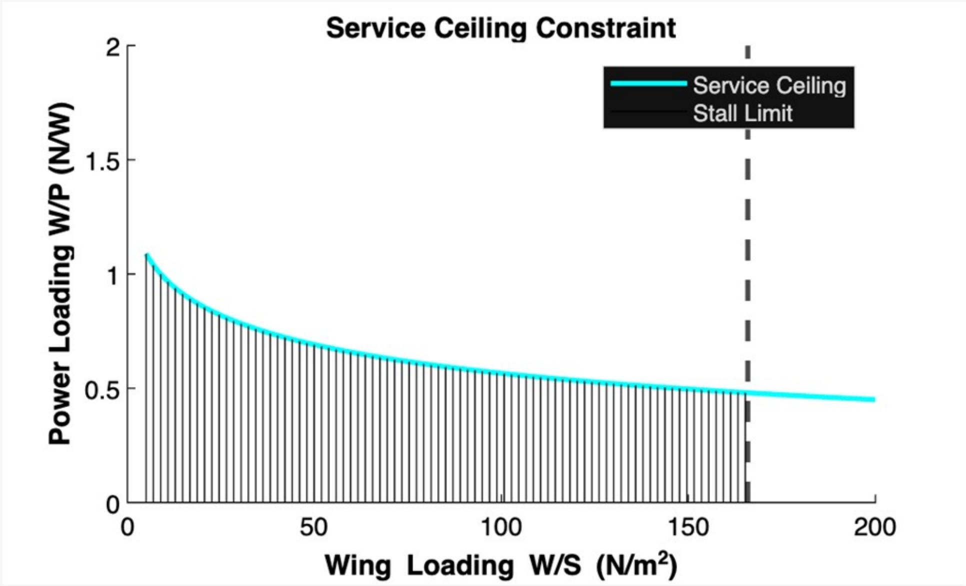


FIGURE 4.5: Service ceiling constraint on the power loading v. Wing-loading curve

Therefore, even though there might be a constraint of absolute ceiling applied to aircrafts, it is superceded by the service ceiling constraint imposed on aircrafts.

4.1.6 Sample Space of Feasible Aircrafts Design

Upon combining all the constraints applied hitherto, the intersection set of the feasible sample spaces of all the individual constraint diagrams can be said to be the sample space of feasible aircraft design. This overall, cumulative constraint diagram would thus give us the complete sample space of power loading and wing loading pairs in which the designs of feasible aircrafts reside, with each point having its own merits and demerits, advantages and disadvantages, and strengths and tradeoffs/compromises which are applied keeping the overall expected trajectories and controls/responses of the aircraft in mind. Considering the battery chosen, the total energy requirement (including the margin of safety) should not exceeded, in order to cater to the ranges initially envisaged.

Below is the trendline graph of how various parameters affecting the wing loading.

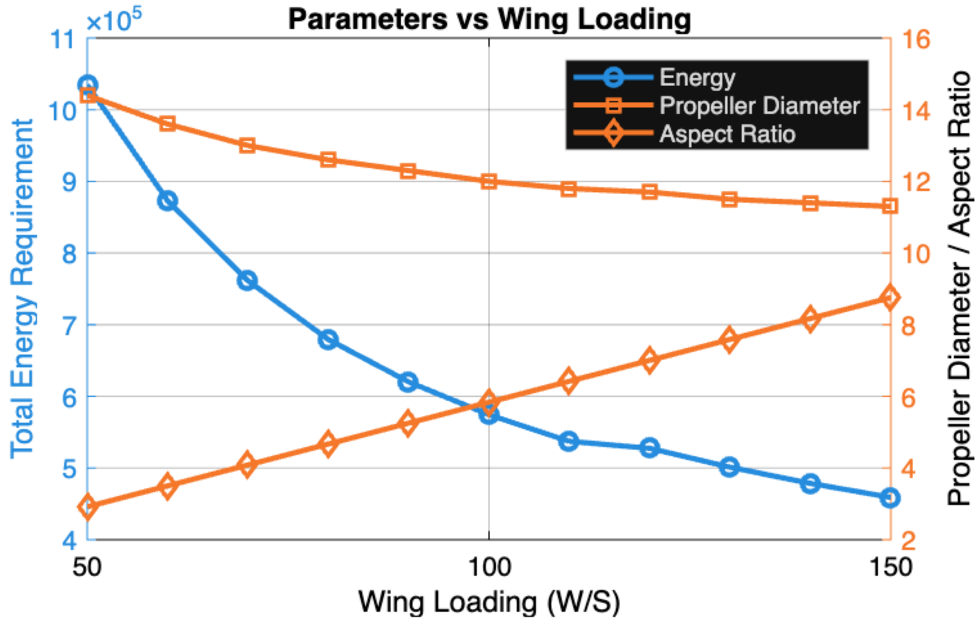


FIGURE 4.6: Parameters affecting the Wing Loading

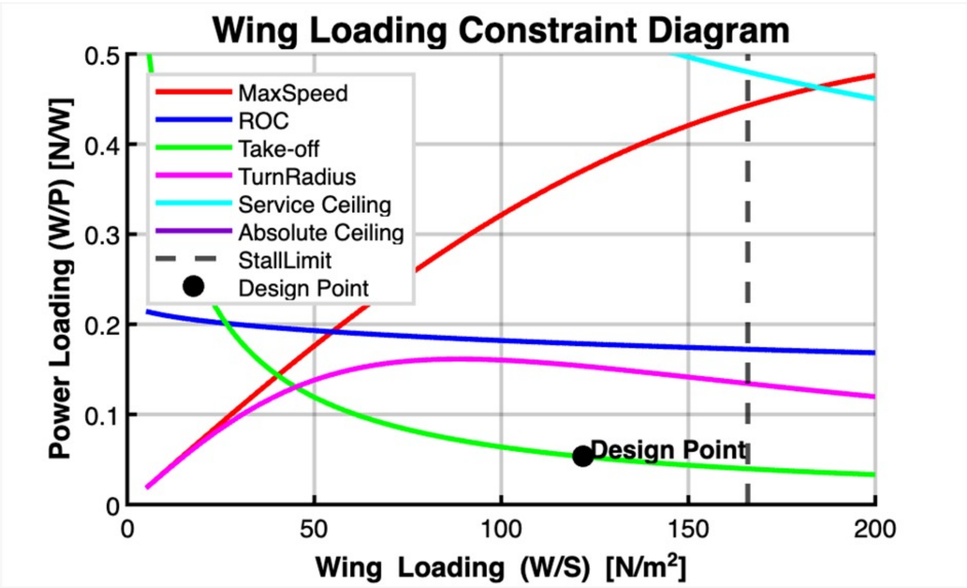


FIGURE 4.7: 'Overall' Wing Loading constraint on the power loading v. Wing-loading plot

The design point was chosen with the following rationale in mind that there is an absolute and strict limit on the span-length of the wing, while preserving the aspect ratio, wing-loading should not be too high in order to conform to material strength properties.

Thus, the area of the aerofoil, and the aspect ratio were found to be:

$$S_{\text{new}} = 0.5623$$
$$AR = 7.1133 \text{ [upon applying wing-loading constraint]}$$

4.2 Aerofoil Selection

Aerofoil selection is a vital step in the design of aircrafts in so far as the wing-design – and the often-times ignored consideration of wing-fabrication – is shaped mainly by the requirement of being easily manufacturable using sheet-metal fabrication, considering the final wing's material of manufacture being 0.5 mm aluminium sheets reinforced with wing-spares, ribs, and (cross-sectional) 3D-printed cell-assemblies.

In view of this strict constraint, it was found that aerofoils having very low camber were suitable, and a vetted list of such aerofoils was made for analysis and analysed using XFLR5 solver.

4.2.1 Eppler Series

The Eppler series is a large family of low-speed aerofoils designed by Richard Eppler (and later Eppler & Somers) specifically to control the boundary layer and achieve very low drag with extensive laminar flow, especially at low to moderate Reynolds numbers (e.g., model aircraft, sailplanes, UAVs, small GA).

This series of aerofoils have a broad low-drag region (a “bucket") in the $C_L - C_D$ polar, corresponding to large fractions of laminar flow on one or both surfaces. This makes them very efficient around their design lift coefficient.

4.2.2 Selig Series

The Selig series (often denoted S-series, e.g., S1223, SD7037) is a family of aerofoils designed by Michael Selig and collaborators primarily for low-Reynolds-number applications like RC models, small UAVs, and micro air vehicles, emphasizing high lift, low drag, and forgiving stall behaviour at $Re \in (5 \times 10^4, 5 \times 10^5)$.

In this sense, even though the Selig had been developed on the basis of the inverse design methodology used by Eppler, and that too for the same applications, viz., low Re flows, encountered in the design of UAVs, the difference is in the maximum lift and stall-predictability characteristics that Eppler lacks, in contradistinction to the theoretical laminar optima that characterise Eppler.

The comparative cross-sections of the aerofoils are given in the following figures:

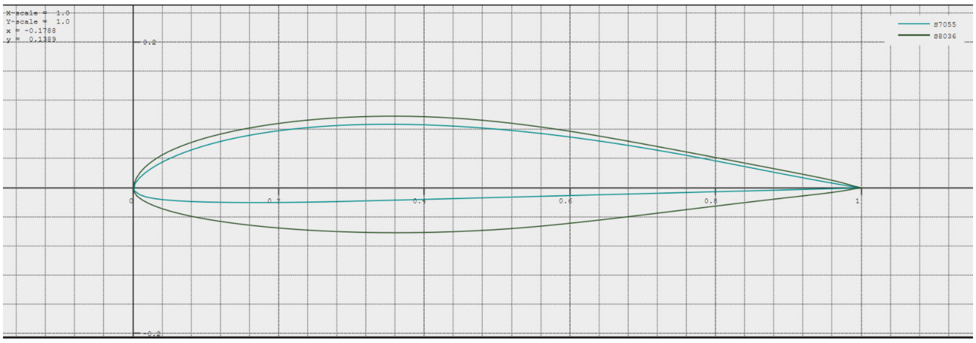


FIGURE 4.8: Selig series: Specimen Aerofoils Shape

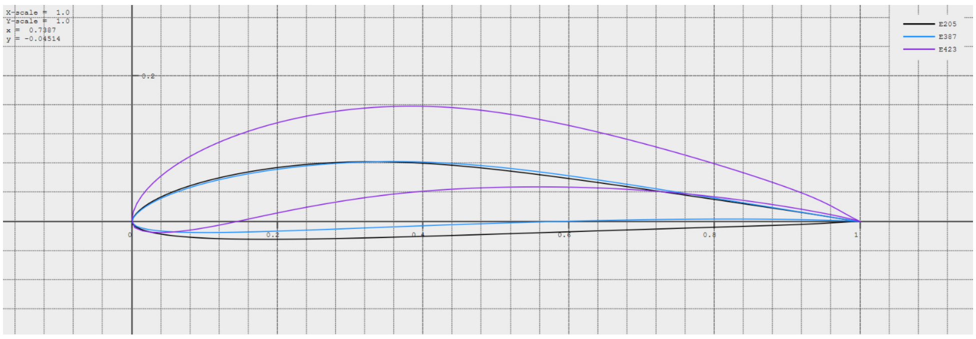


FIGURE 4.9: Eppler Series: Specimen Aerofoils Shape

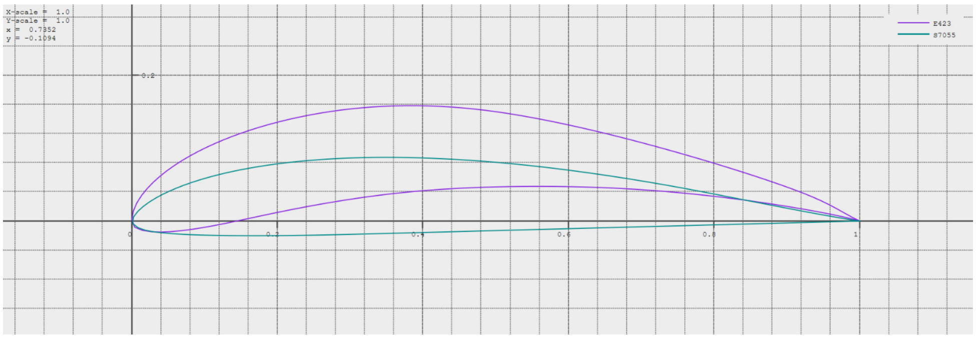


FIGURE 4.10: Eppler-Selig Specimen

4.2.3 Aerofoil Comparative Analysis

Based on the mission parameters and preliminary aerodynamic screening, the following seven aerofoil profiles were shortlisted for detailed comparative analysis:

1. Eppler E205
2. Eppler E387
3. Eppler E423
4. NACA 23015
5. NACA 652-415
6. Selig S7055
7. Selig S8036

The comparative analysis constituted obtaining the plots of the variation of various characteristic parameters of the shortlisted aerofoils such as $C_l - C_d$, $C_l - \alpha$, $C_m - \alpha$, $C_d - \alpha$, $\dfrac{C_l}{C_d} - \alpha$.

From the preceding figure, it may be noted that E423 gives the best $C_l - \alpha$ behaviour with higher C_l values for a given angle of attack; on the other hand, S7055 aerofoil exhibits the best $\dfrac{C_l}{C_d} - \alpha$ behaviour, with its maximum being the maximum amongst the various aerofoils shortlisted. Consideration was also given to $\dfrac{C_l^{1.5}}{C_d} - \alpha$ curve, but the main deciding factor was the former curve whose maximum was found to be most satisfactory.

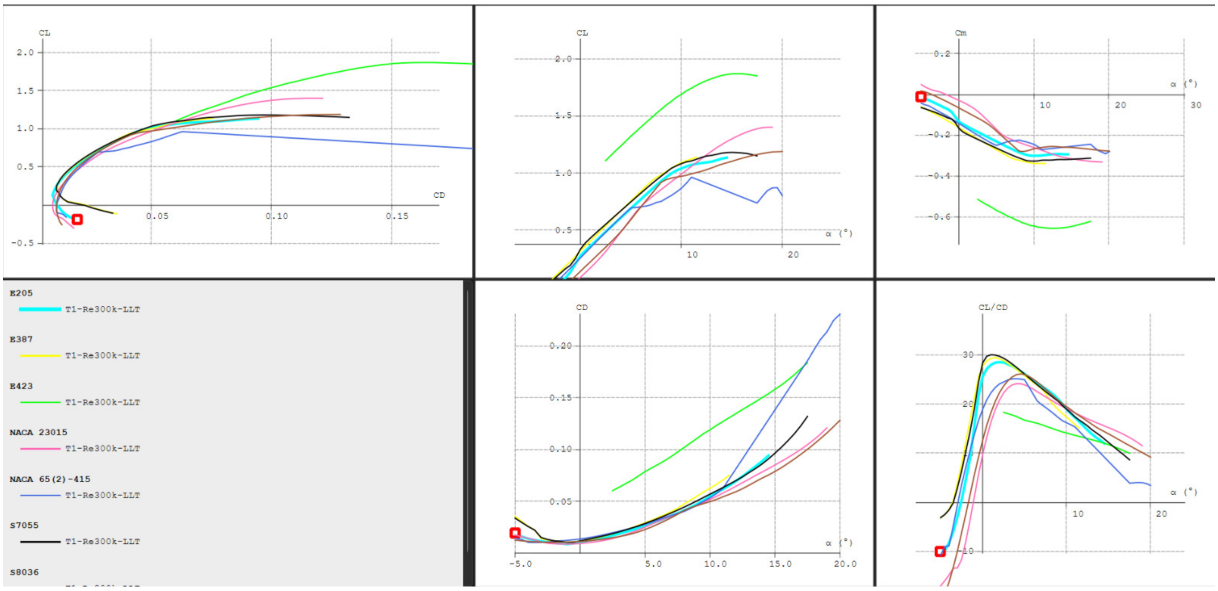


FIGURE 4.11: Comparative Analysis of Shortlisted Aerofoils for Multiple Characteristic Parameters' Variation

4.2.4 Wing design

4.2.4.1 High-Wing Configuration

A high-wing configuration was selected for the UAV due to several aerodynamic, structural, and mission-related considerations. In this configuration, the wing is mounted above the fuselage, which provides inherent stability characteristics that are advantageous for small unmanned aircraft operating at low speeds. When the wing is positioned above the aircraft's center of gravity, the mass of the fuselage and payload tends to hang below the lifting surface, producing a pendulum-like stabilizing effect. This configuration improves lateral stability and helps the aircraft naturally return to equilibrium following small disturbances during flight.

Another major motivation for selecting a high-wing configuration is the integration of the payload delivery mechanism. The UAV is designed to perform payload drop operations, which requires unobstructed space beneath the fuselage. Mounting the wing above the fuselage leaves the underside of the aircraft clear, allowing the payload release system to be installed without interference from the wing structure. This arrangement simplifies the mechanical design of the payload drop mechanism and ensures that the payload can be released cleanly without contacting the wing or other aerodynamic surfaces.

From an operational standpoint, the high-wing layout also improves ground clearance for critical components such as the propeller and payload bay. This is particularly beneficial during take-off and landing operations, especially when operating from uneven or unprepared surfaces. Increased ground clearance reduces the likelihood of damage to the airframe or propulsion system and improves the overall robustness of the UAV during field operations.

In many aircraft designs, lateral stability is further enhanced through the use of wing dihedral. However, in the present UAV design, no dihedral was incorporated into the wing geometry. This decision was made primarily due to manufacturing constraints and the desire to maintain structural simplicity in the wing construction. A straight, rectangular wing without dihedral is easier to fabricate, align, and assemble, particularly when using lightweight materials commonly employed in small UAV structures. Despite the absence of dihedral, the high-wing configuration itself still contributes to the overall stability of the aircraft.

Additionally, the selected wing planform is rectangular, which simplifies structural design and manufacturing. Rectangular wings also provide predictable aerodynamic characteristics and uniform lift distribution along the span, which is suitable for the intended low-speed flight regime of the UAV.

Considering the mission requirements, manufacturing constraints, and stability considerations, the high-wing configuration was determined to be the most suitable choice for the aircraft. This configuration supports the payload drop mechanism, improves operational robustness, and provides sufficient inherent stability for the UAV's intended flight envelope.

4.2.5 Wing Design Parameters

The wing geometry of the UAV was designed to satisfy the aerodynamic performance requirements while also adhering to the geometric and manufacturing constraints imposed on the aircraft. The primary wing parameters used in the design are summarized below:

- Wing area, $S = 0.5307\,m^2$
- Wingspan, $b = 2\,m$
- Aspect ratio, $AR = 7.53$
- Mean aerodynamic chord, $\bar{c} = 0.2653\,m$
- Airfoil: Selig S7055

4.2.5.1 Wing Span and Aspect Ratio

The wingspan of the aircraft was constrained to a maximum value of $2\,m$ due to design and operational limitations. As a result, the aspect ratio of the wing was determined based on the available wingspan and the required wing area rather than being chosen independently. The aspect ratio is defined as

$$AR = \frac{b^2}{S} \tag{4.3}$$

Substituting the design values gives

$$AR = \frac{(2)^2}{0.5307} = 7.53 \tag{4.4}$$

An aspect ratio of approximately 7.5 is suitable for small UAVs operating in the low Reynolds number regime, as it provides a balance between aerodynamic efficiency and structural simplicity.

4.2.5.2 Wing Planform

A rectangular wing planform was selected for the aircraft, corresponding to a taper ratio of

$$\lambda = \frac{c_{tip}}{c_{root}} = 1 \tag{4.5}$$

This choice was primarily motivated by manufacturing considerations. The wing structure is intended to be fabricated using sheet metal, and introducing taper would significantly increase manufacturing complexity. A rectangular wing allows easier fabrication, alignment, and structural assembly while maintaining acceptable aerodynamic performance for the intended flight regime.

The mean aerodynamic chord of the wing is

$$\bar{c} = 0.2653\,m \tag{4.6}$$

which is consistent with the chosen span and aspect ratio.

4.2.5.3 Airfoil Selection

The Selig S7055 airfoil was selected for the wing due to its favourable aerodynamic characteristics at low Reynolds numbers, which are typical of small UAV operations. This airfoil is capable of generating relatively high lift coefficients while maintaining low drag, making it well suited for efficient cruise performance.

The aerodynamic polars obtained from XFLR5 analysis indicate that the airfoil achieves a high lift-to-drag ratio within the operational range of angles of attack considered for the UAV.

4.2.5.4 Sweep Angle

The aircraft operates at a cruise velocity of $18\,m/s$, corresponding to a very low Mach number. At such low speeds, compressibility effects are negligible and there is no aerodynamic requirement for wing sweep. Consequently, the sweep angle was taken to be

$$\Lambda = 0^\circ \tag{4.7}$$

This simplifies both the aerodynamic design and the structural layout of the wing.

4.2.5.5 Dihedral Angle

The dihedral angle primarily contributes to roll stability of an aircraft. However, for the present design the dihedral angle was chosen to be

$$\Gamma = 0^\circ \tag{4.8}$$

This decision was made primarily due to manufacturing constraints and to maintain simplicity in the wing construction. A straight wing without dihedral is easier to fabricate and assemble, particularly when using sheet metal structures. Additionally, the high-wing configuration of the UAV already provides a degree of inherent lateral stability, reducing the need for additional dihedral.

4.2.5.6 Wing Incidence Angle

The wing incidence angle was selected such that the aircraft operates near the condition of maximum aerodynamic efficiency during cruise. From the airfoil polar analysis, the maximum lift-to-drag ratio occurs at approximately 15 or higher for the S7055 airfoil within the considered operating range.

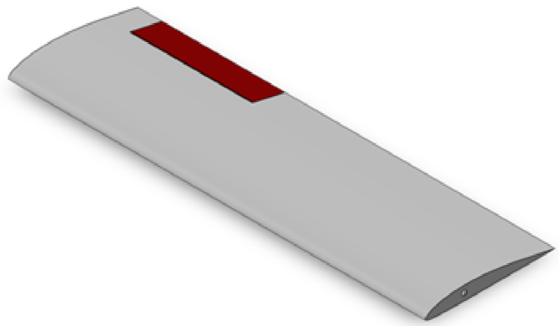


FIGURE 4.12: CAD Model of starboard side wing

To achieve this condition during cruise flight, the wing is mounted with an incidence angle of

$$i_w = 3^\circ \tag{4.9}$$

This allows the aircraft fuselage to remain approximately aligned with the freestream while the wing operates near the angle of attack corresponding to maximum L/D. This improves cruise efficiency and reduces overall drag during steady level flight.

Overall, the wing design reflects a balance between aerodynamic performance, mission requirements, and manufacturing practicality. The selected geometry ensures efficient cruise performance while maintaining a structurally simple and manufacturable wing configuration.

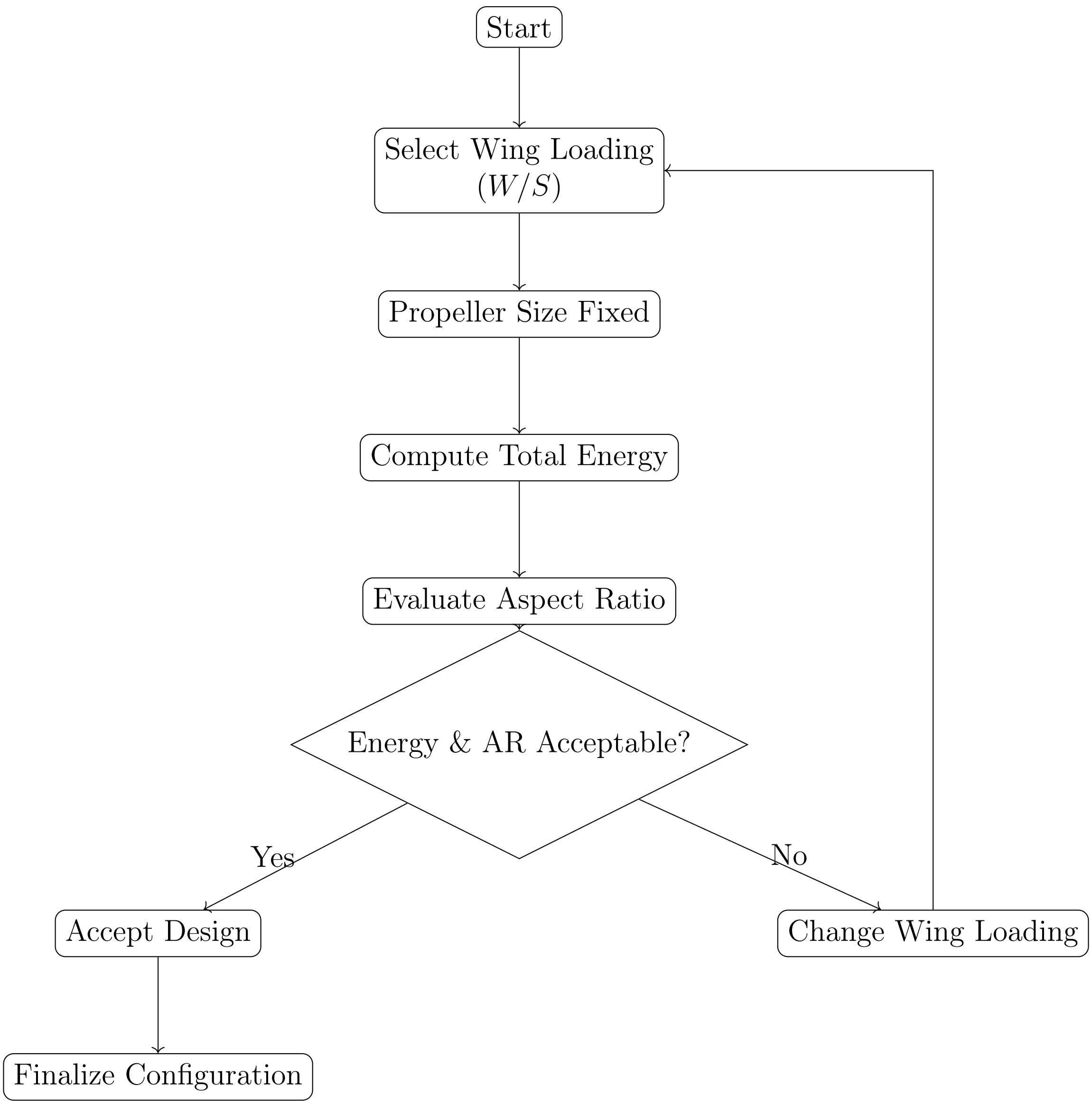
4.2.6 Summary

TABLE 4.1: Key Design Parameters — Chapter 4: Wing Design

Parameter	Symbol	Unit	Previous	Current
Wing geometry (new this chapter)				
Wingspan	b	m	—	2.000
Mean Aerodynamic Chord	$\bar{c}$	m	—	0.2653
Taper Ratio	λ	—	—	1.0 (rectangular)
Sweep Angle	Λ	deg	—	0
Dihedral Angle	Γ	deg	—	0
Wing Incidence Angle	i_w	deg	—	3
Wing Configuration	—	—	—	High-wing
Aerofoil selected				
Aerofoil	—	—	—	Selig S7055
Max. thickness ratio	t/c	—	—	0.105
Wing loading (design)	W/S	$\mathrm{N\,m^{-2}}$	—	121.84
Wing Area	S	$\mathrm{m^2}$	0.413	**0.5307**
Aspect Ratio	AR	—	9.686	**7.53**

Updated from previous chapter

New parameter this chapter


Start
Select Wing Loading (W/S)
Propeller Size Fixed
Compute Total Energy
Evaluate Aspect Ratio
Energy & AR Acceptable?
Yes
No
Accept Design
Change Wing Loading
Finalize Configuration

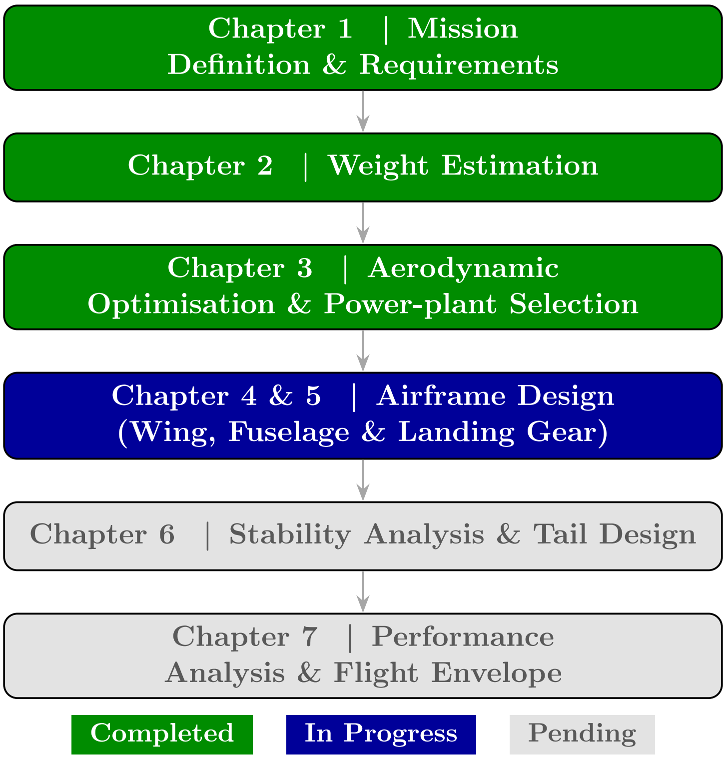


Figure 4.13: Overall UAV Conceptual Design Workflow — Progress after Chapter 4

Chapter 5

Fuselage & Airframe Design

5.1 Fuselage Design

5.1.1 Fuselage Length Estimation

The fuselage length was estimated using a regression-based scaling approach derived from existing UAV data. A dataset containing the maximum takeoff weight (MTOW) and corresponding fuselage lengths of similar UAV platforms was collected and analyzed.

A power-law relationship between fuselage length and aircraft weight was assumed in the form

$$L = aW_0^b \tag{5.1}$$

where:

- L is the fuselage length (m)
- W_0 is the maximum takeoff weight (kg)
- a and b are regression coefficients

To determine the coefficients, the data was transformed into logarithmic form and a linear regression was performed:

$$\ln L = \ln a + b \ln W_0 \tag{5.2}$$

Using least–squares regression on the collected UAV dataset, the following parameters were obtained:

$$a = 1.10, \qquad b = 0.08$$

Name	MTOW (kg)	Fuselage Length (m)
Yangda Mapird Pro [51]	7.5	1.2
Yangda Mapird Plus [26]	11.5	1.4
Yangda Nim-bus Pro	8.6	1.3
Yangda FW-250	12	1.4
Avy Aera [28]	12	1.3
Foxtech Baby Shark 260 [29]	13	1.44
P330 Pro	14	1.21
Foxtech AYK-250 [30]	15	1.26
Wingcopter 178 [33]	18	1.32
FlyDragon FLY-2100 [31]	7	1.2
SkyEye Sierra [34]	12.5	1.5
Foxtech Nimbus [32]	6	1.3

Substituting the design MTOW of the present UAV ($W_0 = 7.021$ kg) into the scaling relation gives

$$L = 1.10 \times (7.021)^{0.08} \Rightarrow L \approx 1.27 \text{ m} \tag{5.3}$$

Figure 5.1 shows the regression fit obtained from the reference UAV data along with the estimated fuselage length for the present design.

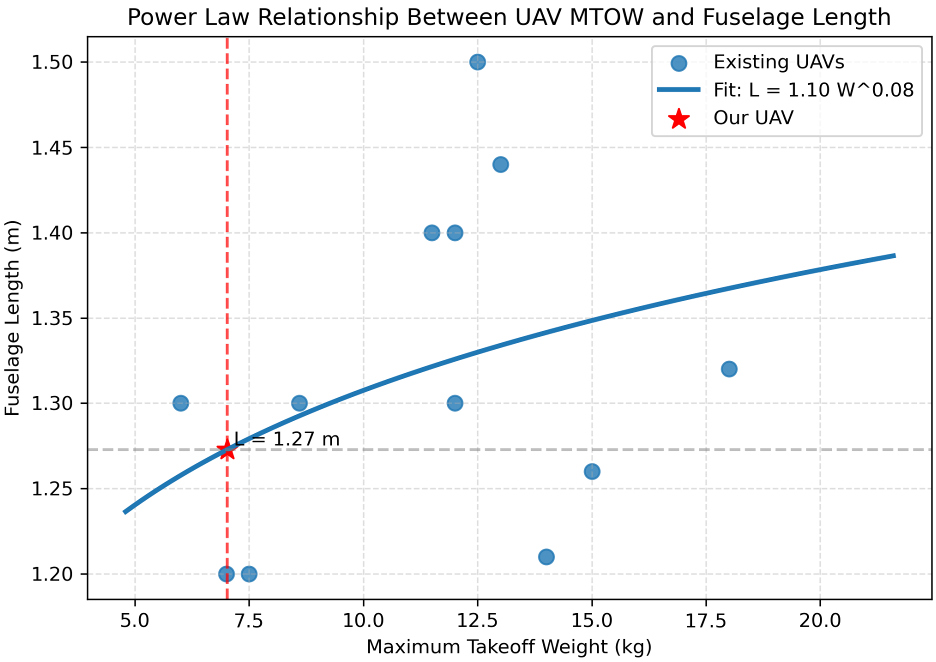


FIGURE 5.1: Power-law regression between UAV MTOW and fuselage length.

5.1.2 Fuselage Design

The fuselage of the UAV serves as the primary structural body of the aircraft and houses critical components such as the payload bay, avionics, battery, and payload release mechanism. The design of the fuselage was carried out with emphasis on structural simplicity, ease of manufacturing, and adequate internal volume for the required subsystems.

The overall fuselage length of the aircraft is 1271 mm (updated to 1088 mm). The forward section of the fuselage was initially designed with a rounded nose profile to provide a smooth aerodynamic shape and reduce pressure drag during flight; however, a better side-profile of the fuselage section's abaft portion would be to have an aerofoil shape, and in conformity with that principle, the fuselage was redesigned to reflect the aerofoil shape of the symmetric aerofoil NACA 0025 with a chord length of 800 mm; symmetric aerofoil was chosen to ensure that there was no asymmetric lift from the fuselage, while also ensuring that the fabricability of the fuselage would not be negatively impacted, while also reducing the wake turbulence from an otherwise boxy design. This region is intended to accommodate components such as avionics, sensors, and battery systems while maintaining a compact and streamlined geometry.

The central fuselage section was initially designed to have a nearly rectangular cross–section with a length of approximately $511\,mm$ before the fuselage design was changed to the aforementioned symmetric aerofoil shape. This section provides the primary internal volume for mounting payload systems and structural members. The geometry has intentionally been kept simple to facilitate manufacturing and assembly using sheet metal structures. While a rectangular configuration allows for easier fabrication, mounting of internal components, and structural reinforcement, the presence of sharp-corners act as locations of stress-concentration which can easily become points of failure; this problem is partially alleviated by using the aerofoil-shaped fuselage.

To further improve aerodynamic performance, the aft section of the fuselage gradually and smoothly tapers towards the tail boom, in keeping with the shape of the aerofoil itself. This tapering reduces flow separation and minimizes form drag by allowing a smoother pressure recovery behind the main fuselage body.

5.1.2.1 Tail Boom

The tail boom is a slender structural member extending from the aft end of the fuselage and connecting the main body of the aircraft to the empennage. In the present design, the tail boom has a circular cross–section with a diameter of approximately $20\,mm$ (updated to $25\,mm$). The circular geometry provides good bending and torsional stiffness while maintaining a lightweight structure.

The primary function of the tail boom is to provide sufficient moment arm between the wing and the tail surfaces. Increasing this distance improves the effectiveness of the horizontal and vertical stabilizers, allowing better control authority and stability of the aircraft. The tail boom therefore plays an important role in maintaining longitudinal and directional stability.

In addition to structural considerations, the use of a slender tail boom also minimizes aerodynamic drag compared to a full fuselage extension. The small cross–section reduces the wetted area and helps maintain overall aerodynamic efficiency.

The tail boom is designed to be lightweight yet sufficiently rigid to withstand aerodynamic loads transmitted from the tail surfaces during flight manoeuvres. This configuration is commonly used in small UAVs as it provides a good balance between structural efficiency, aerodynamic performance, and ease of construction.

Overall, the fuselage configuration provides sufficient internal space for payload integration, supports the structural requirements of the wing and tail, and maintains a streamlined aerodynamic profile suitable for the cruise velocity of $18\,m/s$.

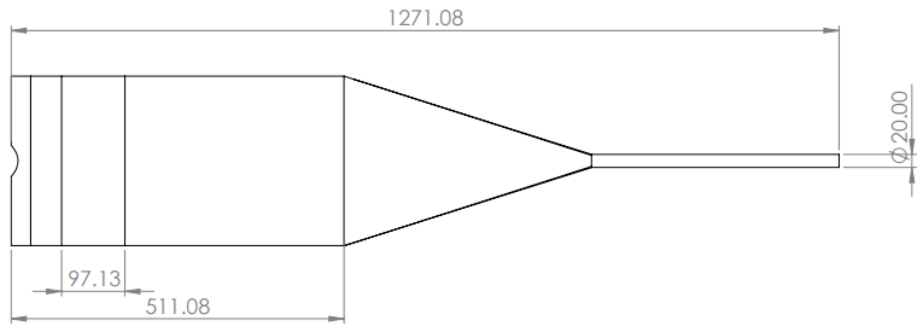


FIGURE 5.2: Top view of the first iteration fuselage design showing primary dimensions

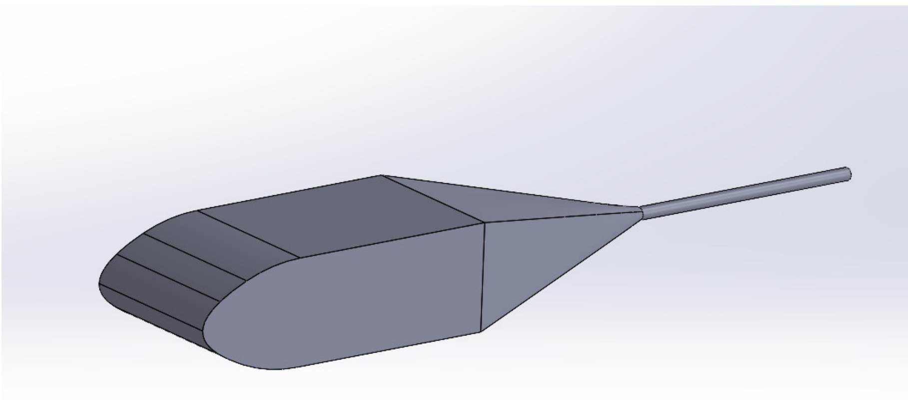


FIGURE 5.3: Isometric view of the first iteration of UAV fuselage and tail boom

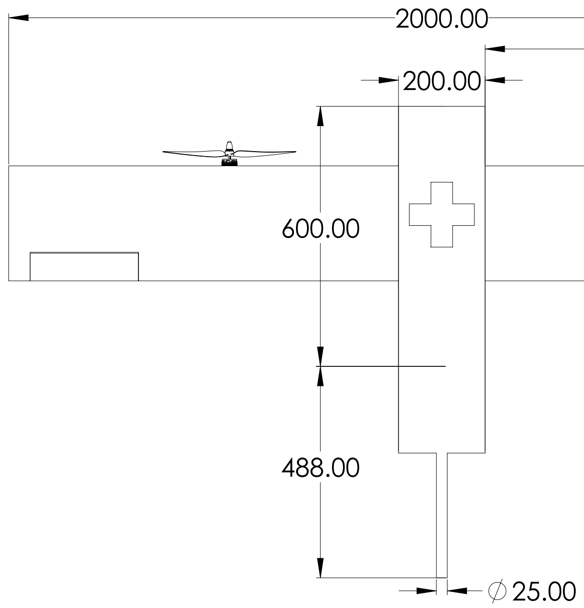


Figure 5.4: Top view of the fuselage showing primary dimensions (in mm)

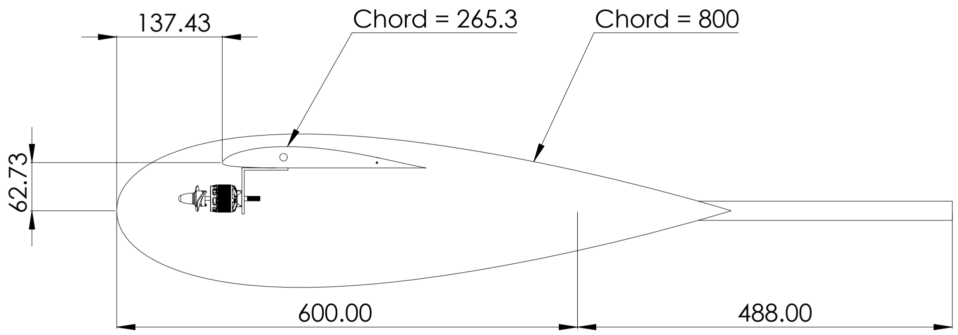


Figure 5.5: Side view of the fuselage showing primary dimensions (in mm)

5.2 Landing Gear Design

The landing gear configuration is a critical factor in ensuring ground stability and structural integrity during the most demanding phases of flight: Take-off and landing. For this design, a tricycle gear arrangement has been selected, as it is the most widely utilized and reliable configuration in modern aviation.

As illustrated in Figure 5.6, this setup consists of two main wheels and a single nose-gear. The main wheels are positioned aft of the aircraft's centre of gravity (CG). Because they are located very close to the CG relative to the forward gear, they are responsible for absorbing the vast majority of the aircraft's weight and landing impact loads. To maintain lateral balance, the two main gears are placed at equidistant intervals from the CG along both the longitudinal (x-axis) and lateral (y-axis) planes, ensuring an even load distribution between the left and right sides.

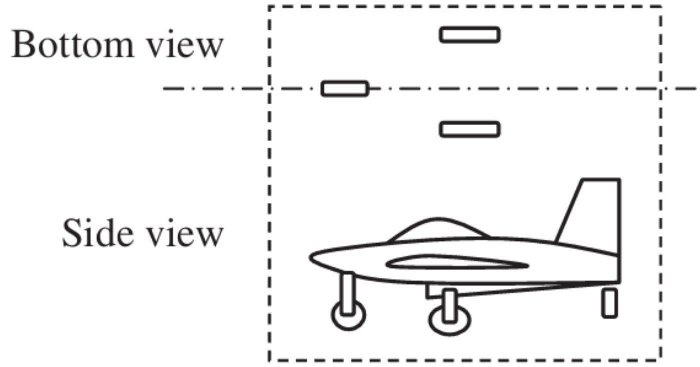


FIGURE 5.6: Top and side schematic views of the tricycle landing gear layout.

The forward gear, commonly referred to as the nose-gear, is positioned much further from the CG than the main gear and consequently carries a significantly smaller portion of the static load.

A primary advantage of this configuration over a tail-dragger setup is its inherent directional stability during ground operations and taxiing. This stability arises from a self-correcting mechanical advantage: If the aircraft begins to yaw slightly while moving on the ground, the rolling and skidding resistance generated by the main gear—acting behind the CG—creates a restoring moment that tends to straighten the aircraft. This characteristic is particularly beneficial during cross-wind landings, as it allows the aircraft to maintain a relatively large crab angle without the risk of a ground loop.

To prevent structural damage during rotation, the landing gear geometry must account for the aircraft's maximum pitch attitude. In this design, the tip-back angle is specifically set to 17°. This value was determined by taking the maximum stall angle of 12°and adding a 5°safety buffer. By establishing this 17°limit, we ensure that even if the pilot rotates the aircraft to its maximum aerodynamic limit during take-off, the tail section will remain clear of the runway surface.

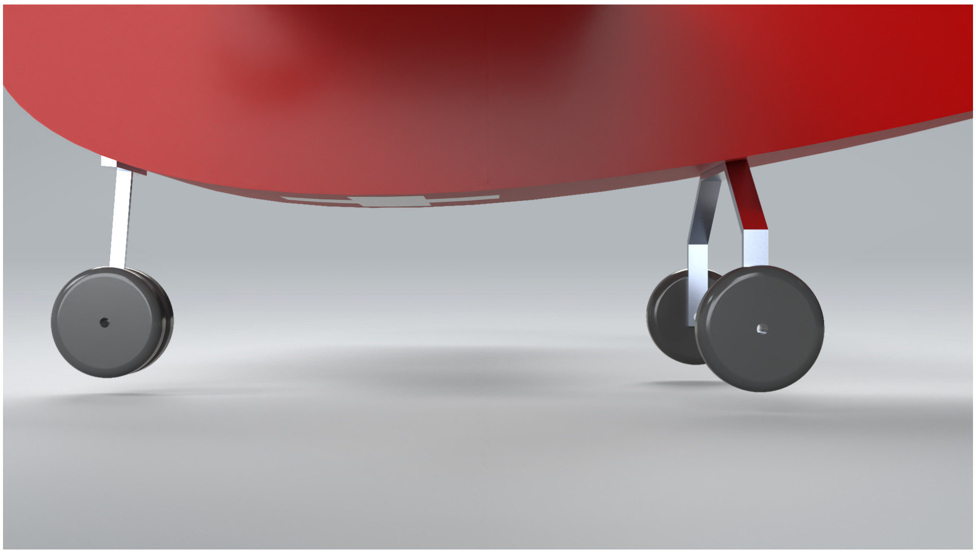


FIGURE 5.7: Landing Gear integration

5.3 Second Weight Estimation (Iteration 2)

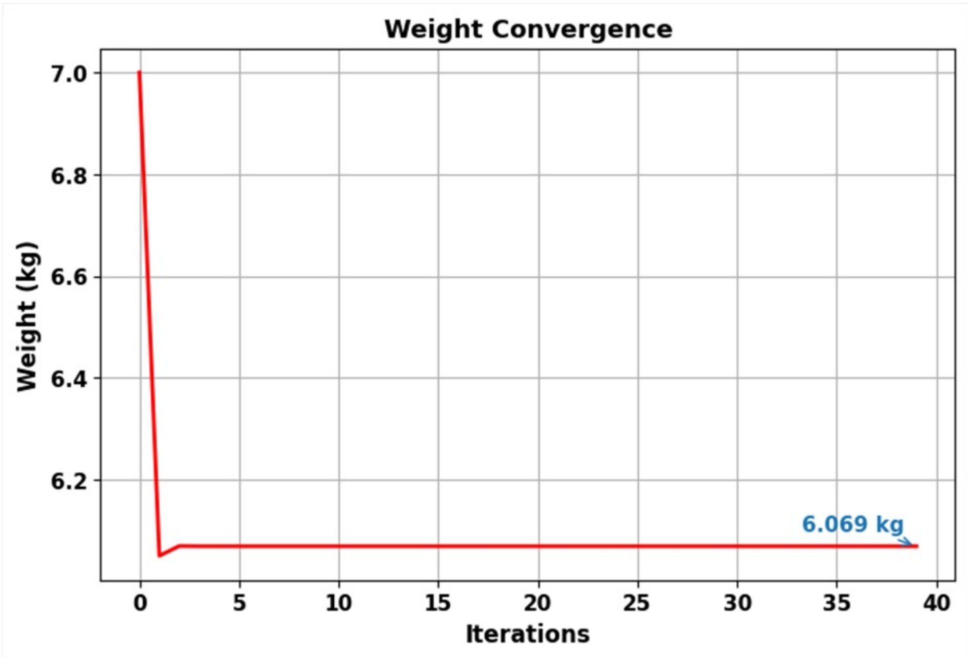


FIGURE 5.8: Second Weight Estimation (Iteration 2 - Initial Phase)

Building upon the initial weight sizing and the convergence methodology established in Section 2.5, a second weight estimation (Iteration 2) was performed. Rather than serving as an isolated calculation, this step acts as the next phase in the iterative design loop, refining the aircraft's mass properties based on updated parameters. This second iteration initially yielded an interim Maximum Take-Off Weight (MTOW) of 6.069 kg, an empty weight of 3.205 kg, and a wing area of 0.5307 m^2.

Considering necessary safety margins—specifically that the UAV cannot always fly at optimal speeds or under ideal conditions—the battery configuration was updated to meet higher sustained power requirements.

The new battery chosen was the **GenX 22.2V 6S 12000MAH 20C / 40C Premium Li-Po Battery** [45]. Because battery weight acts as a driving variable in the total weight loop detailed in Section 2.5, substituting this heavier, higher-capacity battery necessitated a recalculation. Re-initiating the convergence process with the new battery mass resulted in a final updated Maximum Take-Off Weight of **6.592 kg**.

FIGURE 5.9: Updated Battery

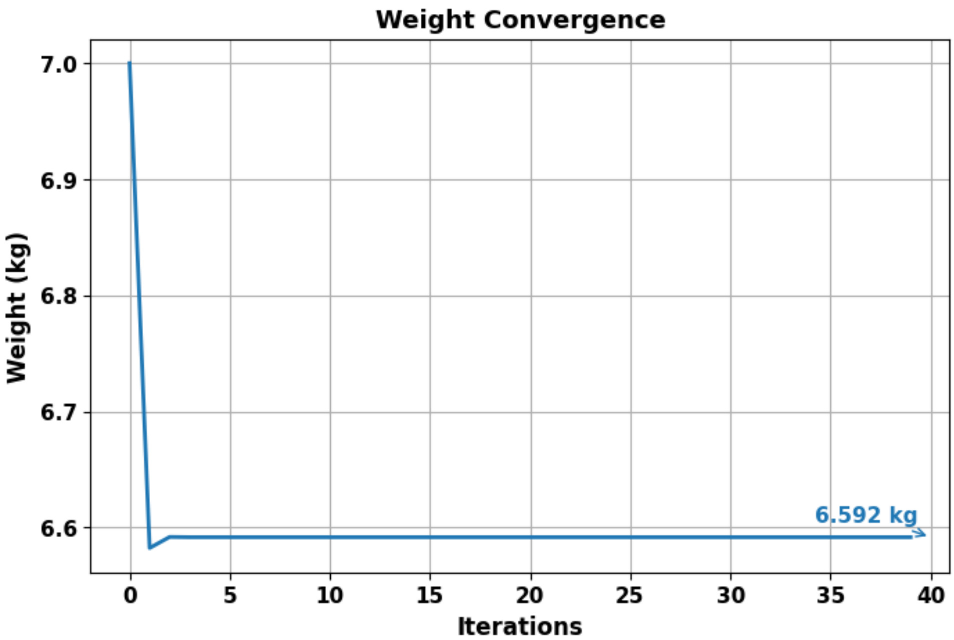
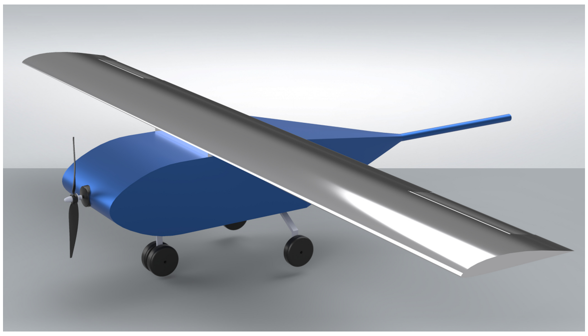


FIGURE 5.10: Second Weight Estimation (Iteration 2 - Final Convergence)

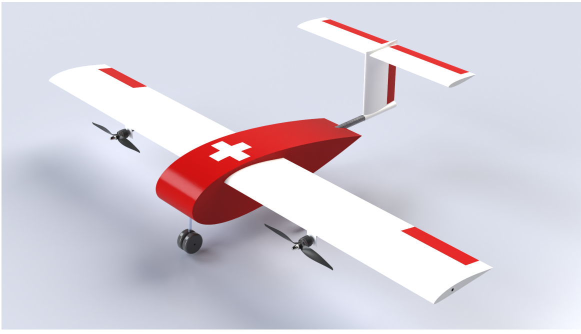
Table 5.1: Iterative Weight Estimation History

Parameter	**Iteration 1** *(Ch. 2)*	**Iteration 2** *(Ch. 5)*	**Final** *(Ch. 5, converged)*
MTOW, W_0 (kg)	6.896	6.069	**6.592**
Empty Weight, W_e (kg)	3.270	3.205	3.205
Battery Mass, W_b (kg)	1.530	—	**1.287**
Payload, W_p (kg)	2.100	2.100	2.100
Wing Area, S (m^2)	0.413	0.5307	0.5307
Aspect Ratio, AR	9.686	7.53	7.53
Wing Loading, W/S (N m^{-2})	—	121.84	121.84
Battery selected	GenX 3S 16 Ah	GenX 6S 12 Ah	GenX 6S 12 Ah
Battery energy (kJ)	657.0	—	959.0
Trigger for next iteration	*Wing-loading constraints*	*Battery upgrade*	*Converged*

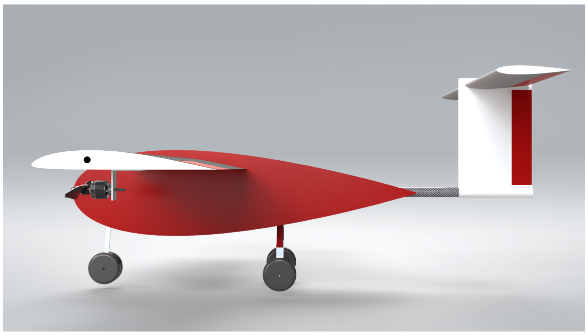
5.4 Updated CAD Model



Isometric View – first iteration

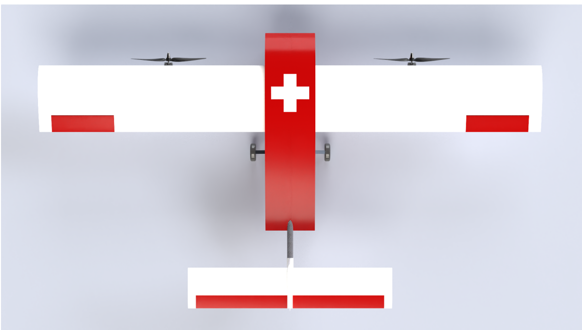


Isometric View – revised iteration

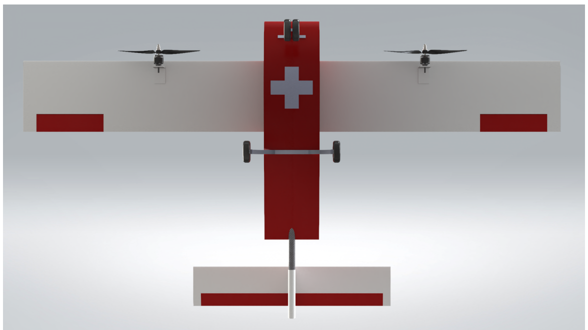


Side View Profile

Front View Profile



Top View Profile



Bottom View Profile

FIGURE 5.11: Different Rendered profile views of the UAV.

5.5 Summary

Table 5.2: Key Design Parameters — Chapter 5: Fuselage & Airframe Design

Parameter	Symbol	Unit	Previous	Current
Fuselage geometry				
Fuselage Length	L_f	m	—	1.200
Cross-section Profile	—	—	—	NACA 0025
Fuselage Chord (profile)	—	mm	—	1200
Central Body Width	—	mm	—	300
Fuselage thickness ratio	t/c	—	—	0.25
Tail boom				
Cross-section	—	—	—	Circular
Boom Diameter	d_{boom}	mm	—	20
Landing gear				
Configuration	—	—	—	Tricycle (nose-gear)
Tip-back Angle Limit	—	deg	—	17
Stall Angle (max)	α_s	deg	—	12
Updated from weight iteration (second estimate)				
Max Take-off Weight	W_0	kg	6.896	**6.592**
Empty Weight	W_e	kg	3.270	3.205
Battery	—	—	GenX 3S 16 Ah	**GenX 6S 12 Ah**
Carried forward unchanged				
Wingspan	b	m	2.000	2.000
Wing Area	S	m^2	0.5307	0.5307

New parameter established this chapter   Updated from previous chapter

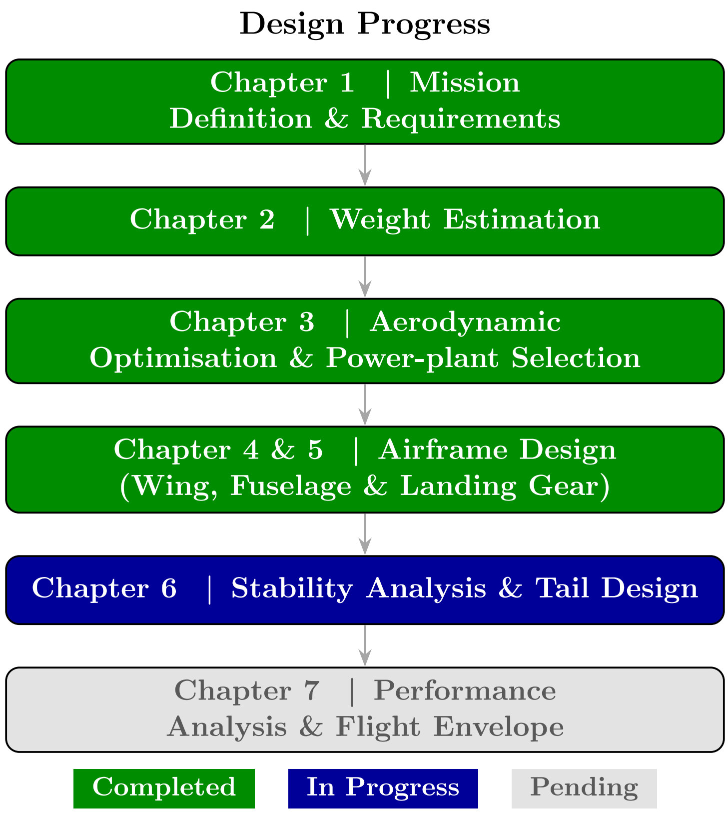


FIGURE 5.12: Overall UAV Conceptual Design Workflow — Progress after Chapter 5

Chapter 6

Stability Determination

6.1 Preliminary Centre of Gravity Estimation

The estimation of the centre of gravity (CG) is a crucial step in the UAV design process, as it directly influences the stability, controllability, and overall flight performance of the aircraft. An accurate prediction of the CG ensures that the UAV operates within safe stability limits and avoids undesirable flight characteristics such as excessive pitching or loss of control.

In this work, the CG of the UAV was determined using a CAD-based mass property analysis in SolidWorks. A detailed three-dimensional model of the UAV was developed, incorporating all major structural and functional components, except the tail, including the fuselage, main wing, propulsion system, and onboard payload.

To achieve a realistic estimation of mass distribution, suitable materials were assigned to each component within the SolidWorks environment. The density of each material was defined using the built-in material library, allowing the software to compute accurate mass properties for individual components as well as for the complete assembly.

6.1.1 Material Assignment and Mass Distribution without tail

Component	Material Assigned	Density (kg/m^3)	Mass (kg)
Fuselage, Tail Boom	Aluminum 6061, Carbon fiber rod	2700, 1780	0.876
Wing, Aileron	Aluminum 6061	2700	1.471
Motor (2Nos)	–	–	0.264
Propeller (2Nos)	–	–	0.096
Battery	–	–	1.530
Landing Gear, Wheel	Aluminum 6061, Rubber	2700, 1000	0.565
Drop Payload	–	–	1.200
Tail	–	–	
Total	–	–	**6.002**

Table 6.1: Material Assignment and Mass Distribution of UAV Components

Once the material properties were assigned, the **Mass Properties** tool in SolidWorks was utilized to evaluate the overall mass, centre of mass, and related inertial properties of the UAV. The CG coordinates were obtained with respect to a predefined reference coordinate system located at the nose. The coordinate axes follow the standard aircraft convention, where the X-axis is positive towards the nose, the Y-axis is positive towards the starboard (right) wing, and the Z-axis is positive downwards.

The computed CG location of the UAV is given by:

- Longitudinal = 221 mm(from the nose of the fuselage)
- Lateral $\approx$ 0.00 mm
- Vertical $\approx$ 0.00 mm

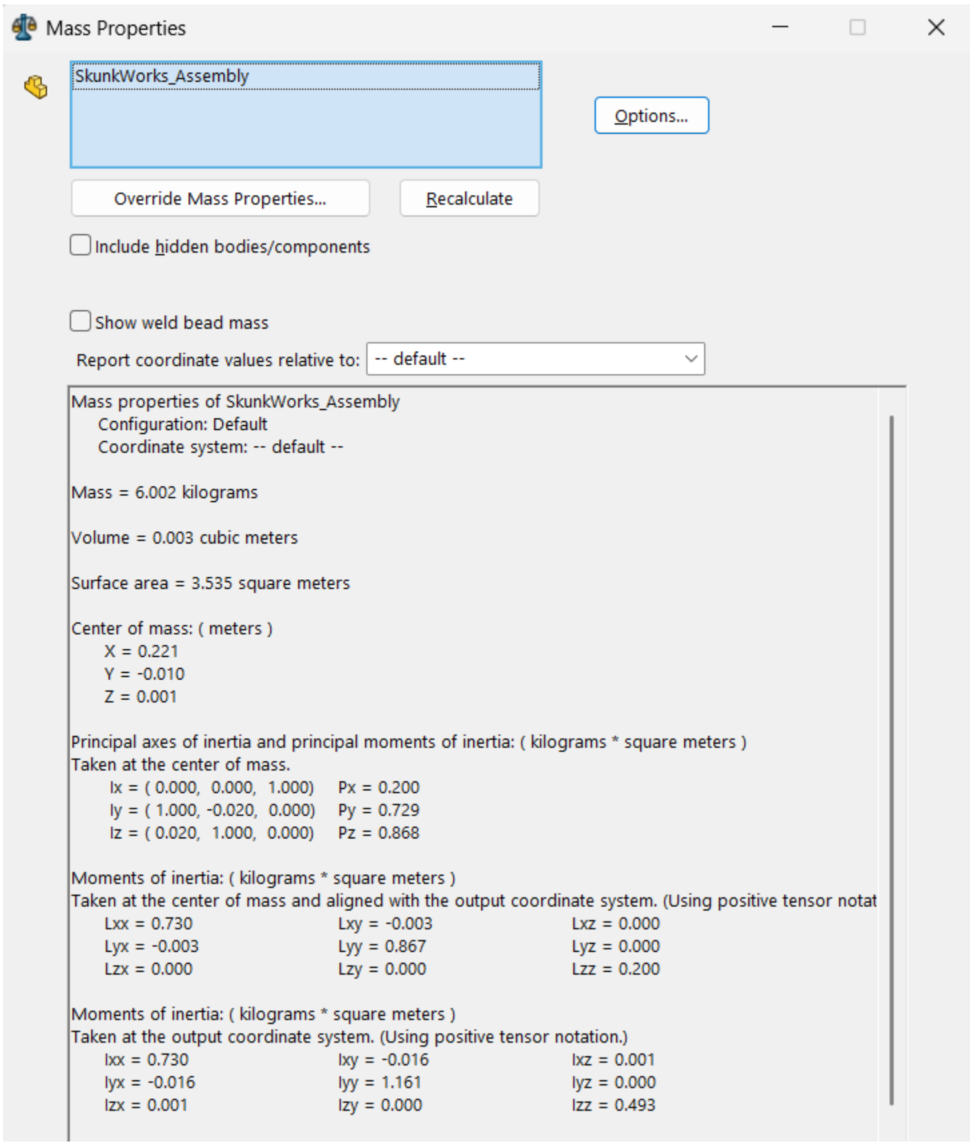


FIGURE 6.1: Mass Properties of CAD from Solidworks

6.2 Longitudinal Static Stability

6.2.1 Definition

Longitudinal static stability refers to the ability of an aircraft to develop a restoring pitching moment when it is disturbed from its equilibrium (trim) condition [52] pg 42. For an aircraft in steady flight, the trim condition is defined by zero pitching moment:

$$C_m = 0 \tag{6.1}$$

If the aircraft experiences a disturbance in angle of attack α, the resulting pitching moment should act to restore the aircraft to its original equilibrium condition.

Consider two aircraft at a trim point. If an upward gust increases the angle of attack to a higher value, a statically stable aircraft will generate a negative (nose-down) pitching moment, tending to return it to equilibrium. Conversely, if the disturbance reduces the angle of attack, a stable aircraft will generate a positive (nose-up) moment, again restoring equilibrium.

6.2.2 Condition for Static Stability

From this behavior, the condition for longitudinal static stability is that the slope of the pitching moment coefficient with respect to angle of attack must be negative:

$$\frac{dC_m}{d\alpha} < 0 \tag{6.2}$$

This condition must hold at the equilibrium (trim) point.

6.2.3 Trim Requirement

In addition to stability, the aircraft must also be trimmable. This requires that the pitching moment coefficient curve intersects the zero-moment axis at a positive angle of attack. Thus:

$$C_{m_0} > 0 \tag{6.3}$$

where C_{m_0} is the pitching moment coefficient at zero angle of attack. This ensures that trim can be achieved at a physically meaningful (positive) angle of attack.

6.2.4 Alternative Formulation

The stability condition can also be expressed in terms of the lift coefficient C_L, since C_L varies with α. In this case, the condition becomes:

$$\frac{dC_m}{dC_L} < 0 \tag{6.4}$$

6.2.5 Contribution of Aircraft Components

The overall pitching moment and static stability characteristics of an aircraft arise from the combined contributions of its components, including the wing, fuselage, horizontal tail, and propulsion system. For analysis purposes, the aircraft is often decomposed into these individual components, and their respective contributions to the pitching moment coefficient are evaluated separately.

These contributions can be estimated using theoretical methods, empirical correlations, or experimental data. Such methods are commonly compiled in stability and control data references and are used for preliminary aircraft design and analysis.

In summary, longitudinal static stability requires:

- A restoring pitching moment after disturbance
- Negative slope of pitching moment curve:

$$\frac{dC_m}{d\alpha} < 0$$

- Ability to trim at a positive angle of attack:

$$C_{m_0} > 0$$

6.3 Different components and Longitudinal Stability

6.3.1 Wing

Wing body alone is an unstable configuration while contributing to changes in pitching moment coefficient with respect to alpha. The moment contribution due to the wing, about the CG of the airplane,comes from the moment due to lift production and the moment independent of the lift.

The following formula is used to estimate the wing contribution to the longitudinal stability [52]

$$C_{m_{cg,w}} = C_{m_{ac,w}} + \left(C_{L_{0,w}} + C_{L_{\alpha,w}}\alpha_w\right)\left(\frac{x_{cg}}{c} - \frac{x_{ac}}{c}\right) \tag{6.5}$$

where

$$C_{L_w} = C_{L_{0,w}} + C_{L_{\alpha,w}}\alpha_w \tag{6.6}$$

Applying the condition for static stability yields

$$C_{m_{0,w}} = C_{m_{ac,w}} + C_{L_{0,w}}\left(\frac{x_{cg}}{c} - \frac{x_{ac}}{c}\right) \tag{6.7}$$

$$C_{m_{\alpha,w}} = C_{L_{\alpha,w}}\left(\frac{x_{cg}}{c} - \frac{x_{ac}}{c}\right) \tag{6.8}$$

Moment independent of lift

The pitching moment coefficient,at zero lift (Cm ac) is **-0.0683**.

This is estimated by running the flow analysis for different angles of attack, the hinge point at which pitching moment coefficient remained constant for different angles of attack was taken to be Aerodynamic Center of the wing. The aerodynamic center of the wing was estimated to be **0.1985m** from the nose of the fuselage.

Moment due to lift

$$C_{m_{\alpha,w}} = C_{L_{\alpha,w}} \left(\frac{x_{cg}}{c} - \frac{x_{ac}}{c} \right) \tag{6.9}$$

The moment due to lift about C.G. is estimated as follows. At angle of attack = 3 degrees, from the XFLR analysis, we found our CL for the finite wing to be 0.378058

$$C_{L_{\alpha,w}} = 3.6107/radians \tag{6.10}$$

$$C_{m_L} = 0.0321 \tag{6.11}$$

Wing contribution to pitching moment coefficient

$$C_{m_w} = C_{m_{Acw}} + C_{m_{Lw}} \tag{6.12}$$

$$C_{m_{cg,w}} = \textbf{-0.014722} \tag{6.13}$$

Now,for the longitudinal stability derivative, we can get

$$C_{m_{\alpha,w}} = C_{L_{\alpha,w}} \left(\frac{x_{cg}}{c} - \frac{x_{ac}}{c} \right) \tag{6.14}$$

$$C_{m_{\alpha,w}} = 3.6107 \left(\frac{0.221}{0.2653} - \frac{0.1985}{0.2653} \right) \tag{6.15}$$

$$C_{m_{\alpha,w}} = \mathbf{0.3062/radians} \tag{6.16}$$

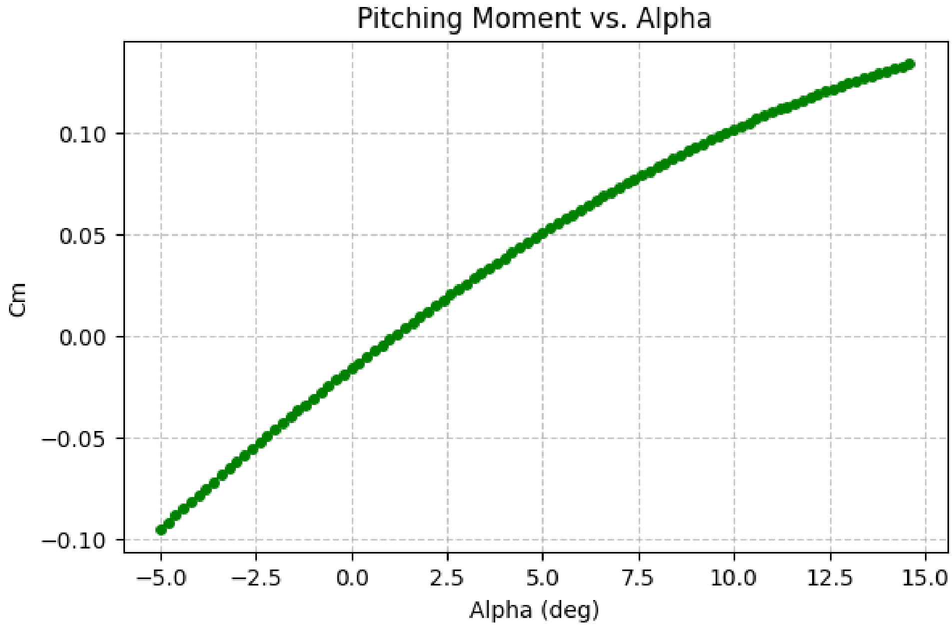


FIGURE 6.2: Cm Vs alpha for wing

This implies that **Wing alone - Unstable configuration**.

6.3.2 Fuselage

The fuselage for our UAV is again a wing configuration .For estimating the stability derivatives,same procedure is carried out as wing's. Fusleage alone is an unstable configuration while contributing to changes in the pitching moment coefficient with respect to alpha. The moment contribution due to the wing, about the CG of the airplane,comes from the moment due to lift production and the moment independent of the lift.

The following formula is used to estimate the contribution of the fuselage to longitudinal stability.

The Fuselage cross sectional is a NACA 0025 aerofoil.It is a symmetric section expecting the aerodynamic center contribution to the pitching moment to be zero.

$$C_{m_{cg,f}} = C_{m_{ac,f}} + \left(C_{L_{0,f}} + C_{L_{\alpha,f}}\alpha_f\right)\left(\frac{x_{cg}}{c} - \frac{x_{ac_f}}{c}\right) \tag{6.17}$$

where

$$C_{L_f} = C_{L_{0,f}} + C_{L_{\alpha,f}}\alpha_f \tag{6.18}$$

Applying the condition for static stability yields

$$C_{m_{0,f}} = C_{m_{ac,f}} + C_{L_{0,f}}\left(\frac{x_{cg}}{c} - \frac{x_{ac_f}}{c}\right) \tag{6.19}$$

$$C_{m_{\alpha,f}} = C_{L_{\alpha,f}}\left(\frac{x_{cg}}{c} - \frac{x_{ac_f}}{c}\right) \tag{6.20}$$

Moment independent of lift

Using Vortex Latex Method from XFLR analysis,the pitching moment coefficient,at zero lift is **0**.

This is estimated by running the flow analysis for different angles of attack, the hinge point at which pitching moment coefficient remained constant for different angles of attack was taken to be Aerodynamic Centre of the wing. Aerodynamic Centre of the fuselage was estimated to be **0.09 m** from the nose of the fuselage.

Moment due to lift

$$C_{m_{\alpha,f}} = C_{L_{\alpha,f}}\left(\frac{x_{cg}}{c} - \frac{x_{ac_f}}{c}\right) \tag{6.21}$$

Moment due to lift about C.G. is estimated as follows.

At angle of attack = 3 degrees, using Vortex Latex Method from XFLR analysis, we found out our C_L for the 3D Fuselage to be 0.061015.

Characteristic Length of Fuselage,c = 0.8 m

$$C_{L_{\alpha,f}} = 0.7952/radians \tag{6.22}$$

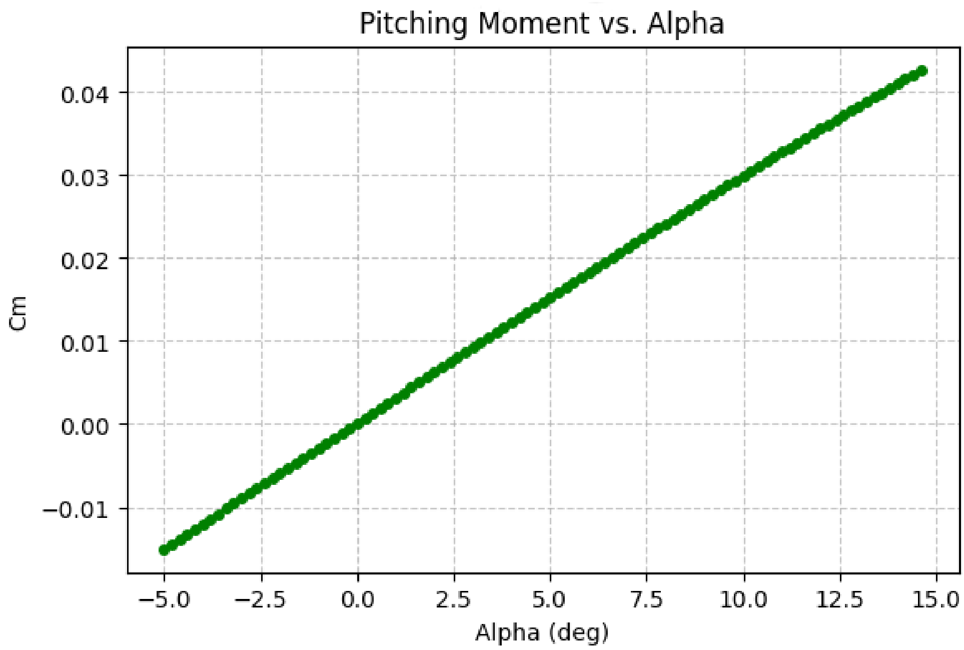


FIGURE 6.3: C_m v. α for fuselage

$$C_{m_{L_f}} = 0.01 \tag{6.23}$$

Fuselage contribution to pitching moment coefficient

$$C_{m_f} = C_{m_{Ac_f}} + C_{m_{Lf}} \tag{6.24}$$

$$C_{m_{cg,f}} = \mathbf{0.01} \tag{6.25}$$

Now,for the longitudinal stability derivative, we can get

$$C_{m_{\alpha,f}} = C_{L_{\alpha,f}} \left(\frac{x_{cg}}{c} - \frac{x_{ac_f}}{c} \right) \tag{6.26}$$

$$C_{m_{\alpha,f}} = \mathbf{0.130214/radians} \tag{6.27}$$

This implies that **Fuselage alone – unstable configuration**.

6.3.3 Tail Contribution and Static Margin Requirements

6.3.3.1 Need for Horizontal Tail Contribution

In the present design iteration, the overall pitching moment coefficient (C_M) of the aircraft is found to be negative under operating conditions. This indicates a net nose-down pitching tendency about the centre of gravity. In order to achieve trim ($C_M = 0$), it is therefore necessary for the horizontal tail to generate a counteracting nose-up moment.

This is typically achieved by designing the tail to produce a negative lift (downward force). Since the tail is located aft of the center of gravity, a downward force in the tail produces a positive pitching moment (nose-up), thus balancing the nose-down contribution from the wing–body combination. The slope of the pitching moment coefficient with respect to the angle of attack (C_{M_α}) is observed to be positive for the current configuration. This implies that any increase in angle of attack results in a further nose-up pitching moment, which drives the aircraft away from equilibrium. This condition corresponds to a longitudinal static instability.

Therefore, the horizontal tail must provide a sufficiently large negative contribution to C_{M_α} to counteract the destabilizing effects of the wing and fuselage. This is achieved through the appropriate selection of the tail size, tail moment arm, and aerodynamic characteristics, which together determine the tail volume coefficient and its effectiveness.

The pitching moment contribution of the horizontal tail about the aircraft CG can be expressed as:

$$C_{m_t} = -\eta_t \, V_H \, C_{L_t} \tag{6.28}$$

and its slope with respect to angle of attack is:

$$C_{m_{\alpha,t}} = -\eta_t \, V_H \, C_{L_{\alpha,t}} \left(1 - \frac{\mathrm{d}\epsilon}{\mathrm{d}\alpha} \right) \tag{6.29}$$

where:

- η_t: tail efficiency = 0.99 (As we are opting for T-Tail Configuration)
- $V_H = \dfrac{S_t l_t}{Sc}$: Horizontal tail volume ratio
- $C_{L_{\alpha,t}}$: Lift curve slope of tail
- $\dfrac{d\epsilon}{d\alpha}$: Downwash gradient (typically $0.3 - 0.5$)

6.3.4 Stability Margin and Horizontal Tail Design

Two parameters that are required to trim the UAV at trim angle of attack and lift coefficient are Tail Arm and Tail Setting Angle.

6.3.4.1 Tail Area

Aspect Ratio of horizontal tail is taken as **4.5**,Horizontal Tail area **St as 25% of wing area** after referencing the book [1]. A Tail arm of 0.7814m is taken to achieve a static margin of 15 percent.

Parameter	**Value**
α_{trim}	3 deg
X_{cg}	0.221 m
$X_{ac,w}$	0.1985 m
$X_{ac,f}$	0.09 m
S_t	0.1327 m^2
η	0.99
C_L	0.378058
$C_{L_{\alpha,t}}$	3.8310 rad^{-1}
AR_{HT}	4.5

TABLE 6.2: Geometric and Aerodynamic parameters used for tail sizing

- Tail Area = 0.1327 m^2
- Horizontal Stabilizer Aspect Ratio = 4.5

6.3.4.2 Tail Setting Angle

The Total pitching moment coefficient about the C.G of the UAV is calculated from below:

$$C_{m_{cg}} = C_{m_{ac,w}} + C_{L_w}(X_{cg} - X_{ac,w}) - \eta\left(\frac{S_t}{S}\right)\left[C_{L_{\alpha,t}}(\alpha + i_t - \frac{d\epsilon}{d\alpha}\alpha)\right]\left(\frac{l_t}{\bar{c}}\right) + C_{m_f} \quad (6.30)$$

Applying the trim condition, we obtain the tail setting angle as

$$i_t = \frac{C_{m_{ac,w}} + C_{L_w}(X_{cg} - X_{ac,w}) + C_{m_f}}{\eta\left(\frac{S_t}{S}\right)C_{L_{\alpha,t}}\left(\frac{l_t}{\bar{c}}\right)} - \alpha + \frac{d\epsilon}{d\alpha}\alpha \quad (6.31)$$

From (6.31) it is evident that we require tail lift curve slope before computing the tail setting angle. A Horizontal Tail of area 0.1327 m^2 and aspect ratio of 4.5 is made into a CAD Model and analyzed using VLM in XFLR.

The aerofoil chosen for horizontal tail is **NACA 0012**.

- This is useful because the tail should generate positive or negative lift depending on trim requirements, **without a built-in lift bias**. For symmetric aerofoils:

  $$C_{m,ac} = 0$$

  So the aerofoil itself does not introduce additional pitching moments.

- Since the aerofoil does not produce lift at zero angle, the required tail force can be adjusted using incidence or elevator deflection, helping reduce unnecessary drag.

- The horizontal tail sometimes needs to produce:

  1. Downward lift (most conventional aircraft)

  2. Upward lift in some flight conditions (manoeuvring or CG shift)

  A symmetric aerofoil works equally well in positive and negative angles of attack. This simplifies the longitudinal stability analysis of the aircraft.

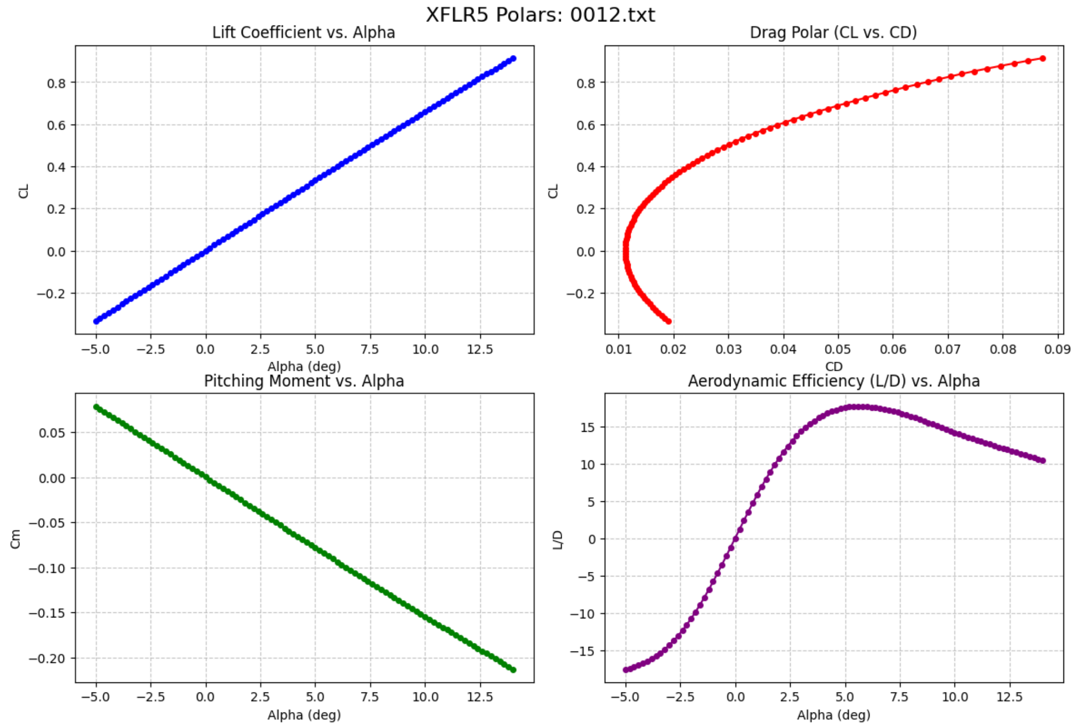


FIGURE 6.4: Aerodynamic forces and moments for NACA 0012 Tail

$$C_{L\alpha_t} = 3.8310/\text{rad} \tag{6.32}$$

6.3.5 Vertical Tail Design

Vertical Tail provides directional stability to the UAV by producing a restoring moment to the UAV. Directional static stability requires the derivative of the yawing moment coefficient with respect to the sideslip angle to be greater than zero.(Sign Conventions are such).

We desire to make a **T-Tail configuration** as

- Places the horizontal tail out of the wake(undisturbed flow) improving horizontal tail efficiency.
- The tail experiences less aerodynamic interference from the wing and fuselage.
- The vertical tail provides structural support to the horizontal tail instead of the tail boom supporting both.

Necessary condition for Directional Stability(WeatherCock Effect):

$$C_{n_\beta} > 0 \tag{6.33}$$

$$C_{n_{\beta v}} = \eta_v \frac{S_v l_v}{Sb} C_{L_{\alpha v}} \tag{6.34}$$

where $C_{L_{\alpha_v}}$ represents the lift curve slope of vertical tail

$$C_{n_\beta} = C_{n_{\beta w}} + C_{n_{\beta f}} + C_{n_{\beta v}} \tag{6.35}$$

For small UAV design, a common design target is

$$\frac{S_v}{S} = 0.09 \tag{6.36}$$

$$S_v = 0.09S \tag{6.37}$$

$$S_v = 0.09 \times 0.5307 = 0.0478\ m^2 \tag{6.38}$$

$$AR_v = \frac{b_v^2}{S_v} \tag{6.39}$$

$$b_v = \sqrt{AR_v S_v} \tag{6.40}$$

$$b_v = \sqrt{1.5 \times 0.0478} = 0.268\ m \tag{6.41}$$

For a rectangular vertical tail,

$$S_v = b_v c \tag{6.42}$$

$$c = \frac{S_v}{b_v} \tag{6.43}$$

$$c = \frac{0.0478}{0.268} = 0.178\ m \tag{6.44}$$

Since the planform is rectangular,

$$c_r = c_t = MAC_v = 0.178\ m \tag{6.45}$$

Also,the vertical tail does not have taper and no dorsal fin attached,we design the vertical tail arm length to be same as horizontal tail arm.
i.e The vertical tail arm is assumed to be equal to the horizontal tail arm since both tails are to be located approximately at the same fuselage station.

$$l_v = l_t \tag{6.46}$$

NACA 0012 is used to design the vertical tail with the above tabulated parameters.

6.4 Centre of Gravity Estimation with Empennage

Assembling all the parts of the UAV with the Tail designed as above and tail arm **0.7m**, again the same procedure to estimate the Centre of Gravity of the UAV was done as stated in section (6.1).

6.4.1 Material Assignment and Mass Distribution

Component	Material Assigned	Density (kg/m^3)	Mass (kg)
Fuselage, Tail Boom	Aluminum 6061, Carbon fiber rod	2700, 1780	0.876
Wing, Aileron	Aluminum 6061	2700	1.471
Motor (2Nos)	–	–	0.264
Propeller (2Nos)	–	–	0.096
Battery	–	–	1.530
Landing Gear, Wheel	Aluminum 6061, Rubber	2700, 1000	0.565
Drop Payload	–	–	1.200
Tail	Aluminum 6061	2700	0.296
Total	–	–	**6.298**

TABLE 6.3: Material Assignment and Mass Distribution of UAV Components

Once again, the **Mass Properties** tool in SolidWorks was utilized to evaluate the overall mass, centre of mass, and related inertial properties of the UAV similar to the preliminary estimation.

The updated CG location of the UAV is given by:

- Longitudinal = 264 mm (from the nose of the fuselage)
- Lateral ≈ 0.00 mm
- Vertical ≈ 0.00 mm

It is clear from the C.G position that the new C.G lies before the old Neutral Point gives glimpse that the current configuration might not require further iterations in designing.

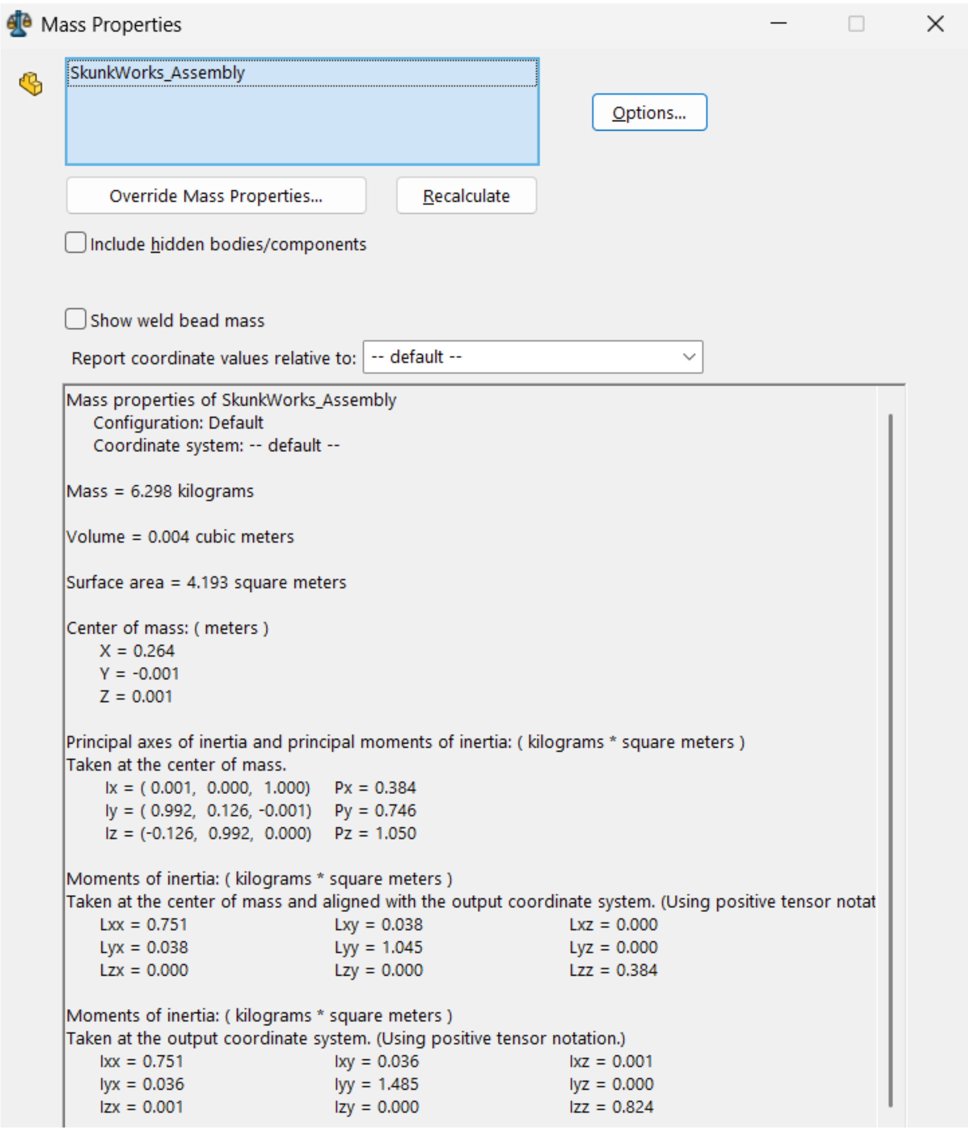


FIGURE 6.5: Mass Properties of CAD from Solidworks

6.4.2 Tail Arm

To get a static margin of around **13 percent** and enough clearance for vertical tail, we estimated the horizontal tail arm as **0.7m** with neutral point 0.29932 from the nose of the fuselage.

$$l_t = \frac{\left[(X_{np} - X_{ac_w}) + \frac{C_{m_f}}{C_{L_\alpha}}\right] S}{\eta \, S_t \left(\frac{C_{L_{\alpha t}}}{C_{L_\alpha}}\right) \left(1 - \frac{d\epsilon}{d\alpha}\right)} = 0.7m \tag{6.47}$$

6.4.3 Total Longitudinal Stability Derivative with Tail

To validate the above estimated parameters,we verified that the Cmcg for the below specified trim conditions (@ $C_{L,\text{trim}}$, α_{trim}, i_t, Tail arm) is found out to be zero by substituting all these in eqn.(6.30).

$$\frac{d\epsilon}{d\alpha} = \frac{2C_{L_\alpha}}{\pi e AR} = 0.3991/\text{rad} \tag{6.48}$$

$$i_t = \frac{C_{m_{ac_w}} + C_{L_w} \frac{X_{cg} - X_{ac_w}}{\bar{c}} + C_{m_f}}{\eta \left(\frac{S_t}{S}\right) C_{L_{\alpha t}} \left(\frac{l_t}{\bar{c}}\right)} - \alpha + \left(\frac{d\epsilon}{d\alpha}\right) \alpha = -0.3126deg \tag{6.49}$$

$$\alpha_t = \alpha + i_t - \epsilon = 0.0260\text{rad} \tag{6.50}$$

$$C_{L_t} = C_{L_{\alpha t}} \, \alpha_t = 0.0996 \tag{6.51}$$

$$C_{m_0} = -C_{m_\alpha} \, \alpha_{\text{trim}} = 0.0252 \tag{6.52}$$

$$C_{m_\alpha} = \left(\frac{X_{cg} - X_{ac_w}}{\bar{c}}\right) C_{L_\alpha} - \eta V_H C_{L_{\alpha_t}} \left(1 - \frac{d\epsilon}{d\alpha}\right) + C_{m_{\alpha f}} = -0.4807/rad \tag{6.53}$$

$$C_m = C_{m_0} + C_{m_\alpha} \alpha_{frl} \tag{6.54}$$

The above eqn.(6.54) is used to plot Cm against α_{frl}

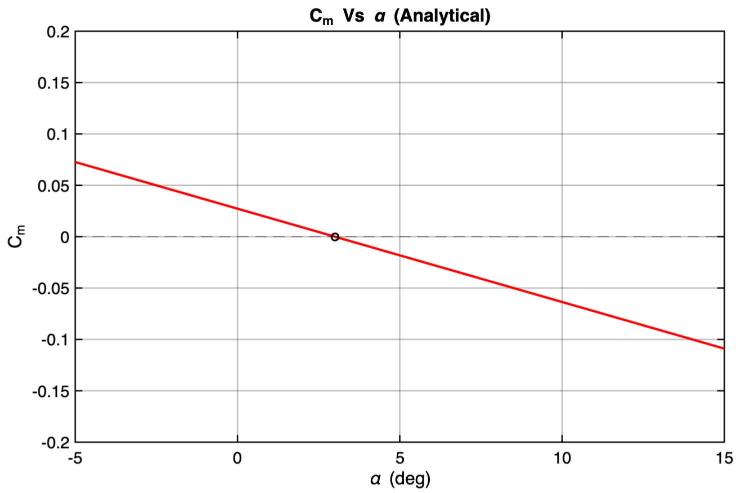


FIGURE 6.6: C_m v. α for UAV

Marker corresponds to the Design Point. Thus we can say that the UAV can be trimmed at $\alpha_{\text{trim}} = 3°$. The wing incidence will therefore be 3°, to maintain zero α_{FRL} at cruise. Thus, the addition of Horizontal Tail to the unstable Wing-Body configuration made sure C_{m_α} is less than zero and C_{m_0} is greater than zero, ensuring fixed stability of the longitudinal static stick. The final tail are summarised in Table 6.4

Parameter	**Value**
Horizontal Tail Area (S_t)	0.1327 m^2
Horizontal Tail Volume Ratio(V_H)	0.66
Horizontal Tail Arm Length (l_t)	0.7 m
Horizontal Tail Aspect Ratio (AR_t)	4.5
Horizontal Tail Span (b_t)	0.773 m
Horizontal Tail Mean Aerodynamic Chord ($\bar{c}_t$)	0.172 m
Horizontal Tail Setting Angle (i_t)	$-0.3126°$
Vertical Tail Area (S_t)	0.0478 m^2
Vertical Tail Arm Length (l_t)	0.7 m
Vertical Tail Aspect Ratio (AR_t)	1.5
Vertical Tail Span (b_t)	0.268 m
Vertical Tail Mean Aerodynamic Chord ($\bar{c}_t$)	0.178 m
Vertical Tail Setting Angle (i_t)	0°
Wing Incidence (i_w)	3°

TABLE 6.4: Tail Geometry and Incidence Parameters

6.5 Primary Control Surfaces Design

Aircraft maneuverability and stability are achieved using three primary control surfaces: the elevator, rudder, and aileron. These control surfaces allow the aircraft to rotate about its three principal axes: pitch, yaw, and roll. Each control surface generates aerodynamic forces by altering the local camber of the lifting surfaces, thereby modifying the pressure distribution around the aircraft.

6.5.1 Elevator

The elevator is located on the horizontal tail and is responsible for controlling the pitch motion of the aircraft. Pitch motion refers to rotation about the lateral axis of the aircraft, which changes the angle of attack of the wing and consequently the lift generated by the aircraft.

When the elevator is deflected upward, the camber of the horizontal tail decreases, producing a downward aerodynamic force on the tail. This downward force creates a nose-up pitching moment about the aircraft's center of gravity. Conversely, when the elevator is deflected downward, the horizontal tail generates an upward force, resulting in a nose-down pitching moment.

The effectiveness of the elevator depends on several parameters including the horizontal tail area, the distance between the center of gravity and the tail (tail moment arm), and the lift curve slope of the tail airfoil. These parameters are often combined into a non-dimensional quantity known as the horizontal tail volume coefficient,$V_H = \frac{S_H l_H}{S_w c_w}$ where S_H is the horizontal tail area, l_H is the tail arm, S_w is the wing area, and c_w is the mean aerodynamic chord.

A sufficiently large horizontal tail volume ensures adequate pitch control authority and trim capability over the entire center of gravity range.

6.5.2 Rudder

The rudder is mounted on the vertical tail and controls the yaw motion of the aircraft, which is the rotation about the vertical axis. Yaw control is essential for directional stability and coordinated turns.

When the rudder is deflected to the right, it generates a lateral aerodynamic force on the vertical tail, causing the aircraft nose to yaw toward the right. A deflection to the left produces the opposite effect. The rudder is particularly important in counteracting asymmetric thrust conditions, such as an engine failure in multi-engine aircraft, and in maintaining directional stability during crosswind operations.

The size of the vertical tail and rudder is determined using the vertical tail volume coefficient,$V_V = \frac{S_V l_V}{S_w b_w}$ where S_V is the vertical tail area, l_V is the distance from the center of gravity to the vertical tail aerodynamic center, S_w is the wing area, and b_w is the wingspan.

A larger vertical tail volume coefficient provides greater directional stability and yaw control authority.

6.5.3 Ailerons

Ailerons are located near the trailing edge of the wing, typically toward the outer portions of the wingspan. They control the roll motion of the aircraft, which is the rotation about the longitudinal axis.

Ailerons operate in pairs: when one aileron deflects upward, the other deflects downward. This differential deflection produces an asymmetric lift distribution between the left and right wings. The wing with the downward-deflected aileron experiences increased lift, while the wing with the upward-deflected aileron experiences reduced lift. This difference in lift generates a rolling moment.

The roll performance of the aircraft is commonly evaluated using the non-dimensional roll rate parameter, $\frac{pb}{2V}$ where p is the roll rate, b is the wingspan, and V is the flight velocity.

The effectiveness of the ailerons depends on their spanwise location, chord ratio, and aerodynamic effectiveness factor. Typically, ailerons occupy the outer 20–40% of the wing span to maximize roll authority while avoiding excessive structural loads.

6.5.4 Interaction of Control Surfaces

Although each control surface primarily controls a single axis of motion, their aerodynamic effects are not completely independent. For example, aileron deflection can introduce adverse yaw due to asymmetric drag, while rudder deflection can induce rolling motion through dihedral effects. Therefore, coordinated use of the elevator, rudder, and ailerons is necessary for stable and efficient aircraft maneuvering.

Proper sizing of these control surfaces ensures that the aircraft remains controllable throughout the flight envelope, including critical conditions such as takeoff, landing, and asymmetric thrust scenarios.

6.5.5 Aircraft Parameters

The primary aircraft parameters used in the analysis are:

- Wing area $S_w = 0.5307\ m^2$
- Wingspan $b_w = 2\ m$
- Mean aerodynamic chord $c_w = 0.2653\ m$
- Aircraft weight $W = 6.592 \times 9.81\ N$
- Stall velocity $V_{stall} = 12.4\ m/s$

- Lift-off velocity

$$V_{LOF} = 1.1 V_{stall} \tag{6.55}$$

- Minimum control speed

$$V_{MC} = 1.2 V_{stall} \tag{6.56}$$

6.5.6 Elevator Deflection Limit and Sizing

6.5.7 Worst Case Condition

The elevator sizing is performed at the **Most aft center of gravity** i.e Beyond Neutral Point. This represents the worst-case condition because the moment arm between the aerodynamic center and CG becomes maximum, requiring larger elevator deflection to generate sufficient pitching moment for trim. Typical elevator limits for small UAVs are

$$-15^\circ \leq \delta_e \leq 25^\circ \tag{6.57}$$

These limits ensure adequate control authority without causing excessive structural loads.

6.6 Elevator Sizing

The elevator provides pitch control by generating a pitching moment through the horizontal tail. The pitching moment derivative with respect to elevator deflection is

$$\frac{dC_m}{d\delta_e} = -V_H \tau_e C_{L_{\alpha_{HT}}} = -1.0108/rad \tag{6.58}$$

where

- V_H = horizontal tail volume coefficient

- τ_e = elevator effectiveness factor = 0.4
- $C_{L_{\alpha_{HT}}}$ = lift curve slope of the horizontal tail = 3.8310/rad

The trim moment required at the center of gravity is

$$C_{m,\mathrm{trim}_{\delta_{e,\mathrm{req}}}} = \left(\frac{x_{cg_{aft}} - x_{ac}}{c}\right) C_L - C_{m_{ac}} = 0.3908/rad \tag{6.59}$$

The required elevator deflection is then

$$\delta_e = \frac{C_{m,trim}}{dC_m/d\delta_e} = -22.15^\circ \tag{6.60}$$

The maximum required deflection by elevator to mitigate the most aft CG is within limits suggested by FAR.

The geometric parameters of the elevator obtained from the control surface sizing are listed below:

- Elevator panel area = 0.13 m^2
- Elevator control surface area = 0.04 m^2
- Elevator span = 0.77 m
- Elevator chord = 0.05 m
- Maximum elevator deflection required = 22.2°

6.7 Rudder Sizing

6.7.1 Worst Case Condition

The rudder is sized at the **minimum control speed** (V_{MC}) with asymmetric thrust. This condition represents the most critical yawing case because:

- Dynamic pressure is lowest
- Engine-out yawing moment is maximum
- Rudder effectiveness is reduced

6.7.2 Rudder Deflection Limit

Typical rudder deflection limits are

$$-30^\circ \leq \delta_r \leq 30^\circ \tag{6.61}$$

The rudder provides directional control and counters asymmetric thrust conditions.

The vertical tail volume coefficient is defined as

$$V_V = \frac{S_V l_V}{S_w b_w} = 0.032 \tag{6.62}$$

where

- S_V = vertical tail area =0.0485 m^2
- l_V = distance from CG to vertical tail = $0.7m$

The yawing moment due to asymmetric thrust is

$$N_{engine} = T_{engine} y_{engine} = 1.5421N \tag{6.63}$$

The rudder deflection required to counter this moment is

$$\delta_r = \frac{N_{required}}{l_V C_{L_{\alpha_{VT}}} \tau_r q S_V} = 15.3^\circ \tag{6.64}$$

where

$$q = \frac{1}{2}\rho V^2 \tag{6.65}$$

is the dynamic pressure. The maximum required deflection by rudder to counter the OEI condition is within limits suggested by FAR

We design the rudder sizing to be

- Rudder panel area = 0.05 m^2
- Rudder control surface area = 0.01 m^2
- Rudder span = 0.23 m
- Rudder chord = 0.05 m
- Maximum rudder deflection required = 15.3°

6.8 Aileron Sizing

6.8.1 Most Demanding Condition

The ailerons are sized at **low flight speeds** near stall. This condition is critical because:

- Dynamic pressure is low
- Roll damping is high
- Aileron effectiveness decreases

Thus larger deflections are required to maintain adequate roll performance.

6.8.2 Aileron Deflection Limit and Sizing

Typical aileron limits are

$$-20^\circ \leq \delta_a \leq 30^\circ \tag{6.66}$$

Ailerons control the roll motion of the aircraft.

The roll performance requirement is expressed using the non-dimensional roll parameter

$$\frac{pb}{2V} = 0.05\text{rad/s} \tag{6.67}$$

where

- p = roll rate
- b = wingspan
- V = flight velocity

Roll rate is considered to take maximum value of 5 rad/sec as mostly we dont consider to do tighter turns. The rolling moment derivative due to aileron deflection is

$$C_{l_{\delta_a}} = \frac{C_{L_\alpha}}{\pi AR}\tau_a \frac{y_2^2 - y_1^2}{S_w b_w/2} \tag{6.68}$$

where aileron effectiveness factor,$\tau_a = 0.45$

The required aileron deflection becomes

$$\delta_a = -\frac{(pb/2V)C_{l_p}}{C_{l_{\delta_a}}} = 26.6^\circ \tag{6.69}$$

The geometric parameters of the aileron obtained from the control surface sizing are listed below:

- Aileron panel area = 0.53 m^2
- Aileron control surface area = 0.04 m^2
- Aileron span = 0.5 m
- Aileron chord = 0.07 m
- Maximum aileron deflection required = 26.6°

6.9 Results

The final sizing results are summarized below.

Parameter	Elevator	Rudder	Aileron
Surface Area (m^2)	0.04	0.01	0.04
Mean Aerodynamic Chord (m)	0.05	0.05	0.07
Surface Span (m)	0.77	0.23	0.5
Required Deflection (deg)	22°	15°	26.6°

6.9.1 Integrated CAD Model

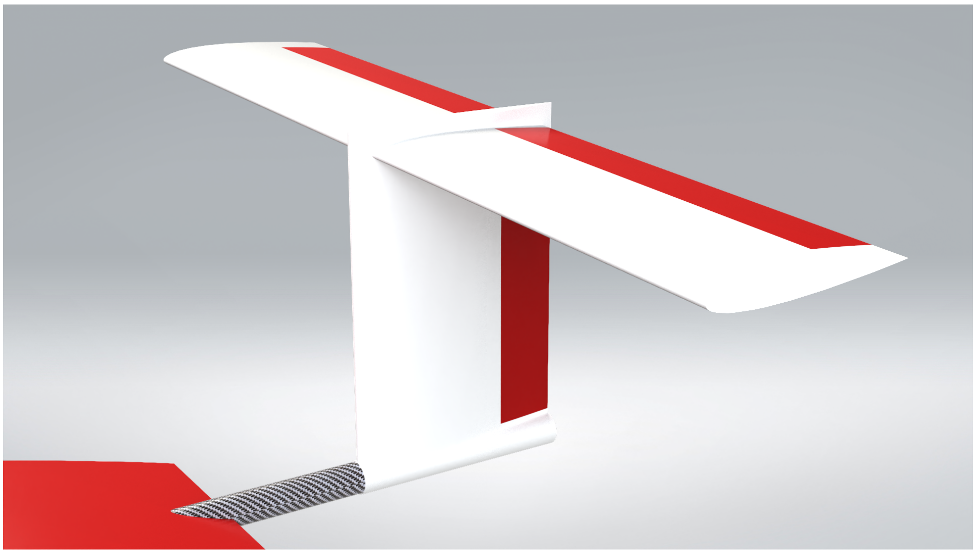


FIGURE 6.7: Zoomed View of the Tail Section

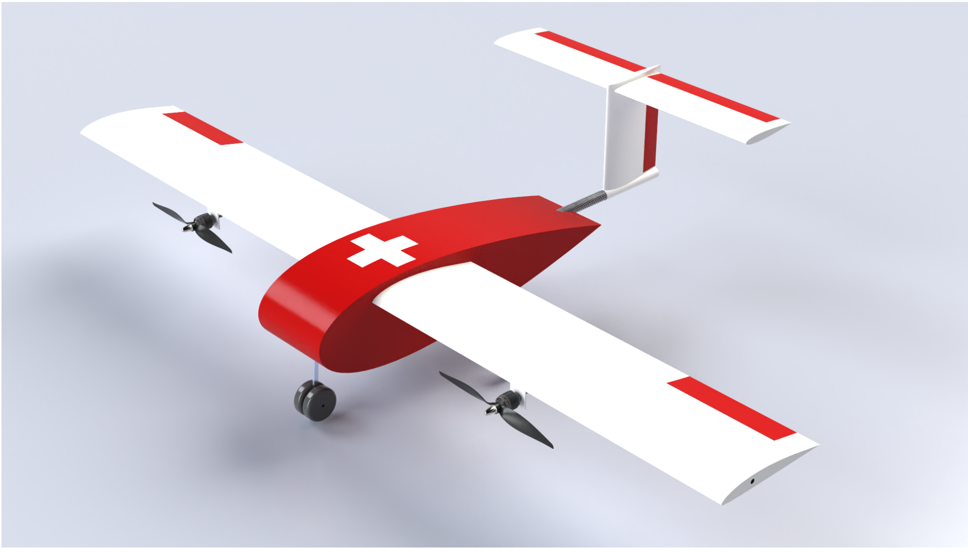


FIGURE 6.8: Full Integrated CAD Model

6.10 Summary

This chapter established the longitudinal and directional static stability characteristics of the UAV. The CG was first estimated without the tail using SolidWorks mass property analysis, yielding a longitudinal CG at 221 mm from the nose. Component-wise pitching moment analysis showed that both the wing ($C_{m_\alpha} = +0.306$ rad^{-1}) and fuselage ($C_{m_\alpha} = +0.130$ rad^{-1}) are individually destabilising. A horizontal tail of area 0.1327 m^2 with a T-tail configuration was designed to restore static stability, achieving a combined $C_{m_\alpha} = -0.4808$ rad^{-1} and a static margin of approximately 13%. The tail setting angle was determined to be $-0.3126°$, and the wing incidence was confirmed at 3° for trim at cruise. A vertical tail of area 0.0478 m^2 was sized to provide directional (weathercock) stability. With the full empennage, the CG shifted to 264 mm from the nose, remaining ahead of the neutral point. Primary control surfaces were subsequently sized under worst-case conditions: the elevator requires a maximum deflection of 22.2° (aft CG), the rudder 15.3° (OEI at V_{MC}), and the aileron 26.6° (low-speed roll), all within FAR-prescribed limits.

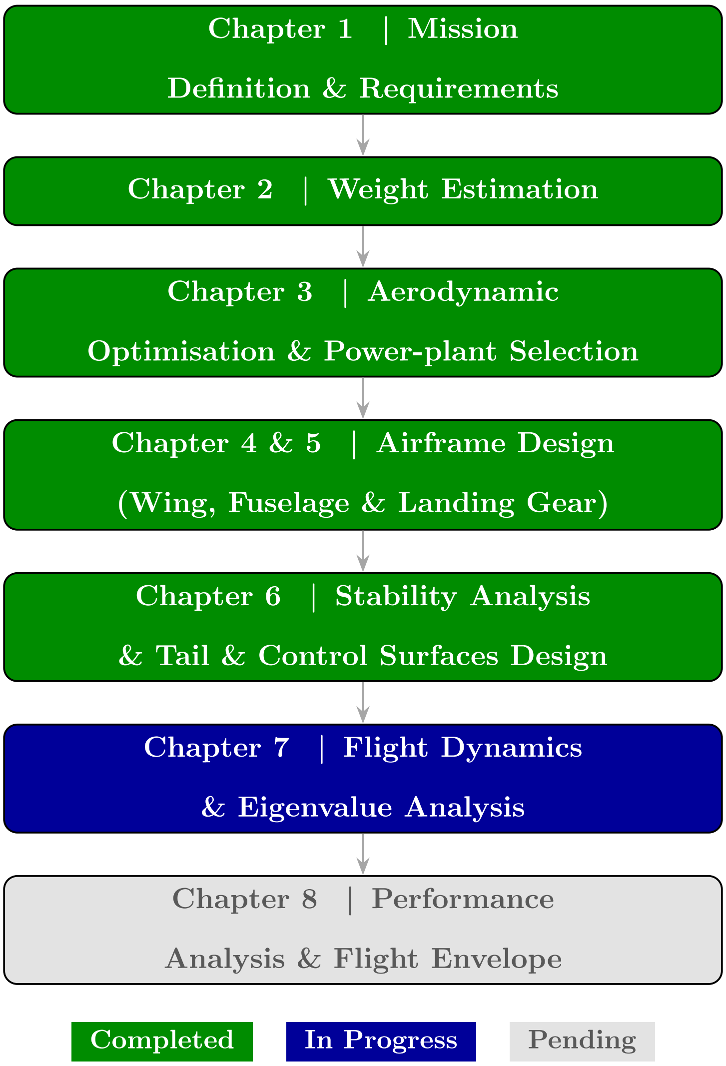


FIGURE 6.9: Overall UAV Conceptual Design Workflow — Progress after Chapter 6

Chapter 7

Flight Dynamics & Eigenvalue Analysis

7.1 Eigenvalue Analysis

This section presents a comprehensive small-perturbation stability analysis of the UAV, following the semi-empirical methodology outlined in the Chapter 9 Appendix of the reference textbook [53]. The analysis is structured to cover: (i) geometry and aerodynamic coefficient estimation, (ii) longitudinal trim, (iii) modal analysis of longitudinal dynamics (short-period and phugoid modes), and (iv) modal analysis of lateral-directional dynamics (roll, Dutch roll, and spiral modes). All numerical results are obtained from the companion Python implementation and validated against closed-form approximate expressions.

7.1.1 Aircraft Parameters

The geometric, mass/inertia, and aerodynamic input parameters used in this analysis are summarised in Tables 7.1–7.2. The configuration is a conventional fixed-wing UAV with a rectangular wing, a conventional empennage, and a tractor propulsion system.

Table 7.1: UAV geometry and mass/inertia data.

Parameter	**Symbol**	**Value**	**Parameter**	**Symbol**	**Value**
Wing ref. area	S_w	$0.53\,\mathrm{m}^2$	Aircraft mass	m	$6.30\,\mathrm{kg}$
Wing span	b_w	$2.00\,\mathrm{m}$	Roll inertia	I_{xx}	$0.75\,\mathrm{kg\,m}^2$
Wing root chord	c_r	$0.27\,\mathrm{m}$	Pitch inertia	I_{yy}	$0.38\,\mathrm{kg\,m}^2$
Wing tip chord	c_t	$0.27\,\mathrm{m}$	Yaw inertia	I_{zz}	$1.05\,\mathrm{kg\,m}^2$
Wing taper ratio	λ	1.00	CG from nose	X_{CG}	$0.26\,\mathrm{m}$
Wing sweep ($c/4$)	$\Lambda_{c/4}$	$0.00°$	Wing incidence	i_w	$3.00°$
Wing dihedral	Γ	$0.00°$	HT incidence	i_t	$-0.42°$
HT planform area	S_{HT}	$0.13\,\mathrm{m}^2$	HT aspect ratio	AR_{HT}	4.5
HT AC from nose	X_{AC}^{HT}	$0.96\,\mathrm{m}$	VT planform area	S_{VT}	$0.05\,\mathrm{m}^2$
VT aspect ratio	AR_{VT}	1.5	VT AC from nose	X_{AC}^{VT}	$0.96\,\mathrm{m}$

Table 7.2: Aerodynamic and flight condition parameters.

Parameter	**Symbol**	**Value**	**Parameter**	**Symbol**	**Value**
Air density	ρ	$1.22\,\mathrm{kg/m}^3$	Speed of sound	a	$340.00\,\mathrm{m/s}$
DPR at tail	η'	0.99	Zero-lift AoA (wing)	$\alpha_{L=0}$	$-3.50°$
Oswald efficiency	e	1.00	Zero-lift drag	C_{D0}	0.02873
Airfoil c_{l_α} (wing)	—	$3.61\,\mathrm{rad}^{-1}$	Airfoil c_{l_α} (HT/VT)	—	$6.24\,\mathrm{rad}^{-1}$
Pitching moment at AC	$C_{m,ac}^{w}$	-0.030	Static margin	SM	14%
Aileron area	S_{δ_a}	$0.04\,\mathrm{m}^2$	Aileron span	l_{δ_a}	$0.50\,\mathrm{m}$
Elevator area	S_{δ_e}	$0.04\,\mathrm{m}^2$	Rudder area	S_{δ_r}	$0.01\,\mathrm{m}^2$

7.1.2 Derived Geometry and Aerodynamic Coefficients

7.1.2.1 Geometric Parameters

From the input data the following geometric quantities are derived. The mean aerodynamic chord is computed using:

$$\bar{c} = \frac{2}{3}\, c_r\, \frac{1+\lambda+\lambda^2}{1+\lambda} = 0.27\,\mathrm{m} \tag{7.1}$$

The remaining derived quantities are listed in Table 7.3.

Table 7.3: Derived geometric parameters.

Parameter	**Symbol**	**Value**	**Unit**
Mean aerodynamic chord	$\bar{c}$	0.2652	m
Wing aspect ratio	AR_w	7.5372	—
HT span	b_{HT}	0.773	m
VT span	b_{VT}	0.268	m
VT AC height above FRL	z_{VT}	0.119	m
HT moment arm	l_{HT}	0.700	m
VT moment arm	l_{VT}	0.700	m
HT volume ratio	$\bar{V}_{HT}$	0.6601	—
VT volume ratio	$\bar{V}_{VT}$	0.0315	—
Non-dim. CG location	h_{CG}	0.996	—
Induced-drag factor	K	0.0422	—

7.1.2.2 Lift-Curve Slopes

Finite-wing lift-curve slopes are obtained from the low-subsonic simplified form:

$$C_{L_\alpha} = \frac{c_{l_\alpha}}{1 + \dfrac{c_{l_\alpha}}{\pi\, AR}} \tag{7.2}$$

Table 7.4: Finite-wing lift-curve slopes.

Surface	**Symbol**	**Value**	**Unit**
Wing	$C_{L_\alpha,w}$	3.133	rad^{-1}
Horizontal tail	$C_{L_\alpha,HT}$	4.329	rad^{-1}
Vertical tail	$C_{L_\alpha,VT}$	2.685	rad^{-1}

The wing lift coefficient at zero angle of attack and the downwash gradient are:

$$C_{L0,w} = C_{L_\alpha,w}\ |\alpha_{L=0}| = 0.1914, \qquad \frac{d\varepsilon}{d\alpha} = \frac{2\, C_{L_\alpha,w}}{\pi\, AR_w} = 0.2646, \qquad \varepsilon_0 = 0.0162 \tag{7.3}$$

7.1.3 Longitudinal Trim Analysis

7.1.3.1 Neutral Point and Aerodynamic Centre

Given an assumed static margin $SM = 14\%\,\bar{c}$, the non-dimensional neutral point and wing-body aerodynamic centre locations are:

$$h_{NP} = h_{CG} + SM = 1.136, \quad X_{NP} = h_{NP}\,\bar{c} = 0.30\,\mathrm{m} \tag{7.4}$$

$$h_{AC}^{wb} = h_{NP} - \bar{V}_{HT}\,\frac{C_{L_\alpha,HT}}{C_{L_\alpha,w}}\left(1 - \frac{d\varepsilon}{d\alpha}\right) = 0.465, \quad X_{AC}^{wb} = 0.12\,\mathrm{m} \tag{7.5}$$

7.1.3.2 Pitching Moment Derivatives and Trim State

The pitching moment coefficient derivatives about the CG are:

$$C_{m_\alpha,w} = \frac{X_{CG} - X_{AC}^{wb}}{\bar{c}} C_{L_\alpha,w} = +1.6629 \text{ rad}^{-1} \tag{7.6}$$

$$C_{m_\alpha,HT} = -\eta' \bar{V}_{HT} C_{L_\alpha,HT} \left(1 - \frac{d\varepsilon}{d\alpha}\right) = -2.0805 \text{ rad}^{-1} \tag{7.7}$$

$$C_{m_\alpha} = C_{m_\alpha,w} + C_{m_\alpha,HT} = -\mathbf{0.4176} \text{ rad}^{-1} \tag{7.8}$$

The negative value of C_{m_α} confirms **positive longitudinal static stability**. The zero-lift pitching moment coefficient is:

$$C_{m0} = C_{m,ac}^{w} + \eta' \bar{V}_{HT} C_{L_\alpha,HT} \varepsilon_0 = 0.0157 \tag{7.9}$$

The complete trim state (for zero elevator deflection) is summarised in Table 7.5.

Table 7.5: Longitudinal trim state of the UAV.

Quantity	**Symbol**	**Value**	**Unit**
Trim angle of attack	α^*	0.0377 (2.16°)	rad (deg)
Whole-a/c C_{L_α}	C_{L_α}	3.9210	rad^{-1}
Whole-a/c C_{L0}	C_{L0}	0.1661	—
Trim lift coefficient	C_L^*	0.3139	—
Trim speed	V^*	24.658	m/s
Trim Mach number	Ma^*	0.0725	—
Trim dynamic pressure	q^*	370.903	N/m^2
Trim drag coefficient	C_D^*	0.0329	—
Thrust required	T_R	6.47	N
Power required	P_R	159.64 (0.21 hp)	W (hp)

7.1.4 Longitudinal Dynamic Stability

7.1.4.1 Stability Derivatives

The key longitudinal stability derivatives evaluated at the trim condition are presented in Table 7.6.

TABLE 7.6: Longitudinal aerodynamic stability derivatives at trim.

Derivative	**Symbol**	**Value**	**Unit**
Pitch-damping	$C_{m_{q_1}}$	-14.9384	rad^{-1}
Lift due to pitch rate	$C_{L_{q_1}}$	5.6584	rad^{-1}
Drag-AoA derivative	C_{D_α}	0.1040	rad^{-1}
Drag due to pitch rate	$C_{D_{q_1}}$	0.1500	rad^{-1}
Lift–Mach derivative	$C_{L_{Ma}}$	0.0229	—
Norm. pitch stiffness	$\bar{q}^* Sc/I_{yy}$	135.9159	s^{-2}
Norm. lift factor	$\bar{q}^* S/W$	3.1859	—
Frequency factor	g/V^*	0.3978	s^{-1}
Non-dim. pitch rate	$\bar{c}/(2V^*)$	0.00538	s

7.1.4.2 Short-Period Mode

Applying the classical second-order approximation (analogous to Eq. 2.33–2.34 of the reference), the short-period modal parameters are:

$$\omega_{n,SP}^2 = -\frac{\bar{q}^* S\bar{c}}{I_{yy}} C_{m_\alpha} \quad \Rightarrow \quad \omega_{n,SP} = 7.53\,\mathrm{rad/s} \tag{7.10}$$

$$2\zeta_{SP}\,\omega_{n,SP} = -\frac{\bar{q}^* S\bar{c}}{I_{yy}} \cdot \frac{\bar{c}}{2V^*}\big(C_{m_{q_1}} + C_{m_{\dot{\alpha}}}\big) \quad \Rightarrow \quad \zeta_{SP} = 0.7245 \tag{7.11}$$

$$T_{SP} = \frac{2\pi}{\omega_{n,SP}} = 0.83\,\mathrm{s} \tag{7.12}$$

The damping ratio $\zeta_{SP} = 0.72$ falls within the MIL-SPEC Level 1 flying-quality corridor ($0.35 \leq \zeta_{SP} \leq 1.30$), indicating satisfactory short-period handling characteristics.

7.1.4.3 Phugoid Mode

The phugoid mode parameters from the approximate relations (Eqs. 5.35–5.36) are:

$$\omega_{n,Ph} = 0.56\,\text{rad/s}, \qquad \zeta_{Ph} = 0.0740, \qquad T_{Ph} = 11.15\,\text{s} \tag{7.13}$$

The positive but small phugoid damping ratio ($\zeta_{Ph} = 0.074$) indicates a **lightly damped but stable** phugoid mode, which is typical for UAVs of this class and consistent with Level 2 flying qualities.

7.1.4.4 Longitudinal State-Space Model

Following the first-order formulation (Eq. 9.7 of the reference), the longitudinal state vector is $\mathbf{x}_{long} = [\Delta V/V^*,\ \Delta\gamma,\ \Delta\alpha,\ \dot{\Delta\alpha}]^\top$ and the linearised system matrix evaluated at trim is:

$$\mathbf{A}_{long} = \begin{bmatrix} -0.0834 & -0.3978 & -0.1318 & -0.0010 \\ 0.7978 & 0.0000 & 4.9699 & 0.0386 \\ 0.0000 & 0.0000 & 0.0000 & 1.0000 \\ 0.0640 & 0.3056 & -54.5499 & -15.2954 \end{bmatrix} \tag{7.14}$$

The eigenvalues of $\mathbf{A}_{long}$ are:

$$\lambda_{Ph} = -0.02781 \pm j\,0.56108 \quad \rightarrow \quad \omega_n = 0.56\,\text{rad/s},\ \zeta = 0.0495 \tag{7.15}$$

$$\lambda_{SP,1} = -9.595 \quad \text{(real)} \tag{7.16}$$

$$\lambda_{SP,2} = -5.729 \quad \text{(real)} \tag{7.17}$$

The eigenvalue analysis confirms all longitudinal modes are **stable** (negative real parts). The phugoid natural frequency from the eigenvalue ($\omega_n = 0.56\,\text{rad/s}$) agrees closely with the approximate value ($0.56\,\text{rad/s}$), validating the analytical approach.

7.1.5 Lateral-Directional Dynamic Stability

7.1.5.1 Aerodynamic Coefficients

The lateral-directional aerodynamic coefficients are estimated following the procedure of Section 9.1.3 of the reference. The sidewash factor computed at trim is:

$$\left(1 + \frac{d\sigma}{d\beta}\right)\eta'_{VT} = 1.0005 \tag{7.18}$$

All lateral-directional stability derivatives are presented in Tables 7.7 and 7.8.

Table 7.7: Lateral-directional static stability derivatives.

Derivative	Symbol	Value	Unit
Side-force due to sideslip	C_{Y_β}	-0.2395	rad^{-1}
Yawing moment due to sideslip	C_{n_β}	$+0.0838$	rad^{-1}
Rolling moment due to sideslip	C_{l_β}	-0.0111	rad^{-1}

The positive $C_{n_\beta} = 0.0838$ rad^{-1} confirms **positive directional static stability** (weathercock stability). The negative $C_{l_\beta} = -0.0111$ rad^{-1} confirms **positive effective dihedral** (roll stability due to sideslip).

Table 7.8: Lateral-directional rate derivatives.

Derivative	Symbol	Value	Unit
Side-force due to roll rate	$C_{Y_{p_2}}$	+0.0063	rad^{-1}
Roll damping	$C_{l_{p_2}}$	−0.6587	rad^{-1}
Yawing moment due to roll rate	$C_{n_{p_2}}$	−0.0372	rad^{-1}
Side-force due to yaw rate	$C_{Y_{r_1}}$	+0.1686	rad^{-1}
Rolling moment due to yaw rate	$C_{l_{r_1}}$	+0.1124	rad^{-1}
Yaw damping (tail term)	$C_{n_{r_1}}$	−0.0594	rad^{-1}
Yaw damping (wing term)	$C_{n_{r_2}}$	−0.0110	rad^{-1}
Roll due to yaw rate (wing)	$C_{l_{r_2}}$	+0.1046	rad^{-1}

7.1.5.2 Dimensional Stability Parameters

The dimensional stability parameters used in the modal approximations are:

$$N_\beta = \frac{\bar{q}^* S b}{I_{zz}} C_{n_\beta} = 31.58\,\text{s}^{-2}, \quad Y_\beta = \frac{\bar{q}^* S}{m} C_{Y_\beta} = -7.49\,\text{m/s}^2 \tag{7.19}$$

$$L_\beta = \frac{\bar{q}^* S b}{I_{xx}} C_{l_\beta} = -5.81\,\text{s}^{-2}, \quad L_{p_2} = \frac{\bar{q}^* S b}{I_{xx}} C_{l_{p_2}} \frac{b}{2V^*} = -14.00\,\text{s}^{-2} \tag{7.20}$$

$$N_{r_1} = \frac{\bar{q}^* S b}{I_{zz}} C_{n_{r_1}} \frac{b}{2V^*} = -0.91\,\text{s}^{-2}, \quad L_{r_1} = \frac{\bar{q}^* S b}{I_{xx}} C_{l_{r_1}} \frac{b}{2V^*} = 2.39\,\text{s}^{-2} \tag{7.21}$$

7.1.5.3 Lateral-Directional Dynamic Modes

Roll Mode. The roll mode is a first-order convergence characterised approximately by:

$$\lambda_r \approx L_{p_2} = -14.00\,\text{s}^{-1} \tag{7.22}$$

The large negative real eigenvalue indicates a **well-damped, fast roll subsidence** mode with time constant $\tau_r = 1/|\lambda_r| = 0.07\,\text{s}$, well within Level 1 flying-quality limits.

Dutch Roll Mode. Applying the approximate coupled lateral-directional formulae:

$$\omega_{n,DR} = 5.68\,\text{rad/s}, \qquad \zeta_{DR} = 0.348, \qquad T_{DR} = 1.11\,\text{s} \tag{7.23}$$

The Dutch roll mode is **stable and adequately damped**. The product $\zeta_{DR}\,\omega_{n,DR} =$ 1.977 rad/s satisfies typical UAV Level 1 flying quality requirements.

Spiral Mode.

$$\lambda_s = +0.06103\ \text{s}^{-1} \quad (\textbf{UNSTABLE}) \tag{7.24}$$

The spiral mode is mildly unstable, as is common in many UAV and GA configurations. The positive eigenvalue implies a slow divergence with time-to-double $T_2 = \ln 2/\lambda_s \approx$ 11.40 s, which is well within the acceptable threshold (typically $T_2 > 20.00$ s for Level 1) and is easily suppressed by a roll-angle stability augmentation system or pilot-in-the-loop correction.

7.1.5.4 Lateral-Directional State-Space Model

Following the first-order formulation (Eq. 9.11 of the reference), with state vector $\mathbf{x}_{lat} = [\Delta\mu,\ \dot{\Delta\mu},\ \Delta\beta,\ \dot{\Delta\beta}]^\top$, the system matrix at trim is:

$$\mathbf{A}_{lat\text{-}dir} = \begin{bmatrix} 0.0000 & 1.0000 & 0.0000 & 0.0000 \\ 0.8849 & -14.0030 & -12.4353 & -2.3901 \\ 0.0000 & 0.0000 & 0.0000 & 1.0000 \\ 0.0666 & 0.3978 & -32.0821 & -3.8852 \end{bmatrix} \tag{7.25}$$

The eigenvalues of $\mathbf{A}_{lat\text{-}dir}$ are:

$$\lambda_{roll} = -14.0185 \quad \text{(roll subsidence)} \tag{7.26}$$

$$\lambda_{spiral} = +0.0604 \quad \text{(spiral, unstable)} \tag{7.27}$$

$$\lambda_{DR} = -1.9650 \pm j\,5.3556 \quad \rightarrow \quad \omega_n = 5.70\,\text{rad/s},\ \zeta = 0.345 \tag{7.28}$$

The eigenvalue results are in excellent agreement with the approximate analytical values, confirming the correctness of the state-space formulation.

7.1.6 Control Effectiveness and Derivatives

Control surface effectiveness parameters are extracted from standard charts (Fig. 7.9 of the reference). The resulting control derivatives are presented in Table 7.9.

Parameter	**Symbol**	**Value**	**Parameter**	**Symbol**	**Value**
Elevator effectiveness	τ_{δ_e}	0.50	Rudder effectiveness	τ_{δ_r}	0.50
Aileron effectiveness	τ_{δ_a}	0.22			
C_{L,δ_e}	—	$+0.5358$ rad^{-1}	C_{m,δ_e}	—	-1.4146 rad^{-1}
C_{l,δ_a}	—	-0.4440 rad^{-1}	C_{n,δ_a}	—	$+0.0128$ rad^{-1}
C_{n,δ_r}	—	-0.0423 rad^{-1}	C_{Y,δ_r}	—	$+0.1209$ rad^{-1}
C_{l,δ_r}	—	$+0.0056$ rad^{-1}			

Table 7.9: Control surface effectiveness and control derivatives at trim.

The negative $C_{m,\delta_e} = -1.4146\,\text{rad}^{-1}$ confirms that a positive (trailing-edge-down) elevator deflection produces a nose-down pitching moment as expected for a conventional tail configuration. The adverse yaw derivative $C_{n,\delta_a} = +0.0128\,\text{rad}^{-1}$ is small relative to C_{n_β}, indicating manageable aileron-induced yaw coupling.

7.1.7 Summary of Results

A consolidated summary of all key stability and performance results is provided in Table 7.10.

TABLE 7.10: Summary of UAV flight dynamics and stability analysis results.

Quantity	**Symbol**	**Value**	**Unit**
Trim Condition			
Trim angle of attack	α^*	2.159	deg
Trim speed	V^*	24.658	m/s
Trim lift coefficient	C_L^*	0.3139	—
Trim drag coefficient	C_D^*	0.0329	—
Thrust required	T_R	6.474	N
Power required	P_R	0.214	hp
Longitudinal Modes			
Short-period nat. frequency (approx.)	$\omega_{n,SP}$	7.534	rad/s
Short-period damping ratio	ζ_{SP}	0.7245	—
Phugoid nat. frequency	$\omega_{n,Ph}$	0.563	rad/s
Phugoid damping ratio	ζ_{Ph}	0.0740	—
Lateral-Directional Modes			
Roll mode eigenvalue	λ_r	-14.003	s^{-1}
Dutch roll nat. frequency	$\omega_{n,DR}$	5.679	rad/s
Dutch roll damping ratio	ζ_{DR}	0.3481	—
Spiral mode eigenvalue	λ_s	$+0.0610$	s^{-1}
Spiral mode stability	—	**Unstable** (mild)	—

7.1.8 Eigenvalue Analysis – Complex Plane Plots

Figure 7.1 shows the eigenvalues of the longitudinal and lateral-directional system matrices plotted in the complex (σ–$j\omega$) plane. All eigenvalues in the left-half plane (LHP) correspond to stable modes. The mildly unstable spiral mode appears as a small positive real eigenvalue in the lateral-directional plot.

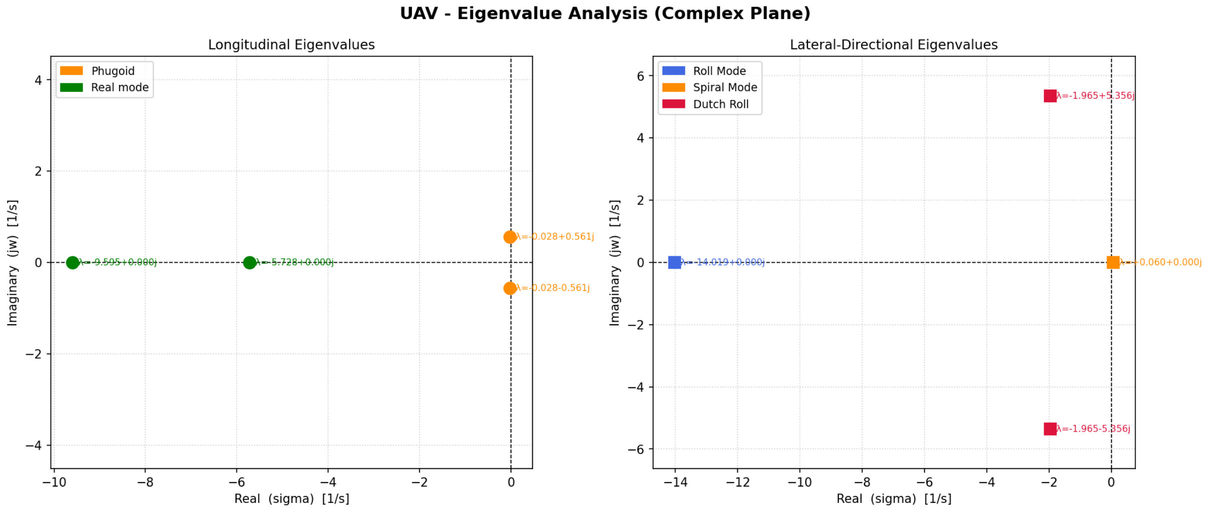


FIGURE 7.1: Eigenvalue plots in the complex plane for the UAV. *Left:* Longitudinal modes — phugoid (orange) and short-period (blue) conjugate pairs plus two real short-period roots. *Right:* Lateral-directional modes — Dutch roll conjugate pair (crimson), roll subsidence (blue) and spiral mode (orange).

7.1.9 Stability Assessment

Table 7.11 provides a qualitative assessment of each mode against standard flying-quality criteria.

TABLE 7.11: Stability mode assessment summary.

Mode	Key Metric	Value	Assessment	Level
Short period	$\zeta_{SP} = 0.72$, $\omega_n = 7.53$ rad/s	Within MIL corridor	Stable, well-damped	1
Phugoid	$\zeta_{Ph} = 0.074 > 0$	$\zeta > 0.04$ (Level 1 min)	Stable, lightly damped	1–2
Roll subsidence	$\tau_r = 0.071$ s $\ll$ limit	Fast convergence	Stable, well-damped	1
Dutch roll	$\zeta_{DR} = 0.348$, $\omega_n = 5.68$ rad/s	$\zeta_{DR}\omega_n = 1.98$	Stable, adequate damping	1
Spiral	$T_2 \approx 11.4$ s	$T_2 > 8$ s (Level 2)	Mildly unstable, slow	2

Overall, the UAV exhibits satisfactory flying qualities across all axes. The only concern is the mildly unstable spiral mode; however, given its slow time-to-double of approximately 11.40 s, this divergence is well within pilot-correctable limits and does not pose a safety risk during normal autonomous or manually-assisted flight. Implementation of a simple roll-angle hold loop in the autopilot would eliminate this instability entirely.

7.2 Conclusion

The control surface sizing was performed using aerodynamic stability principles and conservative flight conditions. Worst-case scenarios such as forward CG trim, asymmetric thrust at minimum control speed, and low-speed roll control were considered to ensure sufficient control authority throughout the flight envelope. These conditions ensure that the UAV remains controllable even under the most demanding operating situations.

7.3 Summary

This chapter presented a comprehensive small-perturbation flight dynamics analysis of the UAV at its trim condition ($V^* = 24.658$ m/s, $\alpha^* = 2.16°$). Longitudinal trim was confirmed with a static margin of 14%, requiring a thrust of 6.47 N and a power of 0.21 hp. Modal analysis of the longitudinal dynamics showed that both the short-period mode ($\zeta_{SP} = 0.72$, $\omega_{n,SP} = 7.53$ rad/s, Level 1) and the phugoid mode ($\zeta_{Ph} = 0.074$, $\omega_{n,Ph} = 0.56$ rad/s, Level 2) are stable, with eigenvalue results validating the analytical approximations. For lateral-directional dynamics, the roll subsidence ($\tau_r = 0.071$ s) and Dutch roll ($\zeta_{DR} = 0.348$, $\omega_{n,DR} = 5.68$ rad/s) modes are both stable and meet Level 1 flying quality requirements. The spiral mode is mildly unstable ($\lambda_s = +0.061$ s^{-1}, $T_2 \approx 11.4$ s), a common characteristic in this class of UAV, and is easily corrected by a roll-angle autopilot hold. Overall, the UAV satisfies all primary stability and handling quality criteria across its flight envelope.

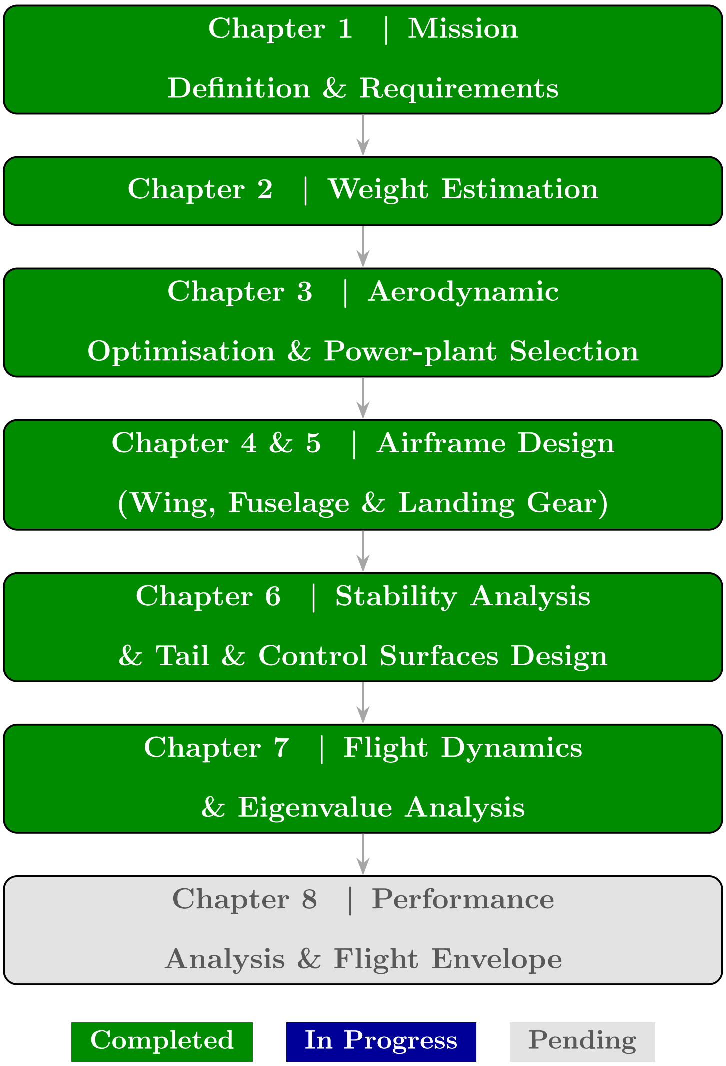


FIGURE 7.2: Overall UAV Conceptual Design Workflow — Progress after Chapter 7

Chapter 8

UAV Performance

8.1 Drag Polar

Parasitic Drag Estimation

The total drag on an aircraft is the resultant of both induced drag due to lift and the skin friction drag which is also known as parasitic drag. Different components of the aircraft such as wing, fuselage and tail etc. contributes to the parasitic drag. The non-dimension coefficient corresponding to this parasitic drag for subsonic flights is given in terms of the sum of individual component contributions as shown below:

$$(C_{Do})_{aircraft} = \frac{\sum(C_{f_c} FF_c Q_c S_{wet_c})}{S_{ref}} \tag{8.1}$$

Where the individual parameters are defined as follows:

- C_{f_c} is the Skin Friction Coefficient corresponding to a certain component.
- FF_c is the component Form Factor that estimates the drag due to viscous separation.
- Q_c is the factor which corresponds to the Interference Effects on the component drag.

- S_{wet_c} is the Wetted Area of the specified component.
- S_{ref} is the wing planform area.

Skin Friction Coefficient (C_f)

Due to the propeller wash and blunt leading edges of the configuration, a fully turbulent boundary layer is assumed across the airframe. The skin friction coefficient for low subsonic turbulent flight is obtained from the below expression:

$$C_f = \frac{0.455}{(\log_{10} Re)^{2.58}} \tag{8.2}$$

Here $Re = \frac{\rho V l}{\mu}$ is the Reynolds's number and the term l is the characteristic length of the specified component.

Form Factor (FF) Modifications for Low-Speed UAV

The component form factor varies for different types of components. Standard empirical formulations for the form factor of lifting surfaces (Wing and Tail) typically include a Mach compressibility multiplier, $1.34M^{0.18}(\cos\Lambda_m)^{0.28}$. However, for this specific UAV operating at a low cruise velocity ($V = 18.8$ m/s, $M \approx 0.053$), the flow is entirely incompressible. Applying the standard compressibility multiplier at this speed mathematically reduces the form factor below 1.0, which physically implies less form drag than a two-dimensional flat plate.

To maintain physical accuracy for a low-speed UAV, the Mach compressibility term is omitted. The corrected baseline form factor for the lifting surfaces is given by:

$$FF_{base} = \left[1 + \frac{0.6}{(x/c)_m}\left(\frac{t}{c}\right) + 100\left(\frac{t}{c}\right)^4\right](\cos\Lambda_m)^{0.28} \tag{8.3}$$

Where (t/c) is the airfoil maximum thickness to chord ratio and $(x/c)_m$ is the chord wise location of the airfoil maximum thickness point. Λ_m is the sweep of the maximum-thickness line.

Fuselage Form Factor and Square-Sided Penalty

Standard formulations treat the fuselage as a slender, conical, or cylindrical body, relying on the maximum fuselage cross-sectional area (A_{max}) and length to calculate the form factor.

Since the central fuselage of this UAV is designed with a thick NACA 0025 cross-section, it behaves aerodynamically as a lifting body rather than a standard cylindrical fuselage. Therefore, the standard fuselage calculation is replaced with the modified lifting-surface equation (FF_{base}) described above, utilizing the $t/c = 0.25$ thickness ratio of the NACA 0025 profile.

Furthermore, the fuselage features a rectangular cross-section with flat sides. A square-sided fuselage has a form factor increment about 30-40% of pressure increment over skin friction drag, due to separation caused by corners. To account for the sharp corners of the 300mm wide center body, a square-sided penalty ($P_{sq} = 0.2$) is applied to the isolated base form factor:

$$FF_{fuselage} = FF_{base} + (FF_{base} - 1)P_{sq} \tag{8.4}$$

This ensures the increased boundary layer separation from the flat side panels is accurately captured in the overall $(C_{Do})_{aircraft}$ summation without artificially decreasing the drag at low velocities.

For a high-wing, mid-wing, or well-filleted low-wing configuration, the interference factor Q is negligible and can be taken as 1.0. The fuselage similarly carries a negligible interference factor ($Q = 1.0$) in most cases, while a conventional tail assumes $Q = 1.04$–1.05

The atmospheric properties at the 200 m altitude are taken as $\rho = 1.22\,\mathrm{kg/m^3}$, a cruising speed of $V = 18.80\,\mathrm{m/s}$, and a dynamic viscosity of $\mu = 1.626 \times 10^{-5}$ $\mathrm{N\,s/m^2}$. The parameters required to estimate the parasitic drag coefficient for each component are tabulated in Table 8.1.

Parameter	**Wing**	**Horizontal Tail**	**Vertical Tail**	**Fuselage**
ℓ	0.27 m	0.15 m	0.18 m	0.80 m
(t/c)	0.105	0.12	0.12	0.25
$(x/c)_m$	0.318	0.3	0.3	0.3
Λ_m	0.0	0.0	0.0	0.0

TABLE 8.1: Required Parameters for C_{D_0} Estimation

The component-level parasitic drag breakdown computed from the flat-plate skin-friction coefficient C_f, the form factor FF, and the interference factor Q is summarised in Table 8.2.

Component	C_f	**Form Factor** (FF)	**Interference** (Q)	S_{wet} $(\mathrm{m^2})$	$C_f \cdot FF \cdot Q \cdot S_{\text{wet}}$
Wing	0.00541	1.210	1.00	0.93	0.00609
Horizontal Tail	0.00605	1.261	1.00	0.22	0.00168
Vertical Tail	0.00587	1.261	1.05	0.09	0.00070
Fuselage	0.00437	2.069	1.00	0.75	0.00678
Total Aircraft C_{D_0}					**0.02873**

TABLE 8.2: Component Parasitic Drag Breakdown

Therefore, the corresponding aircraft parasitic drag coefficient obtained from the above equations is

$$(C_{D_0})_{\text{aircraft}} = \mathbf{0.02873}. \tag{8.5}$$

The corresponding maximum lift-to-drag ratio is

$$\left(\frac{L}{D}\right)_{\max} = \frac{1}{2}\sqrt{\frac{\pi AR\, e}{C_{D_0}}} = 12.55, \tag{8.6}$$

where AR is the wing aspect ratio and e is the Oswald efficiency factor.

The drag polar for the aircraft, relating the lift coefficient C_L to the total drag coefficient C_D, is presented in Figure 8.1.

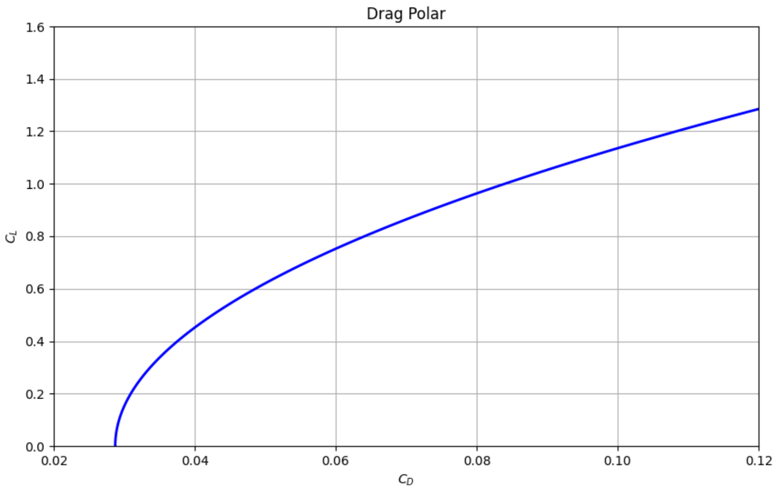


FIGURE 8.1: Drag polar of the aircraft.

8.2 Performance Analysis

8.2.1 Maximum Range and Endurance

From [54], for maximum range (R) and endurance (E), the operating conditions are:

- Maximum range: $(C_L/C_D)_{\max}$, i.e., $C_{D_0} = C_{D_i}$
- Maximum endurance: $(C_L^{3/2}/C_D)_{\max}$, i.e., $3C_{D_0} = C_{D_i}$

The corresponding velocities are given by:

$$V_{R_{\max}} = \left(\frac{2W}{\rho S_{\text{ref}}} \sqrt{\frac{K}{C_{D_0}}} \right)^{1/2} \tag{8.7}$$

$$V_{E_{\max}} = \left(\frac{2W}{\rho S_{\text{ref}}} \sqrt{\frac{K}{3C_{D_0}}} \right)^{1/2} \tag{8.8}$$

The corresponding power required is:

$$P_{R_{\max}} = \rho S_{\text{ref}} C_{D_0} V_{R_{\max}}^3 \tag{8.9}$$

$$P_{E_{\max}} = 2\rho S_{\text{ref}} C_{D_0} V_{E_{\max}}^3 \tag{8.10}$$

For a battery-powered aircraft, the maximum range and endurance are given by:

$$R_{\max} = \eta_p\, \eta_m\, \eta_b\, (\text{Energy}_{\text{Bat}}) \left(\frac{V_{R_{\max}}}{P_{R_{\max}}} \right) \tag{8.11}$$

$$E_{\max} = \frac{\eta_p\, \eta_m\, \eta_b\, (\text{Energy}_{\text{Bat}})}{P_{E_{\max}}} \tag{8.12}$$

where the battery energy capacity is:

$$\text{Energy}_{\text{Bat}} = 959040 \text{ J}$$

The efficiencies are:

$$\eta_p = 0.7, \quad \eta_m = 0.9, \quad \eta_b = 0.9$$

The resulting maximum range and endurance are:

$$R_{\max} = 83.15 \text{ km} \tag{8.13}$$

$$E_{\max} = 2 \text{ hr } 5 \text{ min} \tag{8.14}$$

Thus the designed aircraft is able to achieve it's design parameter in range.

8.2.2 Power, Rate of Climb

Using the value of C_{D_0} obtained, the variation of key performance parameters of power required and rate of climb are evaluated.

8.2.2.1 Power Required

During steady level flight, the power required can be expressed as a function of velocity as:

$$P_{\text{req}} = \frac{1}{2}\rho V^3 S_{\text{ref}} \left(C_{D_0} + \frac{1}{\pi A R e} \left(\frac{2W}{\rho V^2 S_{\text{ref}}} \right)^2 \right) \tag{8.15}$$

The variation of power required with velocity is shown in Fig. 8.2.

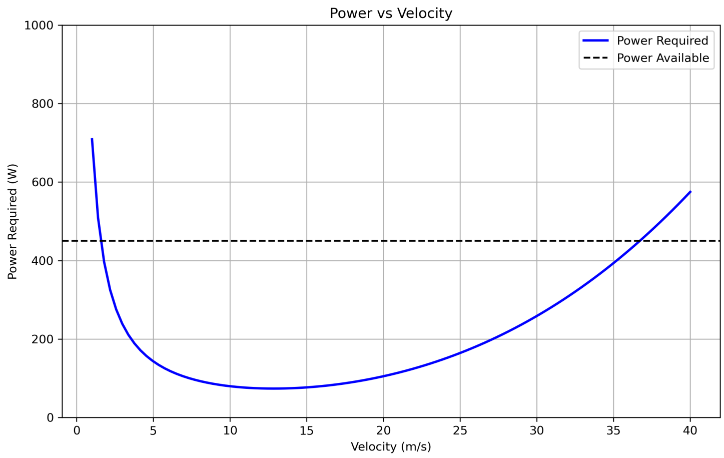


FIGURE 8.2: Power Required vs Velocity

8.2.2.2 Rate of Climb

The rate of climb (ROC) at a given altitude is determined from the excess power available:

$$ROC = \frac{P_{\text{avl}} - P_{\text{req}}}{W} \tag{8.16}$$

The variation of rate of climb with velocity is shown in Fig. 8.3.

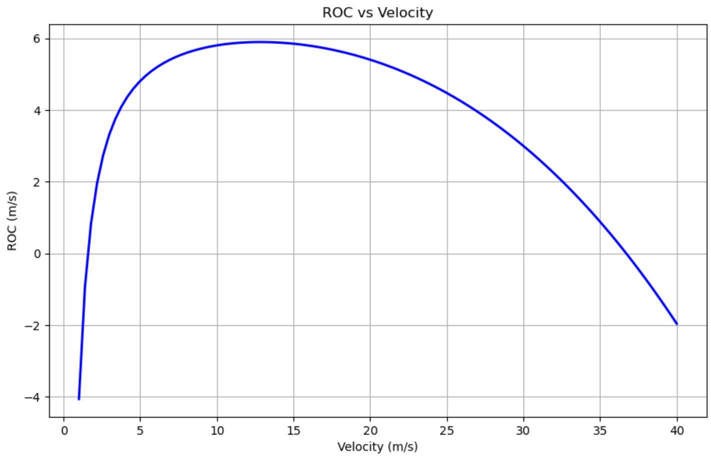


FIGURE 8.3: Rate of Climb vs Velocity

8.2.3 Power, Thrust and Velocity

The updated performance parameters corresponding to the new parasitic drag coefficient and $(L/D)_{max}$ are given in Table 8.3

Segment	Velocity (m/s)	Thrust (N)	Power (W)
Climb-1	22.67435894	11.40205147	258.5342077
Cruise-1	18.78081916	5.080876494	95.42302259
Loiter-1	22.71762403	5.866890823	133.28182
Climb-2	20.47463932	9.297057351	190.353896
Cruise-2	16.9588065	4.142868526	70.25810569
Loiter-2	20.51370695	4.783772517	98.13290753

TABLE 8.3: Recomputed Optimum speeds according to the updated parameters

8.3 Flight Envelope

A flight envelope rules out the unsafe operating limits of the UAV in the context of different speeds and load factor.It defines the combinations of velocity corresponding to different aerodynamic loads within which the aircraft can operate without structural damage or loss of control. The velocity is plotted on the horizontal axis, and the load factor is plotted on the vertical axis.The envelope is constrained by the stall speed, the maneuver speed, the maximum structural load factor, and the maximum design speed

8.3.1 Characteristic Speeds

8.3.1.1 Stall Speed

The stall speed is the minimum speed at which the aircraft can maintain steady level flight at a load factor of unity. It occurs when the wing reaches its maximum lift coefficient.

$$V_s = \sqrt{\frac{2W}{\rho S C_{L_{\max}}}} \tag{8.17}$$

8.3.1.2 Maximum Speed

The maximum speed represents the highest speed that the aircraft can achieve during normal operating conditions. It is generally limited by propulsion capability and aerodynamic drag.

8.3.1.3 Design Dive Speed

The design dive speed represents the maximum allowable structural speed of the aircraft and taken as maximum speed the powerplant produces. Provides a safety margin above the maximum operating speed and is given by

$$V_d = 1.25 V_{max} = 30 m/s \tag{8.18}$$

where V_{max} = 24 m/s

8.3.1.4 Maneuvering Speed

The maneuvering speed is the maximum speed at which full control surface deflection can be applied without exceeding the structural load factor limit.

$$V_a = V_s \sqrt{n_{max}} \tag{8.19}$$

where n_{max} is the maximum positive load factor.

8.3.1.5 Negative Stall Speed

The negative stall speed represents the speed corresponding to the negative load factor limit and is associated with the maximum negative lift coefficient.

$$V_g = \sqrt{\frac{2|n_{min}|W}{\rho S|C_{L_{\mathrm{min}}}|}} \tag{8.20}$$

where n_{min} is the maximum negative load factor and $C_{L_{\mathrm{min}}}$ is the minimum lift coefficient.

8.3.1.6 Load Factor Relation

The load factor corresponding to a given flight speed is expressed as

$$n = \frac{\rho V^2 S C_L}{2W} \tag{8.21}$$

This relation defines the stall boundaries of the V–n diagram and determines the safe operating limits of the aircraft within the flight envelope.

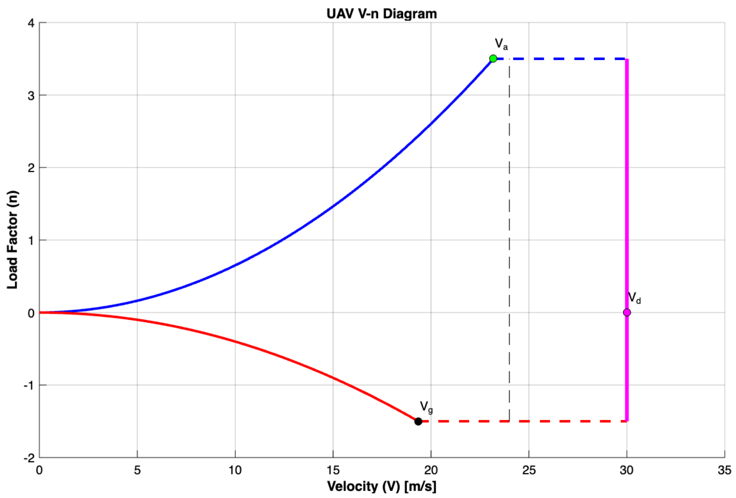


FIGURE 8.4: Flight Envelope before Payload drop

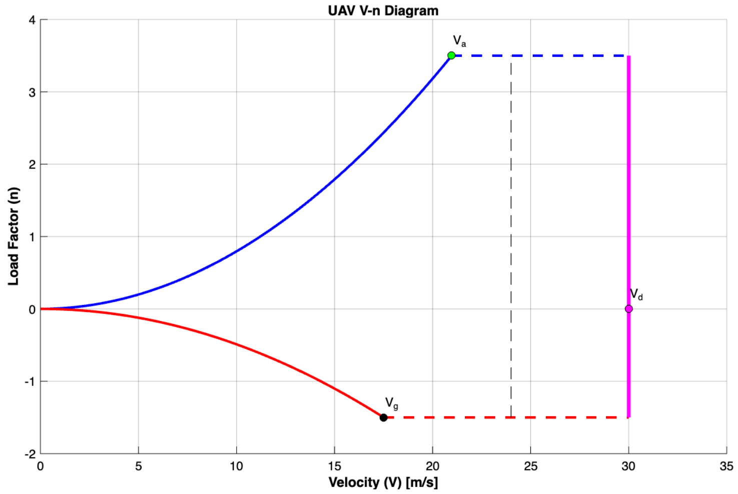


FIGURE 8.5: Flight Envelope after Payload drop

8.4 Performance Comparison

TABLE 8.4: Performance Comparison: Present Design vs. Similar Fixed-Wing UAVs

UAV	MTOW (kg)	Payload (kg)	Span (m)	Cruise ($\mathrm{m\,s^{-1}}$)	Range (km)	Endurance (hr)
Black Swift S2	9.50	2.30	3.00	18.0	110	2.00
Albatross UAV	10.00	4.40	3.00	18.9	250	4.00
Striver	7.00	1.20	2.10	17.0	200	2.40
Believer	5.50	0.70	1.96	20.0	20	2.00
Desert Hawk 3	3.72	0.90	1.37	15.5	15	1.50
Apple White	8.00	2.00	3.30	16.7	20	3.00
Milo Aeromapper Talon	3.50	—	2.00	16.1	30	2.00
Present Design	**6.592**	**1.20**	**2.00**	**18–22**	**83.15**	**2.08**

Note: Payload for the present design refers to the droppable medical payload; total on-board payload (including avionics) is 2.10 kg.

8.5 Summary

Table 8.5: Key Design Parameters — Chapter 7: UAV Performance

Parameter	Symbol	Unit	Previous	Current
Drag polar (updated from Chapter 3)				
Zero-Lift Drag Coeff.	C_{D_0}	—	0.0256	**0.02873**
Max. Lift-to-Drag Ratio	$(L/D)_{max}$	—	15.08	**12.55**
Mission performance				
Maximum Range	R_{max}	km	req. 80–100	**83.15**
Maximum Endurance	E_{max}	hr min	req. 1–2 hr	**2 h 05 min**
Optimum speeds (updated parameters)				
Cruise velocity (opt.)	V_{cr}	$\mathrm{m\,s^{-1}}$	18.74	18.78
Climb velocity (opt.)	V_{cl}	$\mathrm{m\,s^{-1}}$	24.53	22.67
Loiter velocity (opt.)	V_{loi}	$\mathrm{m\,s^{-1}}$	18.74	22.72
Characteristic speeds — before payload drop				
Stall Speed	V_s	$\mathrm{m\,s^{-1}}$	—	12.40
Maximum Speed	V_{max}	$\mathrm{m\,s^{-1}}$	—	24.00
Design Dive Speed	V_d	$\mathrm{m\,s^{-1}}$	—	30.00
Maneuvering Speed	V_a	$\mathrm{m\,s^{-1}}$	—	23.19
Negative Stall Speed	V_g	$\mathrm{m\,s^{-1}}$	—	19.35
Characteristic speeds — after payload drop				
Stall Speed	V_s	$\mathrm{m\,s^{-1}}$	—	11.21
Maneuvering Speed	V_a	$\mathrm{m\,s^{-1}}$	—	20.97
Negative Stall Speed	V_g	$\mathrm{m\,s^{-1}}$	—	17.50
Structural limits				
Max. Positive Load Factor	n_{max}	—	—	+3.5
Max. Negative Load Factor	n_{min}	—	—	−1.5

Updated from previous chapter

New parameter this chapter

Mission requirement met

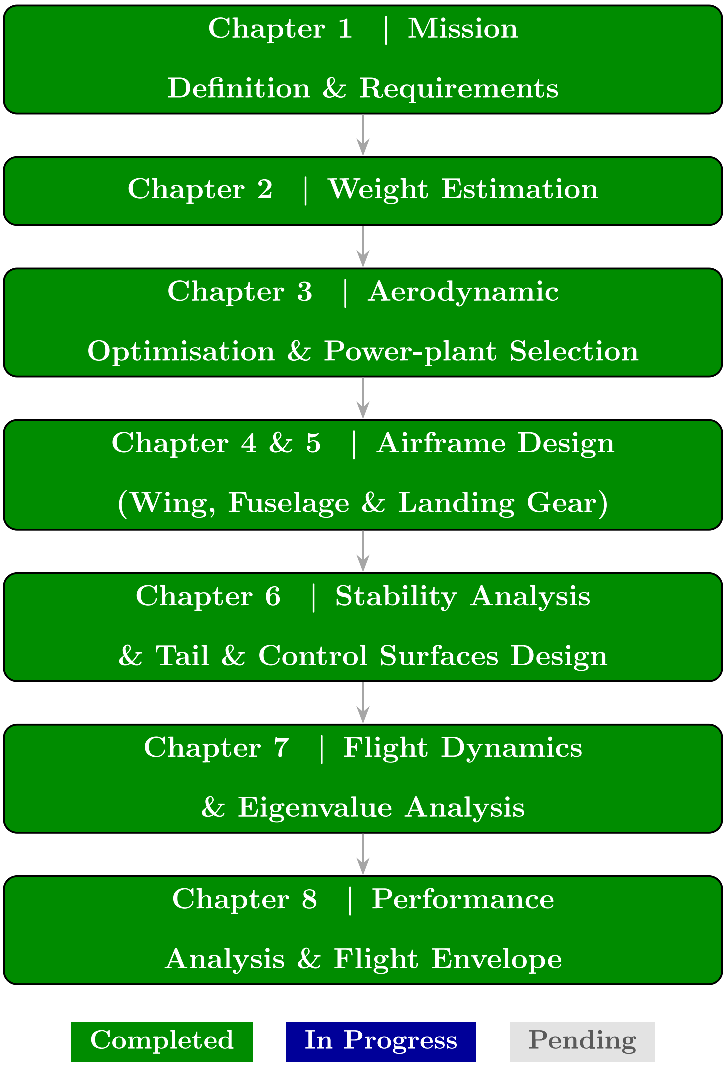


FIGURE 8.6: Overall UAV Conceptual Design Workflow — Progress after Chapter 8

Chapter 9

Conclusion

9.1 Summary of Design

This report has presented the complete conceptual design of a fixed-wing, electrically powered UAV for emergency medical logistics in remote and rural regions of India. The design process followed a structured, iterative methodology progressing from mission analysis and weight estimation through aerodynamic optimisation, airframe sizing, stability verification, and performance analysis. The principal design decisions and their outcomes are documented in the chapter-end parameter tables throughout this report and consolidated below.

9.2 Requirements Compliance

Table 9.1 compares the achieved performance of the present design against the mission specifications established in Chapter 1 (Table 1.1).

Table 9.1: Mission Requirements vs. Achieved Performance

Parameter	Requirement	Achieved	Status
Cruise Speed ($\mathrm{m\,s^{-1}}$)	18 – 25	18 – 22	✓ Met
Take-off Distance (m)	< 150	< 150 (STOL design)	✓ Met
Operational Radius (km)	40 – 50	41.6	✓ Met
Total Range (km)	80 – 100	83.15	✓ Met
Endurance (hr)	1 – 2	2.08	✓ Met
Drop Payload (kg)	1 – 2	1.20	✓ Met
Maximum Wingspan (m)	≤ 2	2.00	✓ Met
Longitudinal Stability	Statically stable	$C_{m_\alpha} = -0.520\,\mathrm{rad}^{-1}$	✓ Met
Static Margin (%)	Positive	≈ 14	✓ Met
Key weight outcomes			
MTOW (kg)	—	6.592	—
Empty Weight (kg)	—	3.205	—
Battery Mass (kg)	—	1.287	—

All primary mission requirements are satisfied by the converged design. The total range of

83.15 km places the UAV comfortably within the 80–100 km envelope, while the endurance of 2 h 5 min exceeds the upper bound of the 1–2 hr requirement, providing reserve capacity for loiter, headwind compensation, and communication link recovery.

9.3 Key Design Trade-offs

Several trade-offs were made during the design process that are worth recording explicitly.

Wing area vs. span constraint. The strict 2 m wingspan limit forced the aspect ratio down from the preliminary value of 9.686 (based on wetted-aspect-ratio regression) to the final value of 7.53, driven by the wing-loading constraint analysis. The resulting increase in induced drag reduced $(L/D)_{max}$ from 15.08 to 12.55, which was partially compensated by careful parasitic drag estimation in Chapter 7.

Battery upgrade and MTOW shift. The second weight iteration necessitated upgrading from a GenX 3S 16 Ah pack to a GenX 6S 12 Ah pack to satisfy the higher power loading imposed by the refined wing area. This increased the total battery energy available from 657.00 kJ to 959.00 kJ, ultimately enabling the range and endurance margins reported above.

Fuselage aerodynamics vs. manufacturability. The initial rectangular fuselage was reshaped to a NACA 0025 symmetric aerofoil profile to reduce wake drag and eliminate stress concentrations at sharp corners, at the cost of marginally increased manufacturing complexity.

Stability vs. propulsion symmetry. The twin-tractor configuration introduces a propulsive moment coupling that was partially accounted for in the stability analysis through the tail volume coefficient sizing. A full propulsive contribution analysis remains an item for the detailed design phase.

9.4 Limitations and Future Work

The present study is a conceptual-level design and several areas require further development before the UAV can proceed to detailed design and fabrication.

1. **Structural analysis.** No finite-element or analytical structural sizing has been performed. The wing spar, rib spacing, tail boom bending loads, and fuselage skin panel stresses must be evaluated against the $\pm 3.5\,\text{g}$ / $-1.5\,\text{g}$ load factor envelope defined in Chapter 7.

2. **Propulsive trim contribution.** The pitching moment contribution of the twin-tractor motors, particularly the thrust-line offset from the aircraft centre of gravity, has not been included in the stability analysis and could shift the trim angle non-trivially at high thrust settings.

3. **Payload delivery accuracy.** The required Circular Error Probable of less than 10.00 m demands a wind-drift compensation algorithm in the drop logic. This has been stated as a requirement but not yet designed or simulated.

4. **Thermal management.** The GenX 6S 12 Ah pack operating at sustained high discharge rates (climb at ≈23 A per motor) will require a thermal management or ventilation strategy, particularly in the ambient temperature conditions of rural South India.

5. **CFD validation.** The aerodynamic coefficients used throughout this report were obtained from XFLR5 vortex-lattice analysis. Higher-fidelity CFD validation of the wing–fuselage junction, tail interference, and propeller slipstream effects is recommended before the prototype build.

9.5 Closing Remarks

The conceptual design presented in this report demonstrates that a fixed-wing UAV with a 2 m wingspan constraint is capable of satisfying the stated emergency medical logistics mission within realistic weight, power, and aerodynamic limits. The iterative design process, grounded in historical UAV regression data and validated against first-principles aerodynamic and stability analyses, produced a self-consistent and physically credible set of design parameters. The design forms a sound baseline for the detailed design, structural sizing, and eventual prototype fabrication phases of this project.

Appendix A

Codes

```
"""
============================================================
UAV Weight Estimation Script

Outputs:
    Final MTOW
    Battery weight
    Empty weight
    Payload weight
    Power requirements
    Energy usage
    Convergence plot (saved)
    Empty weight regression plot (saved)

Author: Abhijeet Mangela (AE25M034)
============================================================
"""

import numpy as np
import matplotlib.pyplot as plt
```

```python
import pandas as pd
import os


# ============================================================
# -------------------- CONSTANTS -----------------------------
# ============================================================

rho = 1.225
g = 9.81

b = 2
AR = 9.686999647
S = b**2 / AR

payload_mass_drop = 1
payload_mass = 2.1

V_climb = 20
V_cruise = 22
V_des = 20
V_loiter = 22

cruise_dist_1 = 40000
cruise_dist_2 = 40000

height_cruise = 200

gamma_climb = 8
gamma_des = 8

```

```python
C_d_0 = 0.03
e = 0.8

energy_density = 3600 * 120

Initial_weight = 8.5


# ==========================================================
# ---------------- POWER MODELS ---------------------------
# ==========================================================


def Power_climb(V_cl, gamma_climb, Weight):
    """Climb power requirement"""
    C_l = Weight * np.cos(np.deg2rad(gamma_climb)) / (0.5 * rho * V_cl**2 * S)
    C_d = C_d_0 + C_l**2 / (np.pi * AR * e)
    D_cl = 0.5 * rho * V_cl**2 * S * C_d
    T_cl = D_cl + Weight * np.sin(np.deg2rad(gamma_climb))
    return T_cl * V_cl


def Power_cruise(V_cr, Weight):
    return Power_climb(V_cr, 0, Weight)


def Power_descent(V_des, gamma_des, Weight):
    return max(Power_climb(V_des, -gamma_des, Weight), 0)


```

```python
def Power_loiter(V_loi, Weight):
    return Power_cruise(V_loi, Weight)


# ==========================================================
# ----------------- TIME CALCULATIONS ----------------------
# ==========================================================

t_cr_1 = cruise_dist_1 / V_cruise
t_cr_2 = cruise_dist_2 / V_cruise
t_loiter = 15 * 60

t_cl_1 = height_cruise / (V_climb * np.sin(np.deg2rad(gamma_climb)))
t_des_1 = height_cruise / (V_des * np.sin(np.deg2rad(gamma_des)))


# ==========================================================
# ----------------- ENERGY MODEL --------------------------
# ==========================================================


def Total_energy(Weight):
    """Total mission energy"""
    energy_climb = Power_climb(V_climb, gamma_climb, Weight) * t_cl_1
    energy_cruise_1 = Power_cruise(V_cruise, Weight) * t_cr_1
    energy_cruise_2 = Power_cruise(V_cruise, Weight - payload_mass_drop * g) * t_cr_2
    energy_descent = (
```

```
        Power_descent(V_des, gamma_des, Weight - payload_mass_drop * g) * t_des_1
    )
    energy_loiter = Power_loiter(V_loiter, Weight) * t_loiter

    return (
        energy_climb
        + energy_cruise_1
        + energy_cruise_2
        + energy_descent
        + energy_loiter
    )


# ========================================================
# ---------------- LOAD EXCEL DATA ----------------------
# ========================================================

print("\nLoading dataset...")
Data = pd.read_excel("Drone_dataset.xlsx")

# Same filtering as notebook
Data = Data[Data["Empty Weight (kg)"] / Data["MTOW (kg)"] <= 0.55]

Mtow_array = Data["MTOW (kg)"].to_numpy()
Empty_weight_array = Data["Empty Weight (kg)"].to_numpy()

Empty_weight_ratios = Empty_weight_array / Mtow_array


# ========================================================
```

```python
# ------------- EMPTY WEIGHT REGRESSION -------------------
# =========================================================

logx = np.log10(Mtow_array)
logy = np.log10(Empty_weight_ratios)

A = np.column_stack((logx, np.ones_like(logx)))
Soln, _, _, _ = np.linalg.lstsq(A, logy, rcond=None)

L = Soln[0]
loga = Soln[1]
a = 10**loga

print("\nRegression results")
print("------------------")
print("a =", a)
print("L =", L)


# =========================================================
# ------------- ITERATIVE WEIGHT SOLVER -------------------
# =========================================================

W_0 = Initial_weight
Weight_array = []

for _ in range(100):
    Weight_array.append(W_0)

    W_b = Total_energy(W_0 * g) / energy_density

```

```python
    W_0 = (payload_mass / (1 - (1.2 * W_b / W_0) - a * W_0**L)).item()


final_weight = Weight_array[-1]


# =========================================================
# ------------- COMPONENT BREAKDOWN -----------------------
# =========================================================

Battery_weight = Total_energy(final_weight * g) / energy_density
Empty_weight = final_weight * a * final_weight**L
Payload = payload_mass

print("\n================ FINAL RESULTS =================")
print(f"Final MTOW         : {final_weight:.3f} kg")
print(f"Battery weight     : {Battery_weight:.3f} kg")
print(f"Empty weight       : {Empty_weight:.3f} kg")
print(f"Payload            : {Payload:.3f} kg")
print(f"Total Energy       : {Total_energy(final_weight*g):.3f} Joules")


# Power reporting
print("\nPower Requirements (W)")
print("----------------------")
print("Climb   :", Power_climb(V_climb, gamma_climb, final_weight * g))
print("Cruise  :", Power_cruise(V_cruise, final_weight * g))
```

```
print("Descent :", Power_descent(V_des, gamma_des, final_weight * g))
print("Loiter  :", Power_loiter(V_loiter, final_weight * g))

print(
    "Climb Thrust T_W  :",
    Power_climb(V_climb, gamma_climb, final_weight * g) / (V_climb * g * final_weight),
)
print(
    "Cruise Thrust T_W:",
    Power_cruise(V_cruise, final_weight * g) / (V_climb * g * final_weight),
)
print(
    "Descent Thrust T_W:",
    Power_descent(V_des, gamma_des, final_weight * g) / (V_climb * g * final_weight),
)
print(
    "Loiter Thrust T_W:",
    Power_loiter(V_loiter, final_weight * g)
    / (V_climb * g * (final_weight - payload_mass_drop)),
)


# ==========================================================
# ----------------- PLOTS --------------------------------
# ==========================================================

os.makedirs("figures", exist_ok=True)
```

```python

# ------------------------------------------------------------
# Convergence Plot
# ------------------------------------------------------------

iters = list(range(1, len(Weight_array) + 1))

plt.figure(figsize=(8, 5))
plt.plot(range(len(Weight_array)), Weight_array, "o-")
plt.xlabel("Iteration")
plt.ylabel("Weight (kg)")
plt.title("Weight Convergence")
plt.annotate(
    f"{final_weight:.3f} kg",
    xy=(iters[-1], final_weight),
    xytext=(-60, 8),
    textcoords="offset points",
    arrowprops=dict(arrowstyle="->", color="tab:blue"),
    fontsize=10,
    color="tab:blue",
)
plt.grid(True)
# plt.show()

plt.savefig("figures/weight_convergence.png", dpi=300)
plt.close()


# ------------------------------------------------------------
# Empty weight regression plot
# ------------------------------------------------------------
```

```
plot_weight = np.linspace(3, 20, 1000)
plot_empt_rat = a * plot_weight**L

plt.figure(figsize=(8, 5))
plt.scatter(Mtow_array, Empty_weight_ratios, label="Data")
plt.plot(plot_weight, plot_empt_rat, label="Fit")
plt.xlabel("MTOW (kg)")
plt.ylabel("Empty Weight Ratio")
plt.title("Empty Weight vs MTOW")
plt.grid(True)
plt.legend()
# plt.show()

plt.savefig("figures/empty_weight_fit.png", dpi=300)
plt.close()


print("\nPlots saved inside: ./figures/")
print("Done!")
```

Listing A.1: Weight estimation code

```
%% (L/D)max estimation of similar UAVs

%% AUTHOR = MONISHA VIJAYAN (AE25M037) & ABHIJEET MANGELA (AE25M034
  )
% UAV Names
e=[];
Name = {
    'Yangda Mapird Plus'
    'Yangda FW-250'
    'Avy Aera'
    'Foxtech Baby Shark 260'
    'Foxtech AYK-250'
    'FlyDragon FLY-2100'
    'Black Swift S2'
    'Foxtech Nimbus'
};
% MTOW (kg)
MTOW = [ ...
     11.5 12.0 12.0 13.0  15.0 ...
     7.0 9.5  6.0 ];
% Wing span(m)
b = [...
     2.43 2.5 2.4 2.5 2.5 ...
     2.1 3 1.8];
% Wing area (m^2)
Area = [ ...
     0.65  0.52 0.96 0.50 0.50 ...
     0.52 0.70  0.38 ];
% Aspect ratio
AR = [ ...
```

```
     9.08  12.02 6.00 12.50 12.50 ...
     8.48 12.86 8.53 ];
% Cruising Altitude (m)
hcruise = [ ...
     3500 3500 3100 3000 2000 ...
     500 4250 1500 ];
% Cruising speed (m/s)
vcruise = [ ...
     19.44 23.33 26.39 22 ...
    25.00 18.5 18.06 20];
%Swet/Sref
sratio =[...
    3.7 3.4 2.20 2.3 2.20...
     3.7 2.3  3.7];
swet=[];
ARwet=[];
for i = 1:length(sratio)
    swt=sratio(i)*Area(i);
    swet(end+1)=swt;
    ARwt=(b(i)^2)/swet(i);
    ARwet(end+1)=ARwt;
end


%Planform type
Planform = {
    'Rectangular'
    'Tapered'
    'Tapered'
    'Tapered'
    'Tapered'
```

```
    'Rectangular'
    'Rectangular'
    'Rectangular'
};


% Taper Ratio
TR=[...
    1, 0.141, 0.287, 0.119, 0.149, 1, 1, 1 ...
];
%------------------------------------------------------------------------

%Oswald Efficiency Factor
for i = 1:length(MTOW)
    if strcmp(Planform{i},'Rectangular')
        eu(i) = 1.78*(1 - 0.045*AR(i)^0.68) - 0.64;

    elseif strcmp(Planform{i},'Tapered')
        d(i) = (1-TR(i))/(2*(1+TR(i)));
        eu(i) = 1/(1+d(i));
    end
    e(end+1)=eu(i);
end
e = max(e, 0.7);
e = min(e, 1.0);
%disp(e)
%------------------------------------------------------------------------

%ISA Parameters
h = hcruise;
rho = NaN(size(h));
```

```
T0 = 288.15;
P0 = 101325;
rho0 = 1.225;
L = 0.0065;
R = 287.05;
g = 9.80665;
%-----------------------------------------------------------------------

%Density Calculation
for i = 1:length(h)
    if ~isnan(h(i))
        T = T0 - L*h(i);
        rho(i) = rho0*(T/T0)^(g/(R*L) - 1);
    end
end

%disp(rho)
%-----------------------------------------------------------------------

%L/D max calculation
L_Dm = [];

for i=1:length(MTOW)
    Cl=(2*MTOW(i)*9.81)/(rho(i)*Area(i)*(vcruise(i)^2));
    L_D=(pi*e(i)*AR(i))/(2*Cl);
    L_Dm(end+1)=L_D;
end
%-----------------------------------------------------------------------

%for our UAV
sratio1=3.6;
```

```
Sref=0.413;
swet=sratio1*Sref;
ARwsk =4/swet;
disp(sqrt(ARwsk))

Table = table(Name(:), MTOW(:), vcruise(:), hcruise(:), ARwet(:),
    L_Dm(:), ...
                'VariableNames', {'Name','MTOW (kg)','cruise velocity
    (m/s)', 'Cruising Height (m)', 'Wetted Aspect Ratio', 'L_D_max'
    });
writetable(Table, 'Dataset_2.xlsx')

our_L_D_max = polyval(fit,sqrt(ARwsk));

x = linspace(1,3,200);

fit = polyfit(ARwet, L_Dm, 1);
y   = polyval(fit, x);

figure('Color','w','Position',[200 200 800 550])   % white bg +
    larger figure

% ---- Scatter ----
scatter(sqrt(ARwet), L_Dm, 70, ...
        'filled', ...
        'MarkerFaceColor',[0.2 0.6 0.2], ...
        'MarkerEdgeColor','k', ...
        'LineWidth',0.8)
hold on

% ---- Fit line ----
```

```
plot(x, y, ...
     'LineWidth', 2.5, ...
     'Color',[0 0.35 0.8])

% ---- Vertical line ----
xline(sqrt(ARwsk), '--r', ...
      'LineWidth', 2, ...
      'Label','Our case', ...
      'LabelVerticalAlignment','bottom')

% ---- Labels ----
xlabel('$\sqrt{AR_{wet}}$','Interpreter','latex','FontSize',14)
ylabel('$L/D_{max}$','Interpreter','latex','FontSize',14)
title('Lift-to-Drag Ratio vs Aspect Ratio','FontSize',15)

% ---- Grid & styling ----
grid on
grid minor
box on

set(gca, ...
    'FontSize',12, ...
    'LineWidth',1.2, ...
    'TickDir','out')

legend({'Data','Linear fit','Reference AR'}, ...
       'Location','northeast')

% cruise
AR=9.686
MTOW=68.516; %MTOW=57.94 AFTER PAYLOAD
```

```
e=1.78*(1-0.045*AR^0.68)-0.64 %for rectangular wing
rho=1.225; % @ cruising altitude = 200m
b=2;
s=(b^2)/AR
vcr=22;
Cl=(2*MTOW)/(rho*vcr^2*s);
k=1/(pi*AR*e);
v0=30;
L_D_max = our_L_D_max
gamma = 8;
Lift_cruise = MTOW;
Drag_cruise = Lift_cruise/L_D_max;
thrust_required_cruise = Drag_cruise;
C_L_cruise = @(v) Lift_cruise/(0.5*rho*s*v^2);
C_d_i_cruise_func = @(v) C_L_cruise(v)^2/(pi*e*AR);
Drag_function_cruise = @(v)(0.5*rho*s*v^2*(2*C_d_i_cruise_func(v)))
    - Drag_cruise;
optimal_velocity_cruise = fsolve(Drag_function_cruise,v0);
Power_required_cruise = thrust_required_cruise*
   optimal_velocity_cruise;
T_W_cruise=thrust_required_cruise/MTOW;
P_W_cruse = Power_required_cruise/MTOW;
CDo = C_d_i_cruise_func(optimal_velocity_cruise);

% climb
gamma = 5;
Lift = MTOW*cosd(gamma);
Drag = 2*Lift/(sqrt(3)*L_D_max);
thrust_required_climb = Drag + MTOW*sind(gamma);
C_L_climb = @(v) Lift/(0.5*rho*s*v^2);
C_d_i_climb_func = @(v) C_L_climb(v)^2/(pi*e*AR);
```

```
Drag_function = @(v)(0.5*rho*s*v^2*(CDo + C_d_i_climb_func(v))) -
    Drag;
optimal_velocity_climb = fsolve(Drag_function,v0);
Power_required_climb = thrust_required_climb*optimal_velocity_climb
T_W_climb=thrust_required_climb/MTOW;
P_W_climb = Power_required_climb/MTOW;

% descent
gamma =-8;
Lift = MTOW*cosd(gamma);
Drag = 2*Lift/(sqrt(3)*L_D_max);
thrust_required_descent = Drag + MTOW*sind(gamma);
C_L_descent = @(v) Lift/(0.5*rho*s*v^2);
C_d_i_descent_func = @(v) C_L_descent(v)^2/(pi*e*AR);
Drag_function = @(v)(0.5*rho*s*v^2*(CDo + C_d_i_descent_func(v))) -
     Drag;
optimal_velocity_descent = fsolve(Drag_function,v0);
Power_required_descent = thrust_required_descent*
    optimal_velocity_descent;
T_W_descent=thrust_required_descent/MTOW;
P_W_climb = Power_required_descent/MTOW;

% loiter

gamma = 0
Lift = MTOW*cosd(gamma);
Drag = 2*Lift/(sqrt(3)*L_D_max);
thrust_required_loiter = Drag + MTOW*sind(gamma);
C_L_loiter = @(v) Lift/(0.5*rho*s*v^2);
C_d_i_loiter_func = @(v) C_L_loiter(v)^2/(pi*e*AR);
```

```
Drag_function = @(v)(0.5*rho*s*v^2*(CDo + C_d_i_loiter_func(v))) -
    Drag;
optimal_velocity_loiter= fsolve(Drag_function,v0);
Power_required_loiter = thrust_required_loiter*
    optimal_velocity_loiter;
T_W_loiter=thrust_required_loiter/MTOW;
P_W_loiter = Power_required_loiter/MTOW;

appox_c = b/AR
```

Listing A.2: Optimum Speeds selection code

```
% CONSTRAINT DIAGRAM TO FINDOUT WING LOADING

% AUTHOR: MONISHA VIJAYAN (AE25M037)

rho_0=1.225;
rho_alt=1.20;
rho_abs=1.19;
MTOW=68.516;
sigma_ceil=(rho_alt/rho_0);
sigma_ceil1=(rho_abs/rho_0);
g=9.81;
V_max=24;
CL_max=1.3;
S=0.413;
V_stall=sqrt((2*MTOW)/(rho_0*CL_max*S));
CD0=0.025;
AR=9.686;
e=0.8;
K=1/(pi*e*AR);
eta_p=0.77;
ROC=24.4720*sind(8);
L_D_max=15.54;
R_min=20;
S_TO=20;
ROC_ceil=0.508;
ROC_ceil1=0;
V_to= 1.2*V_stall;
CL_takeoff = (2*MTOW)/(rho_0*V_to^2*S);
mu=0.04;
WS=linspace(5,200,500);
```

```
% Stalling speed
WS_stall=0.5*rho_0*V_stall^2*CL_max;
% Maximum speed
WP_Vmax=eta_p./((0.5*rho_0*V_max^3*CD0./WS)+(2*K.*WS./(rho_0*V_max)));
% ROC
term_ROC=(1.155/(L_D_max))*sqrt((2.*WS)./(rho_alt*sqrt(3*CD0/K)));
WP_ROC=eta_p./(ROC+term_ROC);
% Takeoff ; CDG = CD0+KCL_to^2-muCL_to
CDG=CD0+K*(CL_takeoff)^2-mu*CL_takeoff;
exp_term=exp(0.6*rho_0*g*CDG*S_TO./WS);
WP_STO=((1-exp_term)./(mu-(mu+CDG/(CL_max/1.1^2)).*exp_term)).*(eta_p/V_to);
% Minimum Turn Radius
n=sqrt((V_max^2/(g*R_min))^2+1);
q_max=0.5*rho_0*V_max^2;
PW_Turn=(V_max/eta_p).*((q_max*CD0./WS)+(K*n^2.*WS./q_max));
WP_Turn=1./PW_Turn;
%Service ceiling
term_ceil=(1.155/(L_D_max*eta_p))*sqrt((2.*WS)./(rho_alt*sqrt(3*CD0/K)));
WP_Ceiling=sigma_ceil./((ROC_ceil/eta_p)+term_ceil);
%Absolute ceiling
term_ceil=(1.155/(L_D_max*eta_p))*sqrt((2.*WS)./(rho_abs*sqrt(3*CD0/K)));
WP_Ceiling1=sigma_ceil1./((ROC_ceil1/eta_p)+term_ceil);

%Maximum Stall Speed Constraint
figure('Color','w')
hold on
```

```
set(gca,'Color','w','XColor','k','YColor','k','GridColor','w');
    grid on
plot(WS,WP_Vmax,'r','LineWidth',2)
idx = WS <= WS_stall;
x = WS(idx);
y = WP_Vmax(idx);
spacing = 5;
for i = 1:spacing:length(x)
    plot([x(i) x(i)], [0 y(i)], 'k','LineWidth',0.5)
end
xline(WS_stall,'--k','LineWidth',2)
xlabel('Wing Loading W/S (N/m^2)','FontWeight','bold')
ylabel('Power Loading W/P (N/W)','FontWeight','bold')
title('Maximum Speed Constraint','Color','k')
legend('Max Speed','Stall Limit')
ylim([0 0.5])
hold off

% Stalling Speed Constraint
figure('Color','w')
hold on
set(gca,'Color','w','XColor','k','YColor','k','GridColor','w');
    grid on
xline(WS_stall,'--r','LineWidth',2)
spacing = 5;
x = WS(WS<=WS_stall);
for i = 1:spacing:length(x)
    plot([x(i) x(i)], [0 0.5], 'k','LineWidth',0.5)
end
xlabel('Wing Loading W/S (N/m^2)','FontWeight','bold')
ylabel('Power Loading W/P (N/W)','FontWeight','bold')
```

```
title('Stall Speed Constraint','Color','k')
legend('Stall Limit')
ylim([0 0.5])
xlim([0 max(WS)])
hold off

% ROC max constraint
figure('Color','w')
hold on
set(gca,'Color','w','XColor','k','YColor','k','GridColor','w');
   grid on
plot(WS,WP_ROC,'b','LineWidth',2)
idx = WS <= WS_stall;
x = WS(idx);
y = WP_ROC(idx);
spacing = 5;
for i = 1:spacing:length(x)
    plot([x(i) x(i)], [0 y(i)], 'k','LineWidth',0.5)
end
xline(WS_stall,'--k','LineWidth',2)
xlabel('Wing Loading W/S (N/m^2)','FontWeight','bold')
ylabel('Power Loading W/P (N/W)','FontWeight','bold')
title('Rate of Climb Constraint','Color','k')
legend('ROC','Stall Limit')
ylim([0 0.5])
hold off
% Takeoff constraint
figure('Color','w')
hold on
set(gca,'Color','w','XColor','k','YColor','k','GridColor','w');
   grid on
```

```
plot(WS,WP_STO,'g','LineWidth',2)
idx = WS <= WS_stall;
x = WS(idx);
y = WP_STO(idx);
spacing = 5;
for i = 1:spacing:length(x)
    plot([x(i) x(i)], [0 y(i)], 'k','LineWidth',0.5)
end
xline(WS_stall,'--k','LineWidth',2)
xlabel('Wing Loading W/S (N/m^2)','FontWeight','bold')
ylabel('Power Loading W/P (N/W)','FontWeight','bold')
title('Takeoff Constraint','Color','k')
legend('Takeoff','Stall Limit')
ylim([0 0.5])
hold off
% Turn radius constraint
figure('Color','w')
hold on
set(gca,'Color','w','XColor','k','YColor','k','GridColor','w'); grid on
plot(WS,WP_Turn,'m','LineWidth',2)
idx = WS <= WS_stall;
x = WS(idx);
y = WP_Turn(idx);
spacing = 5;
for i = 1:spacing:length(x)
    plot([x(i) x(i)], [0 y(i)], 'k','LineWidth',0.5)
end
xline(WS_stall,'--k','LineWidth',2)
xlabel('Wing Loading W/S (N/m^2)','FontWeight','bold')
ylabel('Power Loading W/P (N/W)','FontWeight','bold')
```

```
title('Turn Radius Constraint','Color','k')
legend('Turn Radius','Stall Limit')
ylim([0 0.5])
hold off
% Service ceiling constraint
figure('Color','w')
hold on
set(gca,'Color','w','XColor','k','YColor','k','GridColor','w');
   grid on
plot(WS,WP_Ceiling,'c','LineWidth',2)
idx = WS <= WS_stall;
x = WS(idx);
y = WP_Ceiling(idx);
spacing = 5;
for i = 1:spacing:length(x)
    plot([x(i) x(i)], [0 y(i)], 'k','LineWidth',0.5)
end
xline(WS_stall,'--k','LineWidth',2)
xlabel('Wing Loading W/S (N/m^2)','FontWeight','bold')
ylabel('Power Loading W/P (N/W)','FontWeight','bold')
title('Service Ceiling Constraint','Color','k')
legend('Service Ceiling','Stall Limit')
ylim([0 2])
hold off
% Absolute ceiling constraint
figure('Color','w')
hold on
set(gca,'Color','w','XColor','k','YColor','k','GridColor','w');
   grid on
plot(WS,WP_Ceiling1,'Color',[0.5 0 0.8],'LineWidth',2)
idx = WS <= WS_stall;
```

```
x = WS(idx);
y = WP_Ceiling1(idx);
spacing = 5;
for i = 1:spacing:length(x)
    plot([x(i) x(i)], [0 y(i)], 'k','LineWidth',0.5)
end
xline(WS_stall,'--k','LineWidth',2)
xlabel('Wing Loading W/S (N/m^2)','FontWeight','bold')
ylabel('Power Loading W/P (N/W)','FontWeight','bold')
title('Absolute Ceiling Constraint','Color','k')
legend('Absolute Ceiling','Stall Limit')
ylim([0 2])
hold off

%All Constraints
figure
fig=figure('Color','w');set(gcf,'InvertHardcopy','off');
ax=gca;ax.Color='w';ax.XColor='k';ax.YColor='k';ax.GridColor='k';ax.GridAlpha=0.2;ax.LineWidth=1.5;hold on;grid on;
plot(WS,WP_Vmax,'r','LineWidth',2,'DisplayName','MaxSpeed');
plot(WS,WP_ROC,'b','LineWidth',2,'DisplayName','ROC');
plot(WS,WP_STO,'g','LineWidth',2,'DisplayName','Take-off');
plot(WS,WP_Turn,'m','LineWidth',2,'DisplayName','TurnRadius');
plot(WS,WP_Ceiling,'c','LineWidth',2,'DisplayName','Service Ceiling');
plot(WS,WP_Ceiling1,'Color',[0.5 0 0.8],'LineWidth',2,'DisplayName','Absolute Ceiling');
xline(WS_stall,'--k','LineWidth',2,'DisplayName','StallLimit');
xlabel('Wing Loading (W/S) [N/m^2]','FontWeight','bold');
ylabel('Power Loading (W/P) [N/W]','FontWeight','bold');
```

```
title('Wing Loading Constraint Diagram','FontSize',14,'Fontweight',
    'bold','Color','k');
ylim([0,0.5])
WS_design = 121.844;
WP_design = interp1(WS,WP_STO,WS_design);
plot(WS_design,WP_design,'ko','MarkerFaceColor','k','MarkerSize',8,
    'DisplayName','Design Point')
text(WS_design,WP_design+0.01,' Design Point','Color','k','
    FontWeight','bold')
legend('Location','northwest','TextColor','k','Color','w');
hold off;

% NEW SURFACE AREA
b=2;
W_S=121.844;
W=68.516;
S_new=W/(W_S)
AR=(b^2)/S_new
```

Listing A.3: Wing Loading

```python
## Fuselage Sizing

## AUTHOR = ABHIJEET MANGELA (AE25M034)

import numpy as np
import matplotlib.pyplot as plt
import pandas as pd

Data = pd.read_excel("Similar UAV.xlsx")

Weights = Data["MTOW (kg)"].to_numpy()
Fuselage_lengths = Data["Fuselage Length (m)"].to_numpy()

# Power law fit
a, b = np.polyfit(np.log(Weights), np.log(Fuselage_lengths), 1)

# Smooth curve
x_array = np.linspace(min(Weights)*0.8, max(Weights)*1.2, 200)
y_array = np.exp(b) * x_array**a

# Our UAV
Our_weight = 7.021
Our_length = np.exp(b) * Our_weight**a

plt.figure(figsize=(7,5))

# Data points
plt.scatter(Weights, Fuselage_lengths,
            s=60, alpha=0.8, label="Existing UAVs")

```

```
# Fit curve
plt.plot(x_array, y_array,
         linewidth=2.5,
         label=f"Fit: L = {np.exp(b):.2f} W^{a:.2f}")

# Our UAV point
plt.scatter(Our_weight, Our_length,
            color='red', s=120, marker='*',
            label="Our UAV")

# Vertical line at our weight
plt.axvline(Our_weight, linestyle="--", color="red", alpha=0.7)

# Horizontal line at predicted length
plt.axhline(Our_length, linestyle="--", color="gray", alpha=0.5)

# Annotate value
plt.text(Our_weight*1.02, Our_length,
         f"L = {Our_length:.2f} m",
         verticalalignment='bottom')

plt.xlabel("Maximum Takeoff Weight (kg)")
plt.ylabel("Fuselage Length (m)")
plt.title("Power Law Relationship Between UAV MTOW and Fuselage
   Length")

plt.grid(True, linestyle="--", alpha=0.4)
plt.legend()
plt.tight_layout()

plt.show()
```

Listing A.4: Fuselage Sizing

```
% HORIZONTAL TAIL DESIGN AND STABILITY ANALYSIS
% AUTHOR : MONISHA VIJAYAN (AE25M037)

clear
clc
close all
X_cg = 0.264;
c_bar = 0.2653;
X_ac_w = 0.1985;
X_ac_f = 0.09;
S = 0.5307;
S_t = 0.25*S;
eta = 0.99;
AR = 7.53;
e = 0.7648;
CL_w = 0.378058;
CLalpha_t = 3.8310;
CLalpha = 3.6107;
Cmac_w = -0.0683;
Cmac_f = 0;
CL_f = 0.061015;
Cm_alpha_f = 0.1310;
l_t = 0.7;
de_da = (2*CLalpha)/(pi*e*AR)
Cm_f = Cmac_f + CL_f*(X_cg-X_ac_f)/c_bar;
V_H = (S_t*l_t)/(S*c_bar);
X_np = X_ac_w + c_bar*( eta*V_H*(CLalpha_t/CLalpha)*(1-de_da) -
   Cm_alpha_f/CLalpha );
Kn = (X_np - X_cg)/c_bar;
fprintf('Neutral Point = %.5f m\n',X_np)
```

```
fprintf('Static Margin = %.5f\n\n',Kn)
alpha = 3*pi/180;
func=@(it) Cmac_w + CL_w*(X_cg-X_ac_w)/c_bar ...
          -eta*(S_t/S)*(CLalpha_t*(alpha+it-de_da*alpha))*(l_t/
   c_bar) ...
          +Cm_f;
it=fzero(func,-3*pi/180);
fprintf('Tail incidence angle = %.4f deg\n\n',it*180/pi)
epsilon = de_da*alpha;
alpha_t = alpha + it - epsilon
CL_t = CLalpha_t*alpha_t
Cm_cg = Cmac_w ...
        + CL_w*(X_cg-X_ac_w)/c_bar ...
        - eta*(S_t/S)*CL_t*(l_t/c_bar) ...
        + Cm_f;
fprintf('Total pitching moment about CG = %.8f\n\n',Cm_cg)
Cm_alpha=((X_cg-X_ac_w)/c_bar)*CLalpha ...
          -eta*V_H*CLalpha_t*(1-de_da) ...
          +Cm_alpha_f
fprintf('Cm_alpha = %.6f\n\n',Cm_alpha)
alpha_trim = 3*pi/180;
Cm0 = -Cm_alpha*alpha_trim
alpha=linspace(-5,15,100)*pi/180;
Cm=Cm0+Cm_alpha*alpha;
figure("Color",'white')
plot(alpha*180/pi,Cm,'LineWidth',2,'Color','r')
title('C_m Vs \alpha (Analytical)','Color','k')
xlabel('\alpha (deg)')
ylabel('C_m')
ylim([-0.2 0.2])
ax=gca;
```

```
ax.Color='w';
ax.XColor='k';
ax.YColor='k';
set(gca,'FontSize',14,'LineWidth',1.2)
hold on
plot(alpha_trim*180/pi,0,'ko','MarkerSize',6,'LineWidth',1.5)
yline(0,'k--')
grid on

%[appendix]{"version":"1.0"}
%---
%[metadata:view]
%   data: {"layout":"onright","rightPanelPercent":40.8}
%---
```

Listing A.5: Horizontal Tail Design

```python
"""
============================================================
  Flight Dynamics & Stability Analysis
  Author: Abhijeet Mangela
  Based on: Chapter 9 Appendix -- Case Studies
  Case 1 : UAV aircraft.
============================================================
All formulas are taken directly from the supplied PDF.
"""

import numpy as np
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches

D2R = np.pi / 180.0   # degrees -> radians


def section_header(title):
    print("\n" + "=" * 60)
    print(f"  {title}")
    print("=" * 60)


def sub_header(title):
    print(f"\n--- {title} ---")




def aircraft():
```

```
section_header("CASE 1 -- UAV")

sub_header("A) Input Parameters")

#  Geometry
g         = 9.81          # m/s^2  -- gravitational acceleration
S_w       = 0.5307          # m^2   -- wing planform (reference) area
b_w       = 2          # m      -- wing span
c_r       = 0.2653          # m      -- wing root chord (NACA 23018)
c_t       = 0.2650          # m      -- wing tip chord  (NACA 23012)
lam       = c_t / c_r    # taper ratio  (= 0.5)
Lambda_c4_deg = 0     # deg    -- wing quarter-chord sweep angle
Gamma_deg     = 0     # deg    -- wing dihedral angle
i_w       = 3            # deg    -- wing incidence setting angle

S_HT      = 0.1327          # m^2   -- horizontal tail planform area
AR_HT     = 4.5           # --      -- HT aspect ratio
i_t       = -0.423           # deg    -- HT incidence setting angle
X_AC_HT   = 0.964          # m      -- HT aerodynamic-centre from nose

S_VT      = 0.0478          # m^2   -- vertical tail planform area
AR_VT     = 1.5           # --      -- VT aspect ratio
X_AC_VT   = 0.964          # m      -- VT AC from nose
```

```
    l_f       = 1.12          # m      -- fuselage length
    S_f       = 1.11          # m^2    -- fuselage side projected area
    D_f       = 2*(np.sqrt(0.3*0.25*1.2/np.pi))          # m      -- max fuselage diameter

    S_da      = 0.05          # m^2    -- aileron surface area
    l_da      = 0.76          # m      -- aileron span
    S_de      = 0.04          # m^2    -- elevator surface area
    S_dr      = 0.01          # m^2    -- rudder surface area

    #  Mass & Inertia
    m         = 6.298       # kg     -- aircraft mass
    I_xx      = 0.751       # kgm -- roll inertia
    I_yy      = 0.384       # kgm -- pitch inertia
    I_zz      = 1.045       # kgm -- yaw inertia

    X_CG      = 0.264           # m      -- CG location from nose

    #  Aerodynamic parameters
    alpha_L0_deg = -3.5       # deg    -- wing zero-lift AoA
    Cm_ac_w      = -0.03    # --      -- residual pitching moment at AC
    SM           = 0.14       # --      -- assumed static margin (15%)
    e_oswald     = 1        # --      -- Oswald efficiency factor
    C_D0         = 0.02873     # --      -- zero-lift drag coefficient
    # Airfoil lift-curve slopes (rad)
    cl_alpha_w    = 3.6107       # /rad  -- NACA 23018 airfoil (0.1/deg)
```

```
cl_alpha_HT  = 6.24       # /rad  -- NACA 0012 (0.109/deg)
cl_alpha_VT  = 6.24       # /rad  -- NACA 0012

#  Flight conditions
rho          = 1.22       # kg/m -- air density at flight altitude
a_sound      = 340.0      # m/s   -- speed of sound
eta          = 0.99       # --      -- dynamic pressure ratio at tail

#  Engine/propeller data
P_max        = 450  # Nm/s -- max engine power (360 hp)
N_blades     = 4          # --      -- number of propeller blades
D_prop       = 0.3048       # m      -- propeller diameter
l_prop       = 0.12957       # m      -- prop disc to CG distance

# vertical location of wing root q-chord above CG line (for lat-dir)
z_w          = 0.06      # m

print("  Inputs loaded.")

sub_header("B) Derived Geometry Parameters")

# Mean aerodynamic chord  (Eq. from PDF)
c_bar = (2/3) * c_r * (1 + lam + lam**2) / (1 + lam)

# Wing aspect ratio
AR_w = b_w**2 / S_w

# Tail spans
```

```
    b_HT = np.sqrt(AR_HT * S_HT)
    b_VT = np.sqrt(AR_VT * S_VT)

    # VT aerodynamic centre height above fuselage reference line
    z_VT = (4/9) * b_VT

    # Propeller disc area
    S_prop = np.pi * D_prop**2 / 4

    # Moment arm lengths
    l_HT = X_AC_HT - X_CG
    l_VT = X_AC_VT - X_CG

    # Tail volume ratios
    V_HT = (S_HT * l_HT) / (S_w * c_bar)
    V_VT = (S_VT * l_VT) / (S_w * b_w)

    # Non-dimensionalised CG location
    h_CG = X_CG / c_bar

    # Induced-drag factor
    K = 1.0 / (np.pi * AR_w * e_oswald)

    print(f"  c_bar    = {c_bar:.4f} m")
    print(f"  AR_w     = {AR_w:.4f}")
    print(f"  lam      = {lam:.2f}")
    print(f"  b_HT     = {b_HT:.3f} m,   b_VT = {b_VT:.3f} m")
    print(f"  z_VT     = {z_VT:.3f} m")
    print(f"  l_HT     = {l_HT:.3f} m,   l_VT = {l_VT:.3f} m")
    print(f"  V_HT     = {V_HT:.4f},      V_VT = {V_VT:.4f}")
    print(f"  h_CG     = {h_CG:.3f}")
```

```
    print(f"  K        = {K:.4f}")

    sub_header("C) Aerodynamic Coefficients")

    # Finite-wing lift-curve slopes  (low-subsonic simplified form, PDF Eq.)
    CL_alpha_w  = cl_alpha_w  / (1 + cl_alpha_w  / (np.pi * AR_w))
    CL_alpha_HT = cl_alpha_HT / (1 + cl_alpha_HT / (np.pi * AR_HT))
    CL_alpha_VT = cl_alpha_VT / (1 + cl_alpha_VT / (np.pi * AR_VT))

    print(f"  CL_alpha_w   = {CL_alpha_w:.3f} /rad")
    print(f"  CL_alpha_HT  = {CL_alpha_HT:.3f} /rad")
    print(f"  CL_alpha_VT  = {CL_alpha_VT:.3f} /rad")

    # Wing lift coefficient at zero AoA
    CL0_w = CL_alpha_w * (alpha_L0_deg * D2R) * (-1)   # note: alpha_L0 is negative
    # PDF formula: CL0_w = CL_alpha_w  |alpha_L=0|  pi/180
    CL0_w = CL_alpha_w * abs(alpha_L0_deg) * D2R
    print(f"  CL0_w        = {CL0_w:.4f}")

    # Downwash gradient  de/da
    eps_alpha = 2 * CL_alpha_w / (np.pi * AR_w)
    eps_0     = 2 * CL0_w / (np.pi * AR_w)
    print(f"  de/da        = {eps_alpha:.4f}")
    print(f"  eps_0        = {eps_0:.4f}")

    sub_header("D) Trim Calculations")

    # Neutral-point location (non-dim)
    h_NP = h_CG + SM
```

```
    X_NP = h_NP * c_bar

    # Wing-body AC location (from NP and tail contribution)
    h_AC_wb = h_NP - V_HT * (CL_alpha_HT / CL_alpha_w) * (1 - eps_alpha)
    X_AC_wb = h_AC_wb * c_bar

    print(f"  h_NP          = {h_NP:.3f},   X_NP   = {X_NP:.3f} m")
    print(f"  h_AC_wb       = {h_AC_wb:.3f},  X_AC_wb = {X_AC_wb:.3f} m")

    # Pitching moment derivatives
    Cm_alpha_w   = ((X_CG - X_AC_wb) / c_bar) * CL_alpha_w
    Cm_alpha_HT = -eta * V_HT * CL_alpha_HT * (1 - eps_alpha)
    Cm_alpha    = Cm_alpha_w + Cm_alpha_HT

    Cm0_HT = eta * V_HT * CL_alpha_HT * (i_t * D2R * (-1) + eps_0)
    Cm0    = Cm_ac_w + eta * V_HT * CL_alpha_HT * (eps_0)

    # Trim AoA (zero elevator deflection)
    alpha_trim_rad = -Cm0 / Cm_alpha
    alpha_trim_deg = alpha_trim_rad / D2R

    # Whole-aircraft CL_alpha and CL0
    CL0 = (CL_alpha_w * abs(alpha_L0_deg) * D2R
           + eta * (S_HT / S_w) * CL_alpha_HT * (i_t * D2R - eps_0))
    CL_alpha = (CL_alpha_w
                + eta * (S_HT / S_w) * CL_alpha_HT * (1 - eps_alpha))

```

```python
    # Trim lift coefficient
    CL_trim = (CL_alpha_w * (alpha_trim_deg - alpha_L0_deg) * D2R
               + eta * (S_HT / S_w) * CL_alpha_HT
               * (alpha_trim_rad * (1 - eps_alpha) - eps_0 + i_t *
D2R))

    # Trim speed from L = W
    V_trim = np.sqrt(2 / rho * (m * g / S_w) * (1 / CL_trim))

    # Dynamic pressure at trim
    q_trim = 0.5 * rho * V_trim**2

    # Mach number
    Ma_trim = V_trim / a_sound

    # Trim drag coefficient
    CD_trim = C_D0 + K * CL_trim**2

    # Thrust and power required
    T_req = q_trim * S_w * CD_trim
    P_req = T_req * V_trim

    print(f"  Cm_alpha_w    = {Cm_alpha_w:.4f} /rad")
    print(f"  Cm_alpha_HT   = {Cm_alpha_HT:.4f} /rad")
    print(f"  Cm_alpha      = {Cm_alpha:.4f} /rad")
    print(f"  Cm0           = {Cm0:.4f}")
    print(f"  CL0           = {CL0:.4f}")
    print(f"  CL_alpha      = {CL_alpha:.4f} /rad")
    print(f"  Alpha_trim    = {alpha_trim_rad:.4f} rad  ({
alpha_trim_deg:.2f} deg)")
    print(f"  CL_trim       = {CL_trim:.4f}")
```

```
    print(f"  V_trim       = {V_trim:.3f} m/s")
    print(f"  Ma_trim      = {Ma_trim:.4f}")
    print(f"  q_trim       = {q_trim:.3f} N/m")
    print(f"  CD_trim      = {CD_trim:.4f}")
    print(f"  Thrust req.  = {T_req:.2f} N")
    print(f"  Power req.   = {P_req:.2f} W  ({P_req/745.7:.1f} hp)"
  )

    # (E) LONGITUDINAL STABILITY DERIVATIVES

    sub_header("E) Longitudinal Stability Derivatives")

    # Pitch-rate & alpha-dot derivatives
    Cm_q1 = -2 * eta * V_HT * CL_alpha_HT * (l_HT / c_bar)
    CL_q1 = 2 * eta * V_HT * CL_alpha_HT
    # alpha-dot term (PDF sets Cm_alpha_dot & CL_alpha_dot = 0 in
  this example)
    Cm_adot = 0.0
    CL_adot = 0.0

    # Drag derivatives
    CD_alpha = 2 * K * CL_trim * CL_alpha
    CD_q1    = 2 * K * CL_trim * CL_q1

    # Mach-number derivatives (compressibility)
    CL_Ma = CL_trim * Ma_trim / (1 - Ma_trim**2)
    CD_Ma = 0.0     # PDF sets this to zero for this aircraft

    # Useful ratios
    qSc_Iyy  = q_trim * S_w * c_bar / I_yy
    qS_W     = q_trim * S_w / (m * g)
```

```
    g_V       = g / V_trim
    c_2V      = c_bar / (2 * V_trim)

    print(f"  Cm_q1         = {Cm_q1:.4f} /rad")
    print(f"  CL_q1         = {CL_q1:.4f} /rad")
    print(f"  CD_alpha      = {CD_alpha:.4f} /rad")
    print(f"  CD_q1         = {CD_q1:.4f} /rad")
    print(f"  CL_Ma         = {CL_Ma:.4f}")
    print(f"  q*Sc/Iyy      = {qSc_Iyy:.4f} /s")
    print(f"  q*S/W         = {qS_W:.4f}")
    print(f"  g/V*          = {g_V:.4f}")
    print(f"  c/(2V*)       = {c_2V:.5f} s")


    # (F) LONGITUDINAL DYNAMIC MODES (2nd-order approx.)

    sub_header("F) Longitudinal Dynamic Modes (Approximate)")


    # (omega_n)^2_SP = -(q*Sc/Iyy) * Cm_alpha
    wn_SP_sq = -qSc_Iyy * Cm_alpha
    wn_SP     = np.sqrt(wn_SP_sq)
    T_SP      = 2 * np.pi / wn_SP

    # 2*zeta*wn_SP = -(q*Sc/Iyy) * (c/2V*) * (Cm_q1 + Cm_adot)
    two_zeta_wn_SP = -qSc_Iyy * c_2V * (Cm_q1 + Cm_adot)
    zeta_SP   = two_zeta_wn_SP / (2 * wn_SP)


    # (omega_n)^2_Ph = (g/V*)^2 * [(q*S/W) * (Ma* * CL_Ma + 2*CL*)]
    wn_Ph_sq = (g_V**2) * qS_W * (Ma_trim * CL_Ma + 2 * CL_trim)
```

```
    wn_Ph     = np.sqrt(wn_Ph_sq)
    T_Ph      = 2 * np.pi / wn_Ph

    # 2*zeta*wn_Ph = (g/V*) * [(q*S/W) * (Ma* * CD_Ma + 2*CD*)]
    two_zeta_wn_Ph = g_V * qS_W * (Ma_trim * CD_Ma + 2 * CD_trim)
    zeta_Ph   = two_zeta_wn_Ph / (2 * wn_Ph)

    print(f"\n  SHORT PERIOD MODE")
    print(f"    omega_n    = {wn_SP:.4f} rad/s")
    print(f"    zeta       = {zeta_SP:.4f}")
    print(f"    Period     = {T_SP:.4f} s")

    print(f"\n  PHUGOID MODE")
    print(f"    omega_n    = {wn_Ph:.4f} rad/s")
    print(f"    zeta       = {zeta_Ph:.4f}")
    print(f"    Period     = {T_Ph:.4f} s")


    # (G) LONGITUDINAL A-MATRIX (1st-order form, Sec. 9.1.2)

    sub_header("G) Longitudinal State-Space Matrix A_long")

    # State vector: [V/V*, , , ]
    a11 = -g_V * qS_W * (Ma_trim * CD_Ma + 2 * CD_trim)
    a12 = -g_V
    a13 = -g_V * qS_W * CD_alpha
    a14 = -g_V * qS_W * CD_q1 * c_2V

    a21 =  g_V * qS_W * (Ma_trim * CL_Ma + 2 * CL_trim)
    a22 =  0.0
    a23 =  g_V * qS_W * CL_alpha
```

```
    a24 =  g_V * qS_W * CL_q1 * c_2V

    a31 = 0.0
    a32 = 0.0
    a33 = 0.0
    a34 = 1.0

    # k-factors for the  row
    k_alpha_dot   = 1 + g_V * qS_W * CL_q1 * c_2V
    k_gamma       = (g_V**2) * qS_W * (Ma_trim * CL_Ma + 2 * CL_trim)
    k_alpha       = ((g_V**2) * (qS_W**2)
                     * (Ma_trim * CL_Ma + 2 * CL_trim) * CD_alpha
                     + qSc_Iyy * Cm_alpha)
    k_alphadot_r = (qSc_Iyy * c_2V * Cm_q1
                     - g_V * qS_W * CL_alpha
                     + (g_V**2) * (qS_W**2) * CD_q1 * c_2V
                     * (Ma_trim * CL_Ma + 2 * CL_trim))
    k_V_r         = ((g_V**2) * (qS_W**2)
                     * (Ma_trim * CL_Ma + 2 * CL_trim)
                     * (Ma_trim * CD_Ma + 2 * CD_trim)
                     + qSc_Iyy * 0.0)   # Cm_Ma = 0

    a41 = k_V_r        / k_alpha_dot
    a42 = k_gamma      / k_alpha_dot
    a43 = k_alpha      / k_alpha_dot
    a44 = k_alphadot_r / k_alpha_dot

    A_long = np.array([
        [a11, a12, a13, a14],
        [a21, a22, a23, a24],
```

```
        [a31, a32, a33, a34],
        [a41, a42, a43, a44],
    ])

    print("\n  A_long =")
    print(np.array2string(A_long, precision=4, suppress_small=True,
                          formatter={'float_kind': lambda x: f"{x:10.4f}"}))

    evals_long = np.linalg.eigvals(A_long)
    print("\n  Eigenvalues of A_long:")
    for ev in evals_long:
        print(f"     = {ev.real:+.5f}  {'+' if ev.imag >= 0 else '-'}  j{abs(ev.imag):.5f}")

    # Identify modes from eigenvalues
    for ev in evals_long:
        wn = abs(ev)
        if wn > 0:
            zeta_ev = -ev.real / wn if ev.imag != 0 else None
            if abs(ev.imag) > 0.05:
                label = "Short Period" if wn > 1.0 else "Phugoid"
                print(f"    -> {label}: omega_n = {wn:.4f} rad/s, "
                      f"zeta = {(-ev.real/wn):.4f}")


    #  LATERAL-DIRECTIONAL AERODYNAMICS (Sec. 9.1.3)

    sub_header("H) Lateral-Directional Aerodynamic Coefficients")

    Gamma_rad    = Gamma_deg * D2R
```

```
    alpha_t_rad  = alpha_trim_rad

    #  Sidewash factor
    # (1 + dsig/dbeta) * eta_VT = formula from PDF
    Lambda_c4_rad = Lambda_c4_deg * D2R
    sidewash = (0.724
                + 3.06 * (S_VT / S_w) / (2.0)
                + 0.4 * z_w / D_f
                + 0.009 * AR_w)
    # Note: PDF uses (S_VT/S_w)/(1+cos(_c/4)) with cos1 for small sweep
    # Reproduced exactly as in PDF:
    sidewash = (0.724
                + 3.06 * (S_VT / S_w) / (1 + np.cos(Lambda_c4_rad))
                + 0.4 * z_w / D_f
                + 0.009 * AR_w)

    #  Side-force coefficients
    CY_beta_w   = -0.0001 * Gamma_deg * D2R   #  0
    CY_beta_VT = (-eta * CL_alpha_VT * (S_VT / S_w) * sidewash)
    CY_beta    = CY_beta_w + CY_beta_VT

    #  Yawing-moment coefficients
    Cn_beta_w   = -0.075 * Gamma_deg * D2R * CL_trim
    Cn_beta_VT = -CY_beta_VT * (l_VT / b_w)
    Cn_beta    = Cn_beta_w + Cn_beta_VT

    #  Rolling-moment coefficients
    # z = z_VT*cos() -- l_VT*sin()
    z_eff = z_VT * np.cos(alpha_t_rad) - l_VT * np.sin(alpha_t_rad)

```

```
    Cl_beta_w  = -(Gamma_rad * CL_alpha_w * c_r * b_w) / (6 * S_w)
    Cl_beta_VT = (z_eff / b_w) * CY_beta_VT
    Cl_beta    = Cl_beta_w + Cl_beta_VT

    print(f"  Sidewash factor (1+dsig/db)*eta_VT = {sidewash:.4f}")
    print(f"  CY_beta   = {CY_beta:.4f} /rad")
    print(f"  Cn_beta   = {Cn_beta:.4f} /rad")
    print(f"  Cl_beta   = {Cl_beta:.4f} /rad")

    #  Roll-rate derivatives
    CY_p2_VT = (2 / b_w) * (z_eff - z_VT) * CY_beta_VT
    CY_p2    = CY_p2_VT

    Cl_p2_VT = 2 * (z_eff / b_w) * ((z_eff - z_VT) / b_w) *
   CY_beta_VT
    Cl_p2_w  = -(1/6) * (CL_alpha + CD_trim)    # CL_alpha here is
   whole-ac
    Cl_p2    = Cl_p2_VT + Cl_p2_w

    lVT_eff = l_VT * np.cos(alpha_t_rad) + z_VT * np.sin(
   alpha_t_rad)
    Cn_p2_w  = -(1/6) * (CL_trim - CD_alpha)
    Cn_p2_VT = (-(2 / b_w) * lVT_eff
                 * ((z_eff - z_VT) / b_w) * CY_beta_VT)
    Cn_p2    = Cn_p2_w + Cn_p2_VT

    print(f"\n  CY_p2    = {CY_p2:.4f} /rad")
    print(f"  Cl_p2    = {Cl_p2:.4f} /rad")
    print(f"  Cn_p2    = {Cn_p2:.4f} /rad")

    #  Yaw-rate derivatives
```

```
    CY_r1_VT = -(2 / b_w) * lVT_eff * CY_beta_VT
    Cl_r1_w  = CL_trim / 3
    Cl_r1_VT = (-(2 / b_w**2) * lVT_eff * z_eff * CY_beta_VT)
    Cl_r1    = Cl_r1_w + Cl_r1_VT
    Cn_r1_VT = (2 / b_w**2) * lVT_eff**2 * CY_beta_VT
    Cn_r1    = Cn_r1_VT

    # Secondary yaw/roll-rate terms
    Cn_r2    = -CD_trim / 3
    Cl_r2    =  CL_trim / 3

    print(f"  CY_r1    = {CY_r1_VT:.4f} /rad")
    print(f"  Cl_r1    = {Cl_r1:.4f} /rad")
    print(f"  Cn_r1    = {Cn_r1:.4f} /rad")
    print(f"  Cn_r2    = {Cn_r2:.4f} /rad")
    print(f"  Cl_r2    = {Cl_r2:.4f} /rad")


    # (I) LATERAL-DIRECTIONAL DYNAMIC MODES (2nd-order)

    sub_header("I) Lateral-Directional Dynamic Modes (Approximate)"
   )

    qSb_Ixx = q_trim * S_w * b_w / I_xx
    qSb_Izz = q_trim * S_w * b_w / I_zz
    qS_m    = q_trim * S_w / m
    b_2V    = b_w / (2 * V_trim)

    # Dimensional derivatives
    N_beta  = qSb_Izz * Cn_beta
    N_r2    = qSb_Izz * Cn_r2 * b_2V
```

```
    Y_beta   = qS_m    * CY_beta
    L_beta   = qSb_Ixx * Cl_beta
    L_p2     = qSb_Ixx * Cl_p2   * b_2V
    N_r1     = qSb_Izz * Cn_r1   * b_2V
    L_r1     = qSb_Ixx * Cl_r1   * b_2V
    L_r2     = qSb_Ixx * Cl_r2   * b_2V

    print(f"  N_beta   = {N_beta:.4f} /s")
    print(f"  Y_beta   = {Y_beta:.4f} m/s")
    print(f"  L_beta   = {L_beta:.4f} /s")
    print(f"  L_p2     = {L_p2:.4f} /s")
    print(f"  N_r1     = {N_r1:.4f} /s")
    print(f"  L_r1     = {L_r1:.4f} /s")

    # Roll mode eigenvalue  (_r  L_p2)
    lambda_roll = L_p2

    # Dutch roll mode
    wn_DR_sq = (N_beta
                + g_V * (Y_beta * N_r2 + L_beta / L_p2))
    wn_DR    = np.sqrt(abs(wn_DR_sq))
    two_zeta_wn_DR = (-N_r1
                      - g_V * (Y_beta + L_r1 / L_p2))
    zeta_DR  = two_zeta_wn_DR / (2 * wn_DR)

    # Spiral mode eigenvalue
    lambda_spiral = (g_V * (L_beta * N_r2 - N_beta * L_r2)
                     / (lambda_roll * wn_DR_sq))

    print(f"\n  ROLL MODE")
    print(f"    lambda_r  = {lambda_roll:.4f} /s")
```

```python

print(f"\n  DUTCH ROLL MODE")
print(f"    omega_n    = {wn_DR:.4f} rad/s")
print(f"    zeta       = {zeta_DR:.4f}")

print(f"\n  SPIRAL MODE")
print(f"    lambda_s   = {lambda_spiral:.5f} /s   "
      f"({'STABLE' if lambda_spiral < 0 else 'UNSTABLE'})")

# (J) LATERAL-DIRECTIONAL A-MATRIX (Sec. 9.1.4)

sub_header("J) Lateral-Directional State-Space Matrix A_lat-dir")

# State vector: [, , , ]
# from PDF Eq. 9.11
Alat = np.array([
    [0.0,
     1.0,
     0.0,
     0.0],
    [g_V * L_r2,
     L_p2,
     L_beta + g_V * Y_beta * L_r2,
     -L_r1],
    [0.0,
     0.0,
     0.0,
     1.0],
    [-g_V * N_r2,
     g_V,
```

```
        -(N_beta + g_V * Y_beta * N_r2),
        N_r1 + g_V * Y_beta],
    ])

    print("\n  A_lat-dir =")
    print(np.array2string(Alat, precision=4, suppress_small=True,
                          formatter={'float_kind': lambda x: f"{x:10.4f}"}))

    evals_lat = np.linalg.eigvals(Alat)
    print("\n  Eigenvalues of A_lat-dir:")
    for ev in evals_lat:
        print(f"    = {ev.real:+.5f}  {'+' if ev.imag >= 0 else '-'}  j{abs(ev.imag):.5f}")

    # (K) CONTROL DERIVATIVES (Sec. 9.1.4 end)

    sub_header("K) Control Effectiveness & Derivatives")

    # Control effectiveness parameters (from PDF chart Fig 7.9)
    tau_de = 0.5    # elevator effectiveness
    tau_dr = 0.5    # rudder effectiveness
    tau_da = 0.22   # aileron effectiveness

    CL_de   = eta * (S_HT / S_w) * CL_alpha_HT * tau_de
    Cm_de   = -eta * V_HT * CL_alpha_HT * tau_de

    # Aileron (running 50-90% span)
    # Integral: c(y)*y dy from 0.5b to 0.9b with linear taper
    y1, y2  = 0.5 * b_w, 0.9 * b_w
    integ   = ((y2**2 - y1**2) / 2
```

```
                  + 2 * (y2**3 - y1**3) / (3 * b_w) * (lam - 1))
    Cl_da    = -2 * cl_alpha_w * tau_da / (S_w * b_w) * c_r * integ

    Cn_da    = -(CD_alpha / cl_alpha_w) * Cl_da
    CY_da    = 0.0

    Cn_dr    = -V_VT * tau_dr * CL_alpha_VT
    CY_dr    = (S_VT / S_w) * tau_dr * CL_alpha_VT
    Cl_dr    = (S_VT / S_w) * tau_dr * CL_alpha_VT * (z_eff / b_w)

    print(f"  tau_de = {tau_de},  tau_dr = {tau_dr},  tau_da = {tau_da}")
    print(f"  CL_de   = {CL_de:.4f} /rad")
    print(f"  Cm_de   = {Cm_de:.4f} /rad")
    print(f"  Cl_da   = {Cl_da:.4f} /rad")
    print(f"  Cn_da   = {Cn_da:.4f} /rad")
    print(f"  Cn_dr   = {Cn_dr:.4f} /rad")
    print(f"  CY_dr   = {CY_dr:.4f} /rad")
    print(f"  Cl_dr   = {Cl_dr:.4f} /rad")


    # SUMMARY TABLE

    section_header("AIRCRAFT -- SUMMARY OF RESULTS")
    results = {
        "Trim AoA (deg)"            : alpha_trim_deg,
        "Trim Speed V* (m/s)"       : V_trim,
        "Trim CL"                   : CL_trim,
        "Trim CD"                   : CD_trim,
        "Thrust Required (N)"       : T_req,
        "Power Required (hp)"       : P_req / 745.7,
```

```
        "SP omega_n (rad/s) [approx]": wn_SP,
        "SP zeta           [approx]"   : zeta_SP,
        "SP omega_n (rad/s) [eigen]": abs(evals_long[np.argmax(
                                                [abs(e.imag) for e in
    evals_long])]),
        "Phugoid omega_n (rad/s)"   : wn_Ph,
        "Phugoid zeta"              : zeta_Ph,
        "Roll mode  (1/s)"         : lambda_roll,
        "Dutch-roll omega_n (rad/s)": wn_DR,
        "Dutch-roll zeta"           : zeta_DR,
        "Spiral mode  (1/s)"       : lambda_spiral,
    }
    for k, v in results.items():
        print(f"  {k:<40s}: {v:.5f}")


    # (L) EIGENVALUE PLOTS

    section_header("L) Eigenvalue Plots")

    fig, axes = plt.subplots(1, 2, figsize=(14, 6))
    fig.suptitle("UAV - Eigenvalue Analysis (Complex Plane)",
    fontsize=14, fontweight='bold')

    # -- Longitudinal --
    ax = axes[0]
    ax.axhline(0, color='black', linewidth=0.8, linestyle='--')
    ax.axvline(0, color='black', linewidth=0.8, linestyle='--')

    mode_labels_long = []
    for ev in evals_long:
```

```
        wn = abs(ev)
        if abs(ev.imag) > 0.05 and wn > 1.0:
            label, color = "Short Period", 'royalblue'
        elif abs(ev.imag) > 0.05:
            label, color = "Phugoid", 'darkorange'
        else:
            label, color = "Real mode", 'green'
        ax.plot(ev.real, ev.imag, 'o', color=color, markersize=10, zorder=5)
        ax.annotate(
            f"  ={ev.real:+.3f}{ev.imag:+.3f}j",
            (ev.real, ev.imag), fontsize=7.5, color=color, va='center')
        mode_labels_long.append((label, color))

    seen = {}
    for lbl, col in mode_labels_long:
        seen[lbl] = col
    legend_patches = [mpatches.Patch(color=c, label=l) for l, c in seen.items()]
    ax.legend(handles=legend_patches, fontsize=9)
    ax.set_xlabel("Real (sigma) [1/s]", fontsize=10)
    ax.set_ylabel("Imaginary (jw) [1/s]", fontsize=10)
    ax.set_title("Longitudinal Eigenvalues", fontsize=11)
    ax.grid(True, linestyle=':', alpha=0.6)
    ax.set_aspect('equal', adjustable='datalim')

    # -- Lateral-Directional --
    ax = axes[1]
    ax.axhline(0, color='black', linewidth=0.8, linestyle='--')
    ax.axvline(0, color='black', linewidth=0.8, linestyle='--')
```

```
    mode_labels_lat = []
    for ev in evals_lat:
        wn = abs(ev)
        if abs(ev.imag) > 0.05:
            label, color = "Dutch Roll", 'crimson'
        elif abs(ev.real) > 0.5:
            label, color = "Roll Mode", 'royalblue'
        else:
            label, color = "Spiral Mode", 'darkorange'
        ax.plot(ev.real, ev.imag, 's', color=color, markersize=10, zorder=5)
        ax.annotate(
            f"  ={ev.real:+.3f}{ev.imag:+.3f}j",
            (ev.real, ev.imag), fontsize=7.5, color=color, va='center')
        mode_labels_lat.append((label, color))

    seen = {}
    for lbl, col in mode_labels_lat:
        seen[lbl] = col
    legend_patches = [mpatches.Patch(color=c, label=l) for l, c in seen.items()]
    ax.legend(handles=legend_patches, fontsize=9)
    ax.set_xlabel("Real  (sigma)  [1/s]", fontsize=10)
    ax.set_ylabel("Imaginary  (jw)  [1/s]", fontsize=10)
    ax.set_title("Lateral-Directional Eigenvalues", fontsize=11)
    ax.grid(True, linestyle=':', alpha=0.6)
    ax.set_aspect('equal', adjustable='datalim')

    plt.tight_layout()
```

```
    plt.savefig("eigenvalue_plot.png", dpi=150, bbox_inches='tight')
    plt.show()
    print("\n  Eigenvalue plots saved to eigenvalue_plot.png")





# MAIN

if __name__ == "__main__":
    print("=" * 60)
    print("  FLIGHT DYNAMICS & STABILITY ANALYSIS")
    print("  Chapter 9 Appendix -- Case Studies")
    print("=" * 60)

    aircraft()

    print("\n" + "=" * 60)
    print("  ANALYSIS COMPLETE")
    print("=" * 60)
```

Listing A.6: Flight Stability Analysis

```
%% UAV Performance Analysis - Optimum Velocities and Power
%% AUTHOR = ABHIJEET MANGELA (AE25M034)

clear; clc;

%% Aircraft Parameters
Sref = 0.5307;            % Reference area (m^2)
AR = 7.53;                % Aspect ratio
b = 2;                    % Wingspan (m)
e = 0.8;                  % Oswald efficiency factor
C_d_o = 0.02873;          % Zero-lift drag coefficient (given)
L_D_max = 12.55;          % Maximum lift-to-drag ratio

%% Flight Conditions
rho = 1.142;              % Air density at cruise altitude (200m)
g = 9.81;                 % Gravitational acceleration

%% Weight Configurations
W1 = 6.5 * g;             % Weight before payload drop (N)
W2 = 5.3 * g;             % Weight after payload drop (N)

%% Initialize Results Storage
segments = {'Climb-1'; 'Cruise-1'; 'Loiter-1'; 'Climb-2'; 'Cruise-2
    '; 'Loiter-2'};
velocities = zeros(6,1);
thrusts = zeros(6,1);
powers = zeros(6,1);

%% Flight Phase Calculations - Before Payload Drop

```

```
% --- Climb-1 (gamma = 5 deg) ---
gamma_climb = 5;
W = W1;
L_climb = W * cosd(gamma_climb);
D_climb = 2 * L_climb / (sqrt(3) * L_D_max);
T_climb = D_climb + W * sind(gamma_climb);

% Find optimal velocity for climb
v0 = 20;  % Initial guess
C_L_climb_func = @(v) L_climb / (0.5 * rho * Sref * v^2);
C_D_climb_func = @(v) C_d_o + C_L_climb_func(v)^2 / (pi * e * AR);
D_climb_func = @(v) 0.5 * rho * Sref * v^2 * C_D_climb_func(v);
drag_error = @(v) D_climb_func(v) - D_climb;
V_climb1 = fsolve(drag_error, v0, optimset('Display','off'));
P_climb1 = T_climb * V_climb1;

velocities(1) = V_climb1;
thrusts(1) = T_climb;
powers(1) = P_climb1;

% --- Cruise-1 (gamma = 0 deg, level flight) ---
gamma_cruise = 0;
W = W1;
L_cruise = W * cosd(gamma_cruise);
D_cruise = L_cruise / L_D_max;
T_cruise = D_cruise + W * sind(gamma_cruise);

% Find optimal velocity for cruise
C_L_cruise_func = @(v) L_cruise / (0.5 * rho * Sref * v^2);
C_D_cruise_func = @(v) C_d_o + C_L_cruise_func(v)^2 / (pi * e * AR)
   ;
```

```
D_cruise_func = @(v) 0.5 * rho * Sref * v^2 * C_D_cruise_func(v);
drag_error = @(v) D_cruise_func(v) - D_cruise;
V_cruise1 = fsolve(drag_error, v0, optimset('Display','off'));
P_cruise1 = T_cruise * V_cruise1;

velocities(2) = V_cruise1;
thrusts(2) = T_cruise;
powers(2) = P_cruise1;

% --- Loiter-1 (gamma = 0 deg, minimum power) ---
gamma_loiter = 0;
W = W1;
L_loiter = W * cosd(gamma_loiter);
D_loiter = 2 * L_loiter / (sqrt(3) * L_D_max);
T_loiter = D_loiter + W * sind(gamma_loiter);

% Find optimal velocity for loiter
C_L_loiter_func = @(v) L_loiter / (0.5 * rho * Sref * v^2);
C_D_loiter_func = @(v) C_d_o + C_L_loiter_func(v)^2 / (pi * e * AR);
D_loiter_func = @(v) 0.5 * rho * Sref * v^2 * C_D_loiter_func(v);
drag_error = @(v) D_loiter_func(v) - D_loiter;
V_loiter1 = fsolve(drag_error, v0, optimset('Display','off'));
P_loiter1 = T_loiter * V_loiter1;

velocities(3) = V_loiter1;
thrusts(3) = T_loiter;
powers(3) = P_loiter1;

%% Flight Phase Calculations - After Payload Drop

```

```
% --- Climb-2 (gamma = 5 deg) ---
gamma_climb = 5;
W = W2;
L_climb = W * cosd(gamma_climb);
D_climb = 2 * L_climb / (sqrt(3) * L_D_max);
T_climb = D_climb + W * sind(gamma_climb);

% Find optimal velocity for climb
C_L_climb_func = @(v) L_climb / (0.5 * rho * Sref * v^2);
C_D_climb_func = @(v) C_d_o + C_L_climb_func(v)^2 / (pi * e * AR);
D_climb_func = @(v) 0.5 * rho * Sref * v^2 * C_D_climb_func(v);
drag_error = @(v) D_climb_func(v) - D_climb;
V_climb2 = fsolve(drag_error, v0, optimset('Display','off'));
P_climb2 = T_climb * V_climb2;

velocities(4) = V_climb2;
thrusts(4) = T_climb;
powers(4) = P_climb2;

% --- Cruise-2 (gamma = 0 deg, level flight) ---
gamma_cruise = 0;
W = W2;
L_cruise = W * cosd(gamma_cruise);
D_cruise = L_cruise / L_D_max;
T_cruise = D_cruise + W * sind(gamma_cruise);

% Find optimal velocity for cruise
C_L_cruise_func = @(v) L_cruise / (0.5 * rho * Sref * v^2);
C_D_cruise_func = @(v) C_d_o + C_L_cruise_func(v)^2 / (pi * e * AR)
    ;
D_cruise_func = @(v) 0.5 * rho * Sref * v^2 * C_D_cruise_func(v);
```

```
drag_error = @(v) D_cruise_func(v) - D_cruise;
V_cruise2 = fsolve(drag_error, v0, optimset('Display','off'));
P_cruise2 = T_cruise * V_cruise2;

velocities(5) = V_cruise2;
thrusts(5) = T_cruise;
powers(5) = P_cruise2;

% --- Loiter-2 (gamma = 0 deg, minimum power) ---
gamma_loiter = 0;
W = W2;
L_loiter = W * cosd(gamma_loiter);
D_loiter = 2 * L_loiter / (sqrt(3) * L_D_max);
T_loiter = D_loiter + W * sind(gamma_loiter);

% Find optimal velocity for loiter
C_L_loiter_func = @(v) L_loiter / (0.5 * rho * Sref * v^2);
C_D_loiter_func = @(v) C_d_o + C_L_loiter_func(v)^2 / (pi * e * AR);
D_loiter_func = @(v) 0.5 * rho * Sref * v^2 * C_D_loiter_func(v);
drag_error = @(v) D_loiter_func(v) - D_loiter;
V_loiter2 = fsolve(drag_error, v0, optimset('Display','off'));
P_loiter2 = T_loiter * V_loiter2;

velocities(6) = V_loiter2;
thrusts(6) = T_loiter;
powers(6) = P_loiter2;

%% Create Results Table
ResultsTable = table(segments, velocities, thrusts, powers, ...
```

```
    'VariableNames', {'Segment', 'Velocity_m_s', 'Thrust_N', 'Power_W'});

%% Display Results
fprintf('\n========================================\n');
fprintf('UAV PERFORMANCE ANALYSIS RESULTS\n');
fprintf('========================================\n\n');

fprintf('%-12s | %10s | %10s | %10s\n', 'Segment', 'Velocity', 'Thrust', 'Power');
fprintf('%-12s | %10s | %10s | %10s\n', '', '(m/s)', '(N)', '(W)');
fprintf('----------------------------------------------------------\n');

for i = 1:length(segments)
    fprintf('%-12s | %10.2f | %10.2f | %10.1f\n', ...
        segments{i}, velocities(i), thrusts(i), powers(i));
end

fprintf('----------------------------------------------------------\n\n');

% Display the table object
disp(ResultsTable);

%% Export Results to Excel
% Define output filename
output_filename = 'UAV_Performance_Results.xlsx';

% Write the main results table
```

```
writetable(ResultsTable, output_filename, 'Sheet', 'Performance
    Data');

% Create a summary sheet with aircraft parameters
summary_data = {
    'Parameter', 'Value', 'Unit';
    'Reference Area (Sref)', Sref, 'm^2';
    'Aspect Ratio (AR)', AR, '-';
    'Wingspan (b)', b, 'm';
    'Oswald Efficiency (e)', e, '-';
    'Zero-lift Drag Coeff (C_d_o)', C_d_o, '-';
    'Max L/D Ratio', L_D_max, '-';
    'Air Density (rho)', rho, 'kg/m^3';
    'Weight Before Payload Drop', W1, 'N';
    'Weight After Payload Drop', W2, 'N';
    'Payload Mass', (W1-W2)/g, 'kg'
};

% Write summary to second sheet
writecell(summary_data, output_filename, 'Sheet', 'Aircraft
    Parameters');

fprintf('\n*** Results exported to: %s ***\n\n', output_filename);
```

Listing A.7: Performance analysis for different segments

```
% FLIGHT ENVELOPE DIAGRAM
% AUTHOR : MONISHA VIJAYAN(AE25M037)
clear;clc;close all;
W=(6.592)*9.81; %(6.592-1.2) after payload drop
S=0.5307;
rho=1.220;
CL_max_pos=1.3;
CL_max_neg=-0.8;
n_max_pos=3.5;
n_max_neg=-1.5;
V_s=sqrt((2*W)/(rho*S*CL_max_pos));
V_max=24;
V_d=1.25*V_max;
V_a=V_s*sqrt(n_max_pos);
V_g=sqrt((2*abs(n_max_neg)*W)/(rho*S*abs(CL_max_neg)));
V=0:0.1:V_d;
n_curve_pos=(rho.*V.^2.*S.*CL_max_pos)/(2*W);
n_curve_neg=(rho.*V.^2.*S.*CL_max_neg)/(2*W);
n_curve_pos(n_curve_pos>n_max_pos)=n_max_pos;
n_curve_pos(V>V_a)=NaN;
n_curve_neg(V>V_g)=NaN;
figure('Color','w')
ax=gca;ax.Color='w';ax.XColor='k';ax.YColor='k';hold on;grid on;ax.
    GridColor='k';
plot(V,n_curve_pos,'b','LineWidth',2);
plot(V,n_curve_neg,'r','LineWidth',2);
line([V_a V_d],[n_max_pos n_max_pos],'Color','b','LineStyle','--','
    LineWidth',2);
line([V_g V_d],[n_max_neg n_max_neg],'Color','r','LineStyle','--','
    LineWidth',2);
```

```
line([V_max V_max],[n_max_neg n_max_pos],'Color','k','LineStyle','
    --');
line([V_d V_d],[n_max_neg n_max_pos],'Color','m','LineWidth',3);
plot(V_a,n_max_pos,'ko','MarkerFaceColor','g');
plot(V_g,n_max_neg,'ko','MarkerFaceColor','k');
plot(V_d,0,'ko','MarkerFaceColor','m');
xlabel('Velocity (V) [m/s]','Color','k','FontWeight','bold');
ylabel('Load Factor (n)','Color','k','FontWeight','bold');
title('UAV V-n Diagram','Color','k');
text(V_a,n_max_pos+0.2,'V_a','Color','k');
text(V_d,0.2,'V_d','Color','k');
text(V_g,n_max_neg+0.2,'V_g','Color','k')
axis([0 V_d+5 n_max_neg-0.5 n_max_pos+0.5]);
fprintf('Vs = %.2f m/s\n',V_s);
fprintf('Vmax = %.2f m/s\n',V_max);
fprintf('Vd = %.2f m/s\n',V_d);
fprintf('Va = %.2f m/s\n',V_a);
fprintf('Vg = %.2f m/s\n',V_g);
```

Listing A.8: Flight Envelope Diagram

```
% CONTROL SURFACES DESIGN
% AUTHOR : MONISHA VIJAYAN (AE25M037)
clc; clear;
S_w   =0.5307;
b_w   = 2;
c_w   = 0.2653;
AR_w  = b_w^2 / S_w;
rho_SL = 1.225;
rho    = 1.142;
V_stall = 12.4;
V_LOF = 1.1 * V_stall;
V_MC  = 1.2 * V_stall;
V_design = 18;
V_D = 1.25 * V_design;
MTOW  = 6.592;
g  = 9.81;
W = MTOW * g;
CG_aft  = 0.3;
CL_alpha_w  = 3.6107;
CL_alpha_HT = 3.8310;
CL_alpha_VT = 3.83;
d_eps_d_alpha = 0.399;
N_engines  = 2;
T_engine  = 3.144*9.81;
y_engine  = 0.05;
Y_MC_moment = T_engine * y_engine;
V_H_target = 0.6596;
l_HT = 0.7;
CL_max= 1.2;
Cm_AC_w  = -0.0683;
```

```
x_ac= 0.1985;
Cm_trim_needed = ((CG_aft- x_ac)/c_w) * CL_max +Cm_AC_w
tau_e   = 0.4; % ASSUMED ELEVATOR EFFECTIVENESS FOR UAVs
dCm_dde = -V_H_target * tau_e * CL_alpha_HT
delta_e_required_trim = Cm_trim_needed / dCm_dde;
delta_e_deg      = rad2deg(delta_e_required_trim)
S_HT =0.1327;
c_HT = 0.172;
chord_ratio_e = 0.30;
c_e  = chord_ratio_e * c_HT;
b_e  = S_HT / c_HT;
S_e  = c_e * b_e;
Ch_delta_e = -0.0010 * (180/pi); % For UAV typical servo sizing
q_LOF = 0.5 * rho * V_LOF^2;
H_e   = Ch_delta_e * q_LOF * S_e * c_e * abs(delta_e_required_trim)
    ;
V_V_target = 0.032;
l_VT = 0.7;
S_VT = V_V_target * S_w * b_w / l_VT
Y_bank   = W * sind(5);
y_cg_bank = 0;
N_required = Y_MC_moment + Y_bank * 0
q_MC   = 0.5 * rho * V_MC^2;
tau_r  = 0.35;
delta_r_required = N_required / (l_VT * CL_alpha_VT * tau_r * q_MC
    * S_VT);
delta_r_deg  = rad2deg(delta_r_required);
c_VT  = sqrt(S_VT * 0.6);
chord_ratio_r = 0.30;
c_r  = chord_ratio_r * c_VT;
h_VT  = S_VT / c_VT;
```

```
S_r   = c_r * h_VT;
Ch_delta_r = -0.010 * (180/pi);
H_r   = Ch_delta_r * q_MC * S_r * c_r * abs(delta_r_required);
pb_2V_min = 0.06;
V_2    = 1.2 * V_stall;
p_min = pb_2V_min * 2 * V_2 / b_w;
y1_frac   = 0.6;
y2_frac   = 0.98;
y1 = y1_frac * b_w/2;
y2 = y2_frac * b_w/2;
tau_a   = 0.45;
Cl_delta_a    = (CL_alpha_w / (pi * AR_w)) * tau_a * (y2^2 - y1^2) /
    (S_w * b_w/2);
Cl_p = -CL_alpha_w / 6;
delta_a_req   = -pb_2V_min * Cl_p / Cl_delta_a;
delta_a_deg   = rad2deg(delta_a_req);
c_a   = 0.25*c_w;
b_a   = (y2 - y1);
S_a   = c_a * b_a;
q_V2  = 0.5 * rho * V_2^2;
Ch_delta_a = -0.010 * (180/pi);
H_a    = Ch_delta_a * q_V2 * S_a * c_a * abs(delta_a_req);
fprintf('%-30s %12s %12s %12s\n', 'Parameter', 'Elevator', 'Rudder'
   , 'Aileron');
fprintf('%s\n', repmat('-', 1, 68));
fprintf('%-30s %12.2f %12.2f %12.2f\n', 'Panel area [m^2]', S_HT,
   S_VT, S_w);
fprintf('%-30s %12.2f %12.2f %12.2f\n', 'Surface area [m^2]', S_e,
   S_r, S_a*2);
fprintf('%-30s %12.2f %12.2f %12.2f\n', 'Surface chord [m]', c_e,
   c_r, c_a);
```

```
fprintf('%-30s %12.2f %12.2f %12.2f\n', 'Surface span [m]', b_e, h_VT, b_a*2);
fprintf('%-30s %12.1f %12.1f %12.1f\n', 'Max deflection req [deg]', abs(delta_e_deg), abs(delta_r_deg), abs(delta_a_deg));
fprintf('%-30s %12.3f %12.4f %12s\n', 'Volume coefficient', V_H_target, V_V_target, 'N/A');
```

LISTING A.9: Primary Control Surfaces Design

Appendix B

Contributions to Design

Table B.1: Team Contribution – Week 1

Name	Contribution
DS Vivek	Literature Survey and Market Research, Report, PPT
Divyam Pandey	Literature Survey and Market Research, Report, PPT
G.Vishwas	Literature Survey and Market Research, Report, PPT
Manav Agarwal	Literature Survey and Market Research, Report, PPT
Abhijeet Mangela	Literature Survey and Market Research, Report, PPT
Monisha V	Literature Survey and Market Research, Report, PPT

Table B.2: Team Contribution – Week 2

Name	Contribution
DS Vivek	Battery Data Collection, Report, PPT
Divyam Pandey	Similar UAV data collection, CAD Aided Estimation (wing area), Report, PPT
G.Vishwas	Similar UAV data collection, CAD Aided Estimation (wing area), Report, PPT
Manav Agarwal	Similar UAV data collection, Report, PPT
Abhijeet Mangela	MTOW coding, Payload Weight Estimation, Power and Energy Analysis, Report, PPT
Monisha V	Payload Weight Estimation, Report, PPT

Table B.3: Team Contribution – Week 3

Name	Contribution
DS Vivek	Battery selection, Report, PPT
Divyam Pandey	CAD Design, Report, PPT
G.Vishwas	CAD Design, Report, PPT
Manav Agarwal	CAD Aided Estimation (wing area), Report, PPT
Abhijeet Mangela	Motor and Propeller Selection, Report, PPT
Monisha V	L/D max, Optimum Velocity

Table B.4: Team Contribution – Week 4

Name	**Contribution**
DS Vivek	Battery analysis, Avionics, PPT
Divyam Pandey	CAD Design, Report, PPT
G.Vishwas	CAD Design, Report, PPT
Manav Agarwal	CAD Design, Report, PPT
Abhijeet Mangela	Motor, Propeller, Power Analysis, PPT
Monisha V	L/D max, Optimum Velocity, Power Analysis

Table B.5: Team Contribution – Week 5

Name	**Contribution**
DS Vivek	Documentation, report writing, and PPT
Divyam Pandey	Aerodynamic wing analysis (LLT in flow5), CAD modeling, and PPT
G.Vishwas	Aerodynamic wing analysis (LLT in flow5)
Manav Agarwal	CAD modeling and PPT
Abhijeet Mangela	Airfoil data collection, fuselage sizing via regression analysis, and PPT
Monisha V	Wing loading analysis, second weight estimation, and PPT

Week 6: Midterm Presentation

Table B.6: Team Contribution – Week 7

Name	Contribution
DS Vivek	Report, Presentation, Documentation
Divyam Pandey	CG estimation, Presentation, Moment contribution from wing & Fuselage
G.Vishwas	CG estimation, Report
Manav Agarwal	Flow-chart, Landing gear, Presentation
Abhijeet Mangela	Presentation
Monisha V	Fuselage, Flow-chart, Report

Table B.7: Team Contribution – Week 8

Name	Contribution
DS Vivek	Report and Presentation
Divyam Pandey	CAD design, XFLR5, CG Estimation, Drag Polar
G.Vishwas	CAD design, XFLR5, CG Estimation, Drag Polar
Manav Agarwal	Report and Presentation
Abhijeet Mangela	Peformance Analysis
Monisha V	Stability analytical Analysis, Vertical tail design, Flight Envelope

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